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(54) **STERILE BODILY-FLUID COLLECTION
DEVICE AND METHODS**

Publication Classification

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(51) **Int. Cl.**

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(52) **U.S. Cl.**

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(2013.01); **A61B 5/150992** (2013.01); **A61B**

5/154 (2013.01); **A61B 5/157** (2013.01); **A61B**

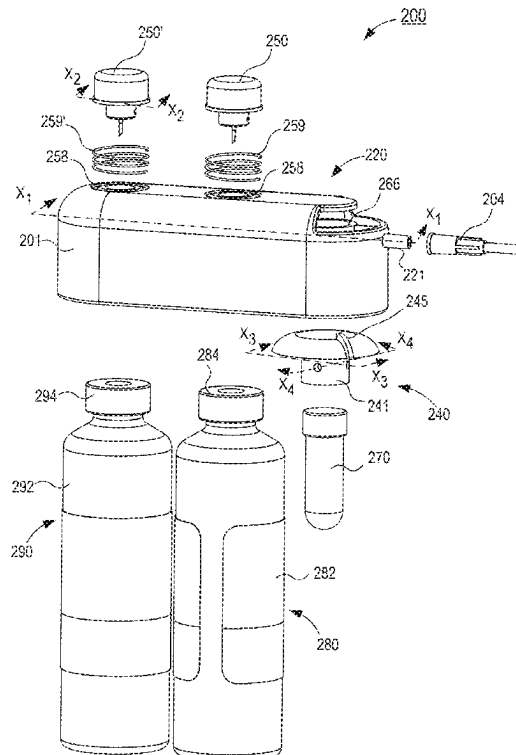
10/0045 (2013.01); **A61B 5/15003** (2013.01);

A61B 5/155 (2013.01)

(57)

ABSTRACT

An apparatus includes a pre-sample reservoir, a diversion mechanism, and a flow metering mechanism. The diversion mechanism has an inlet port couplable to a lumen-defining device to receive bodily-fluids from a patient, a first outlet port fluidically couplable to the pre-sample reservoir, and a second outlet port fluidically couplable to a sample reservoir. The diversion mechanism defines a first fluid flow path and a second flow path that are configured to place the first outlet port and the second outlet port, respectively, in fluid communication with the inlet port. The flow metering mechanism is configured to meter a flow of a predetermined volume of bodily-fluid through the first fluid flow path into the pre-sample reservoir, to meter a flow of a second volume of bodily-fluid through the second fluid flow path into the sample reservoir, and to display a volumetric indicator associated with the predetermined volume and the second volume.



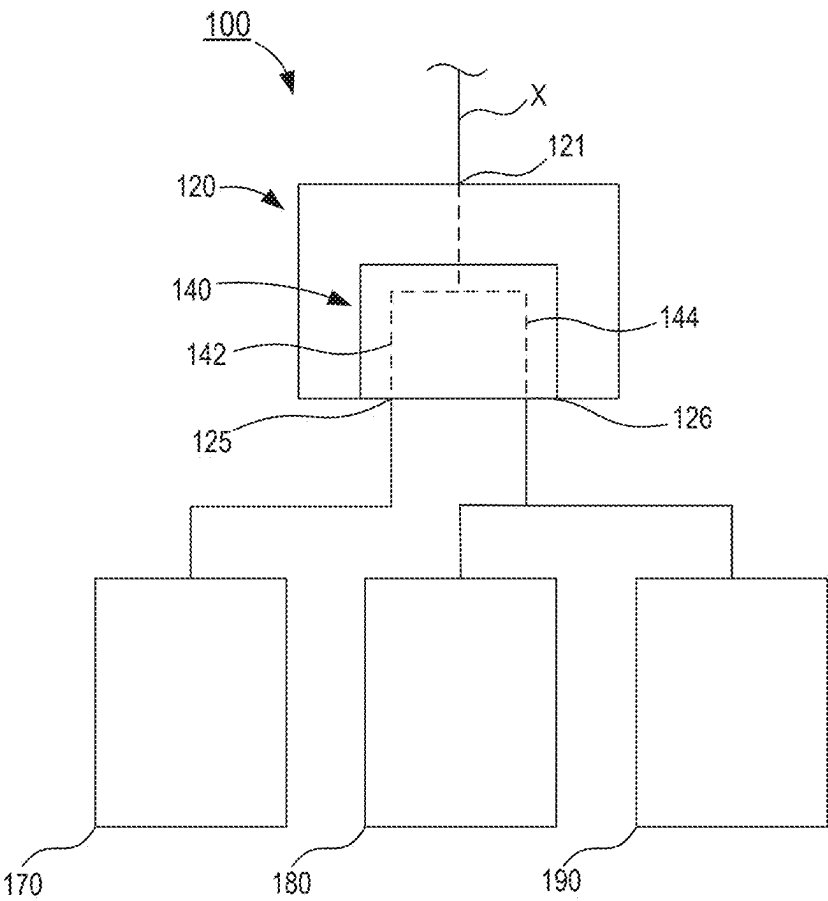


FIG. 1

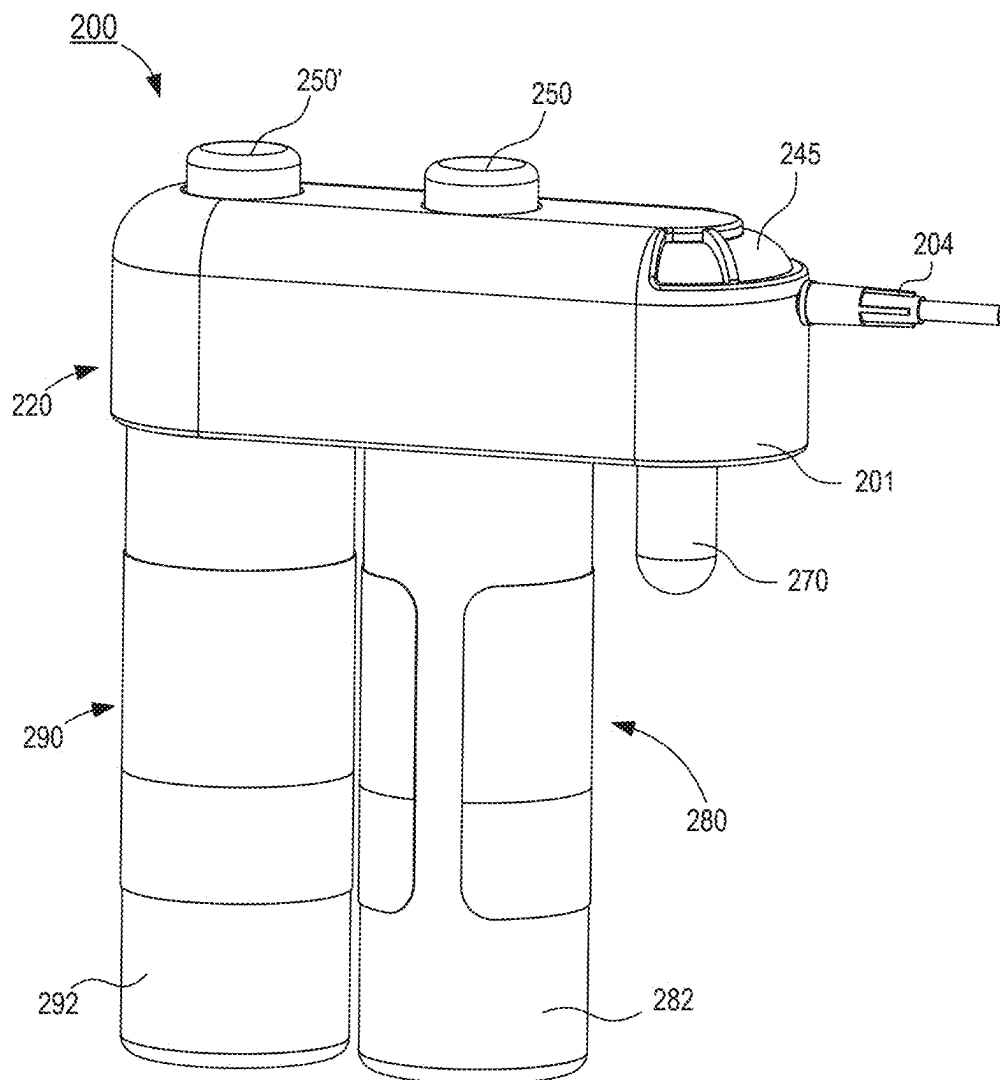
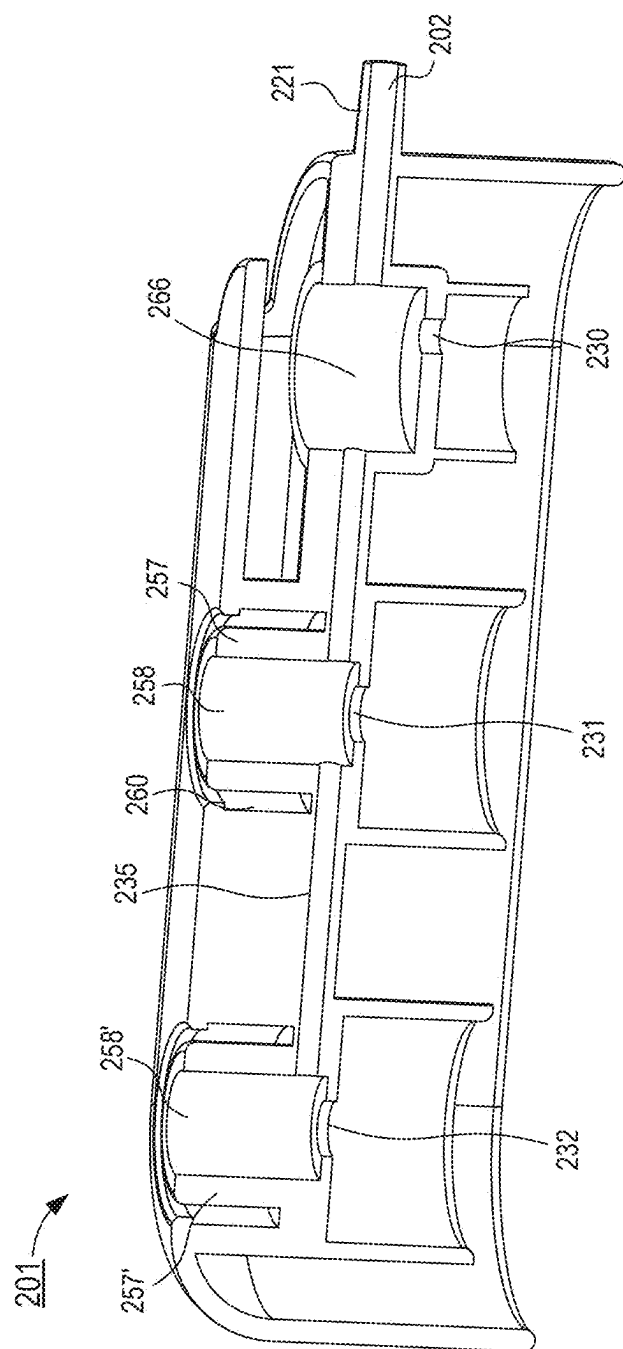


FIG. 2



FIG. 3



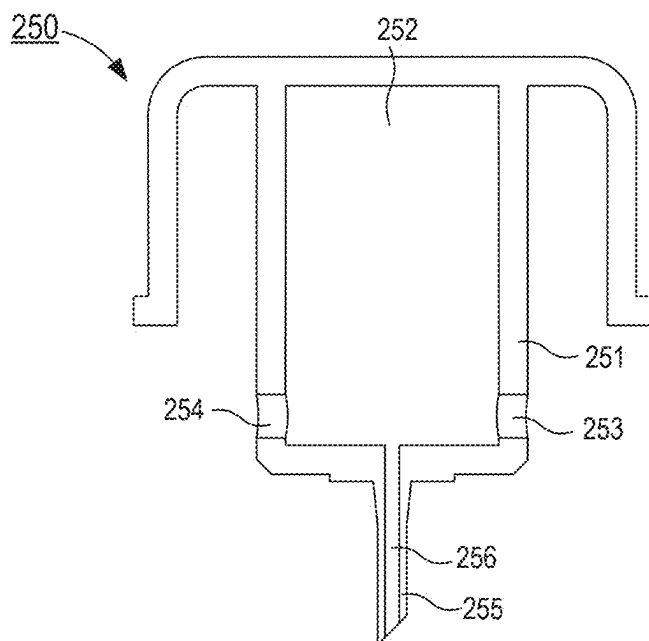


FIG. 5

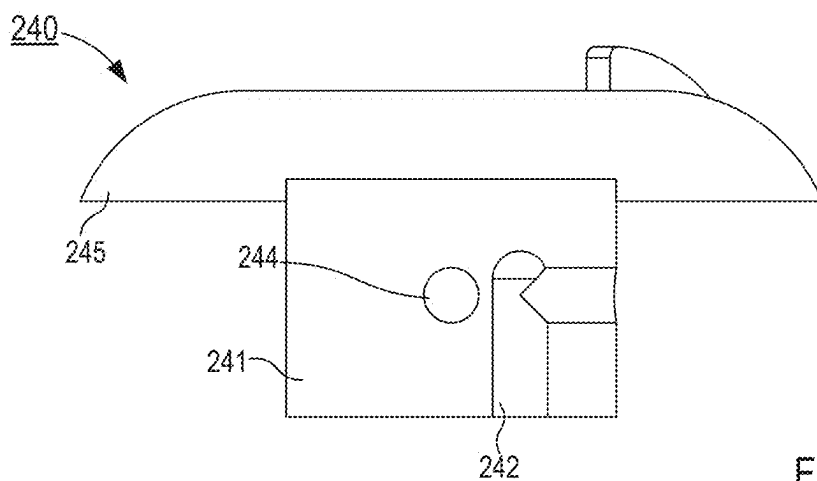


FIG. 6

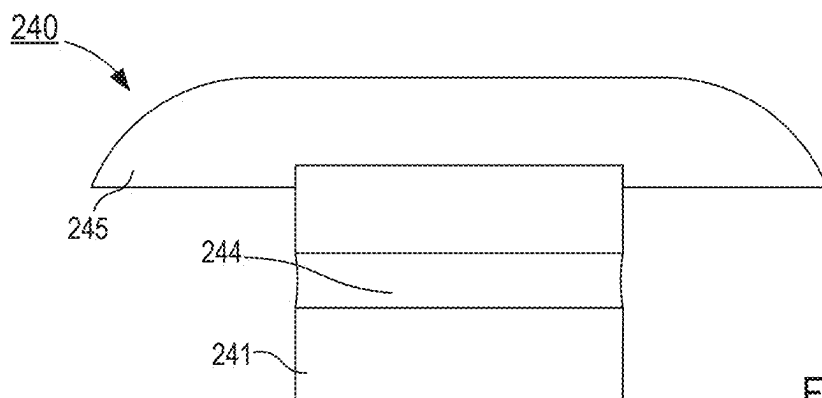


FIG. 7

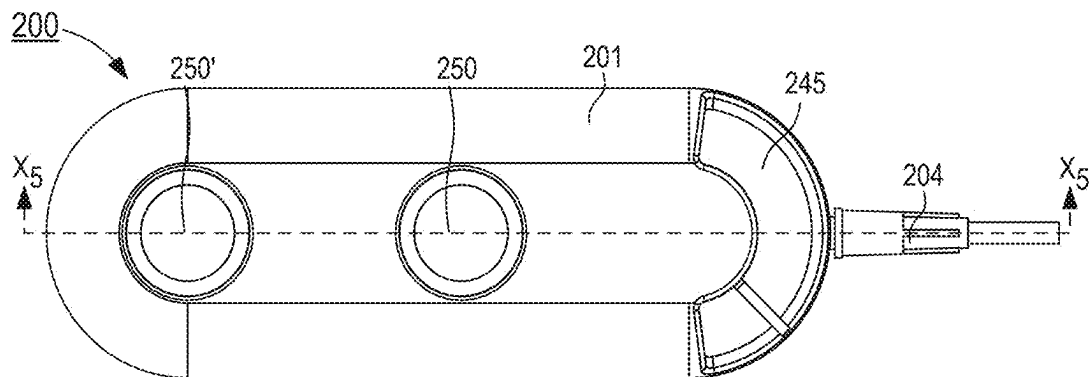


FIG. 8

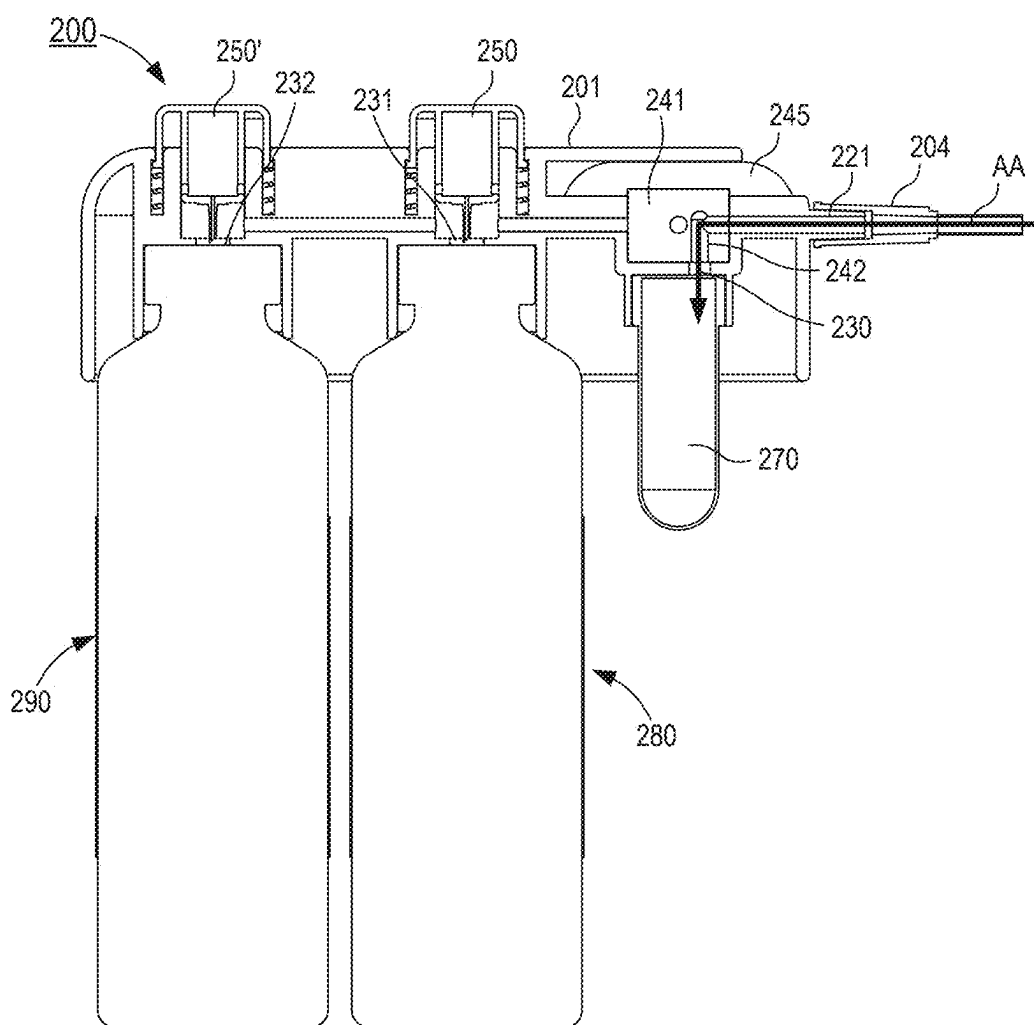


FIG. 9

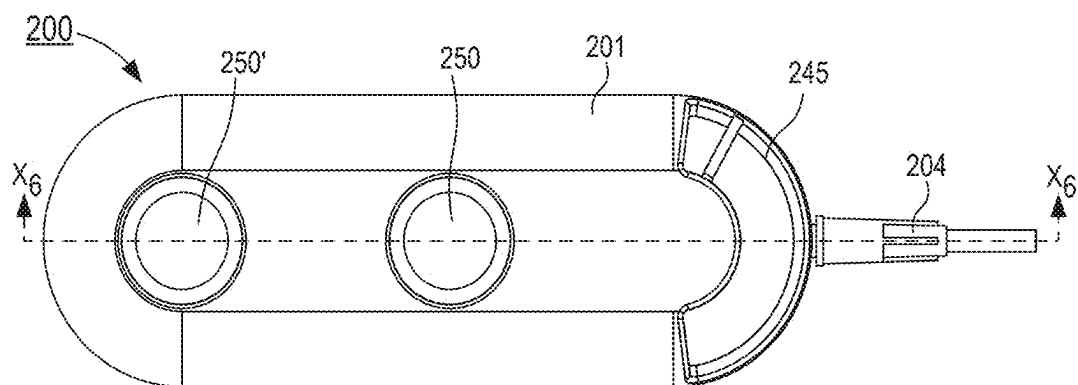


FIG. 10

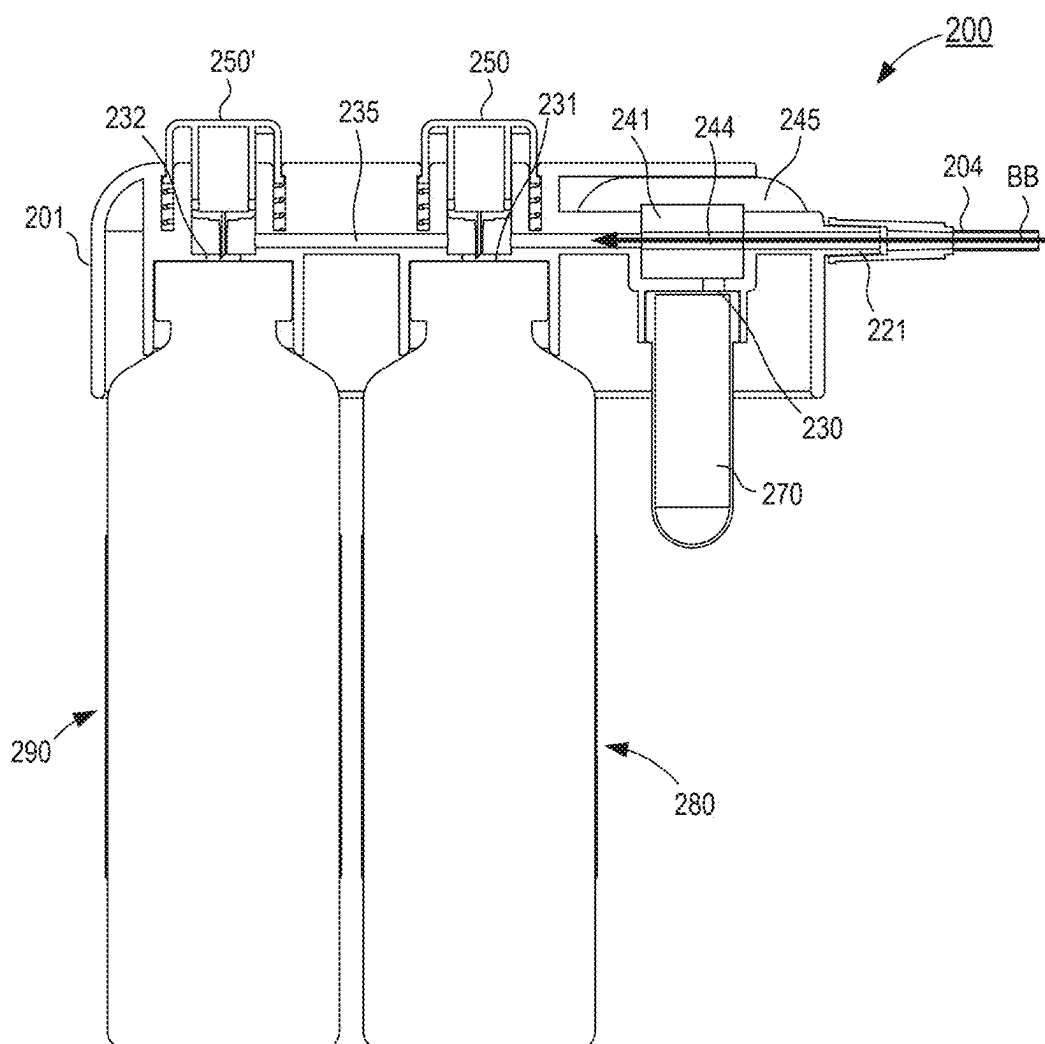


FIG. 11

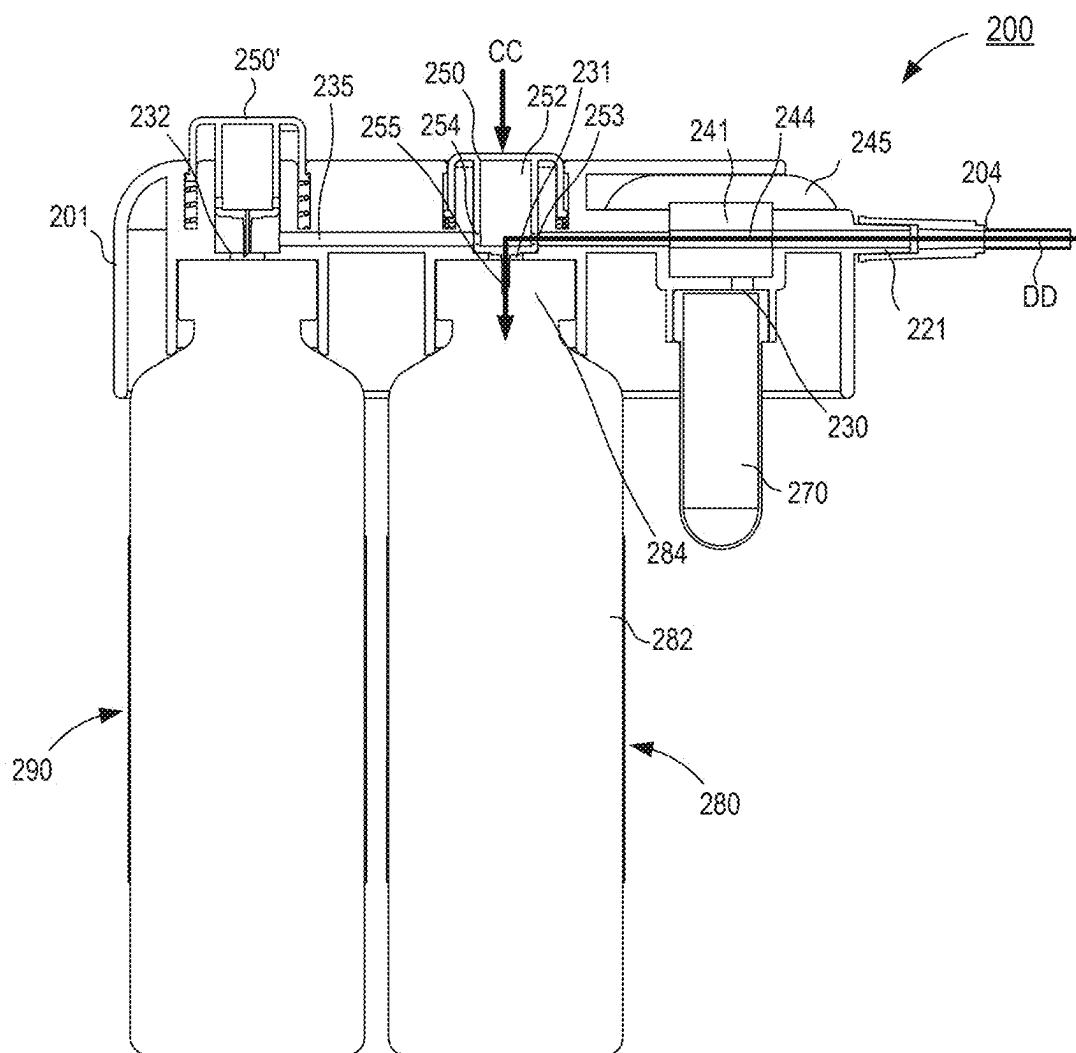


FIG. 12

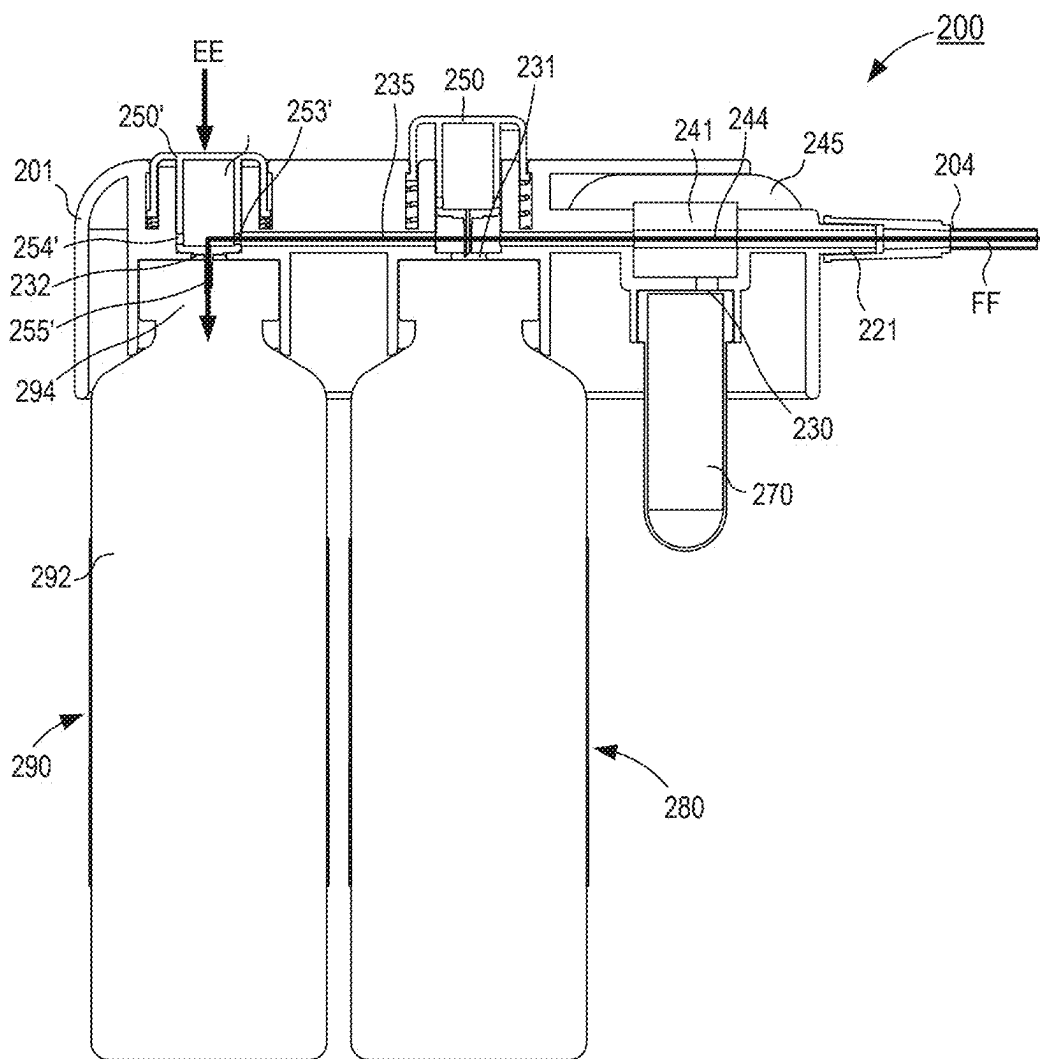


FIG. 13

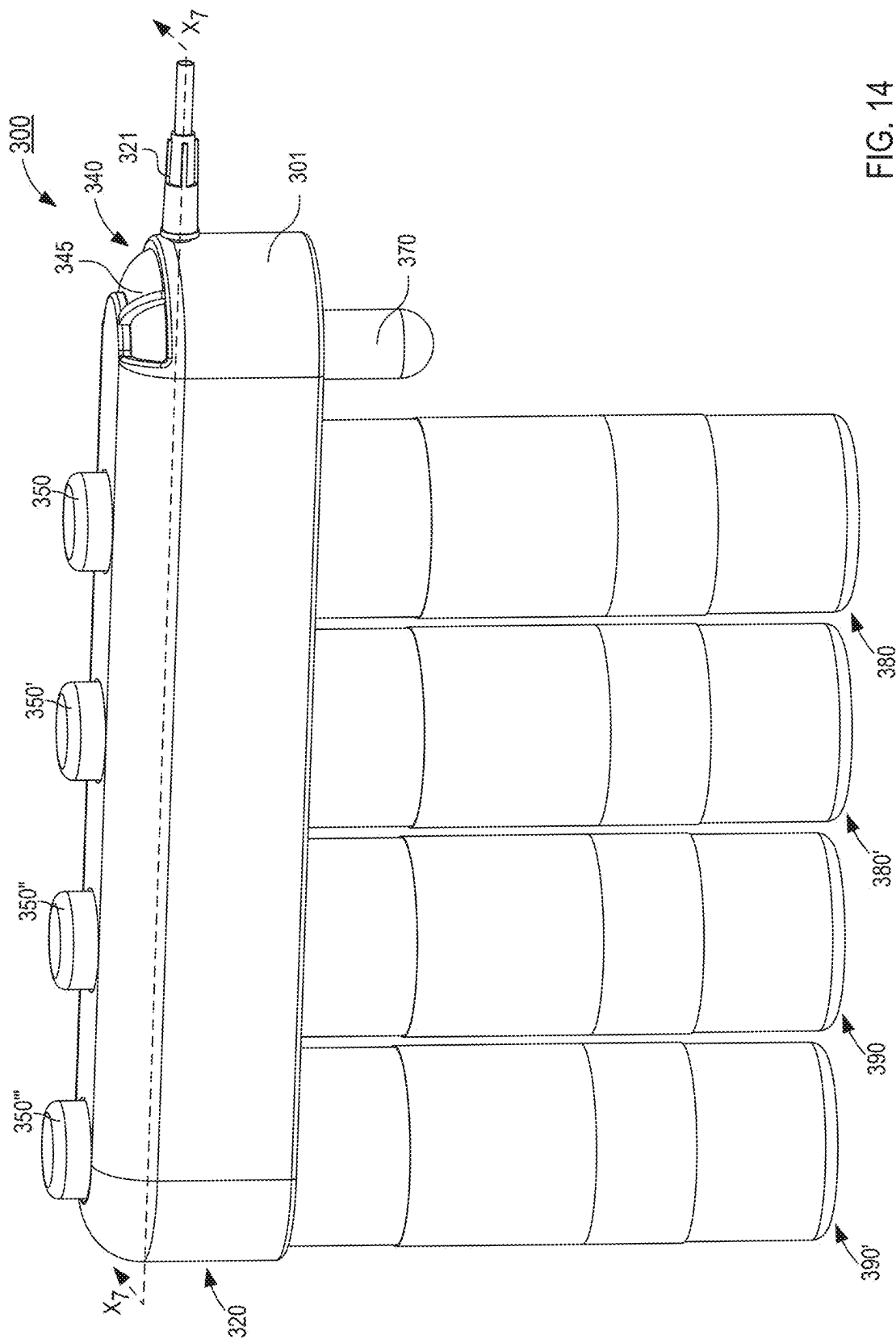


FIG. 14

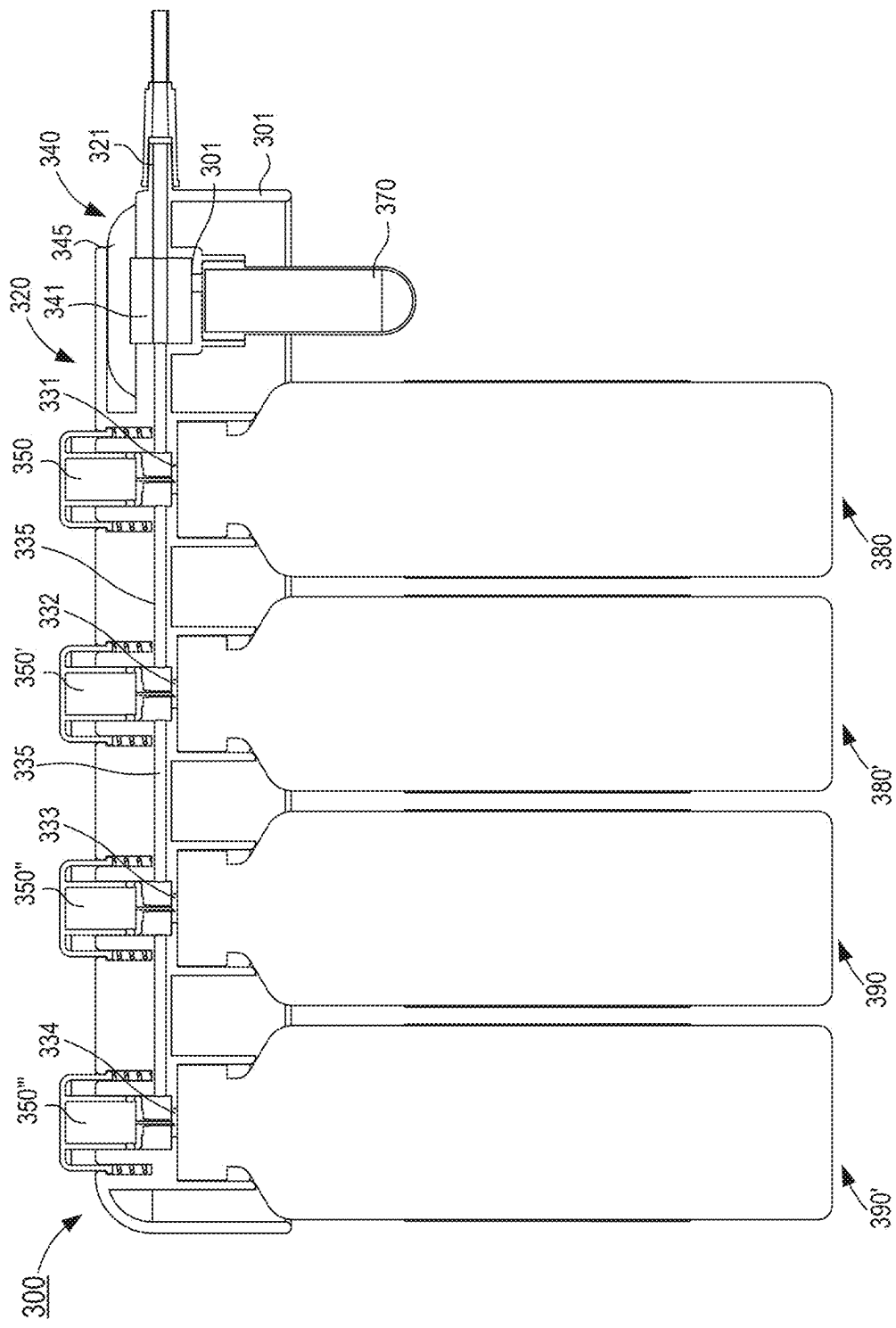


FIG. 15

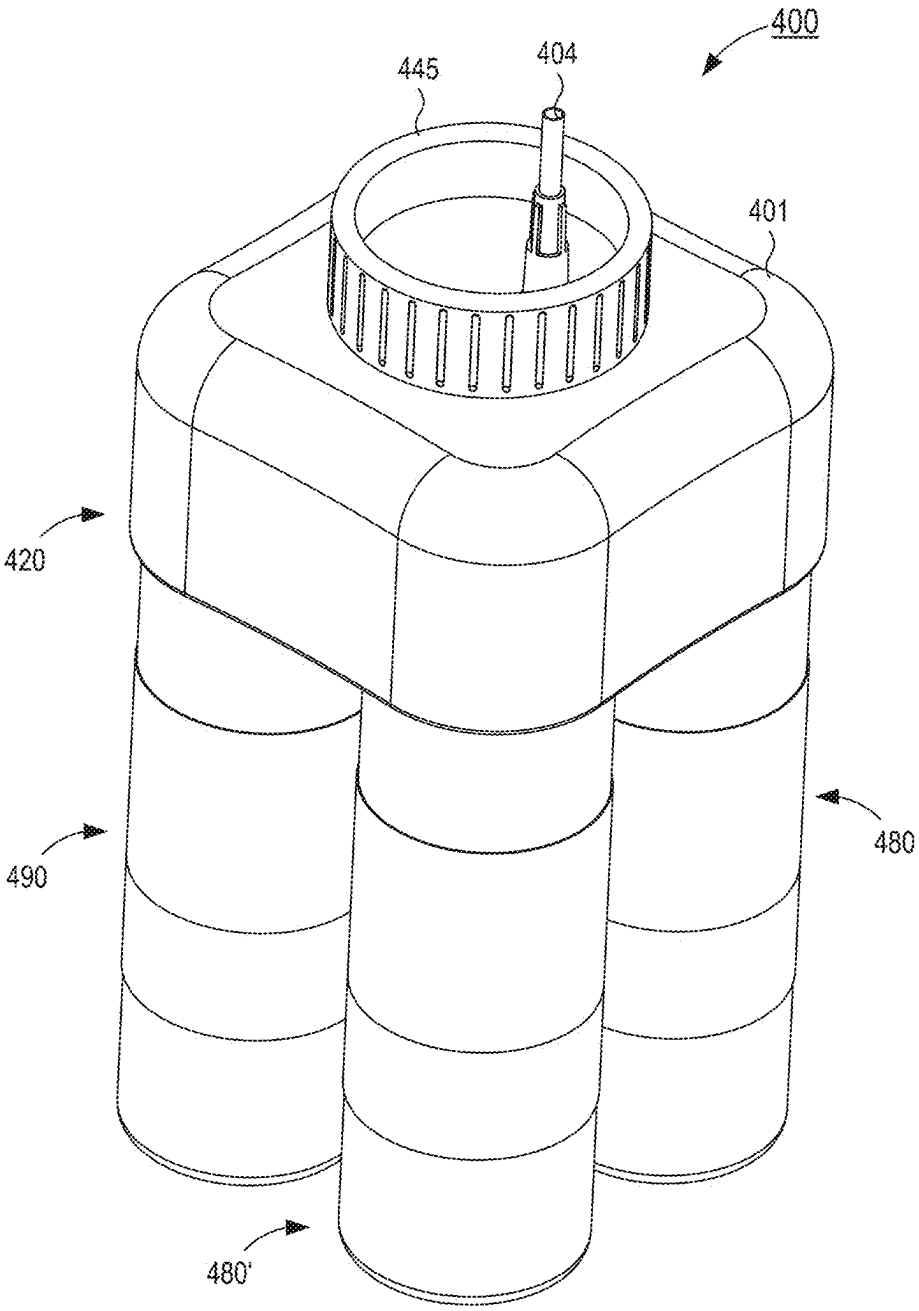


FIG. 16

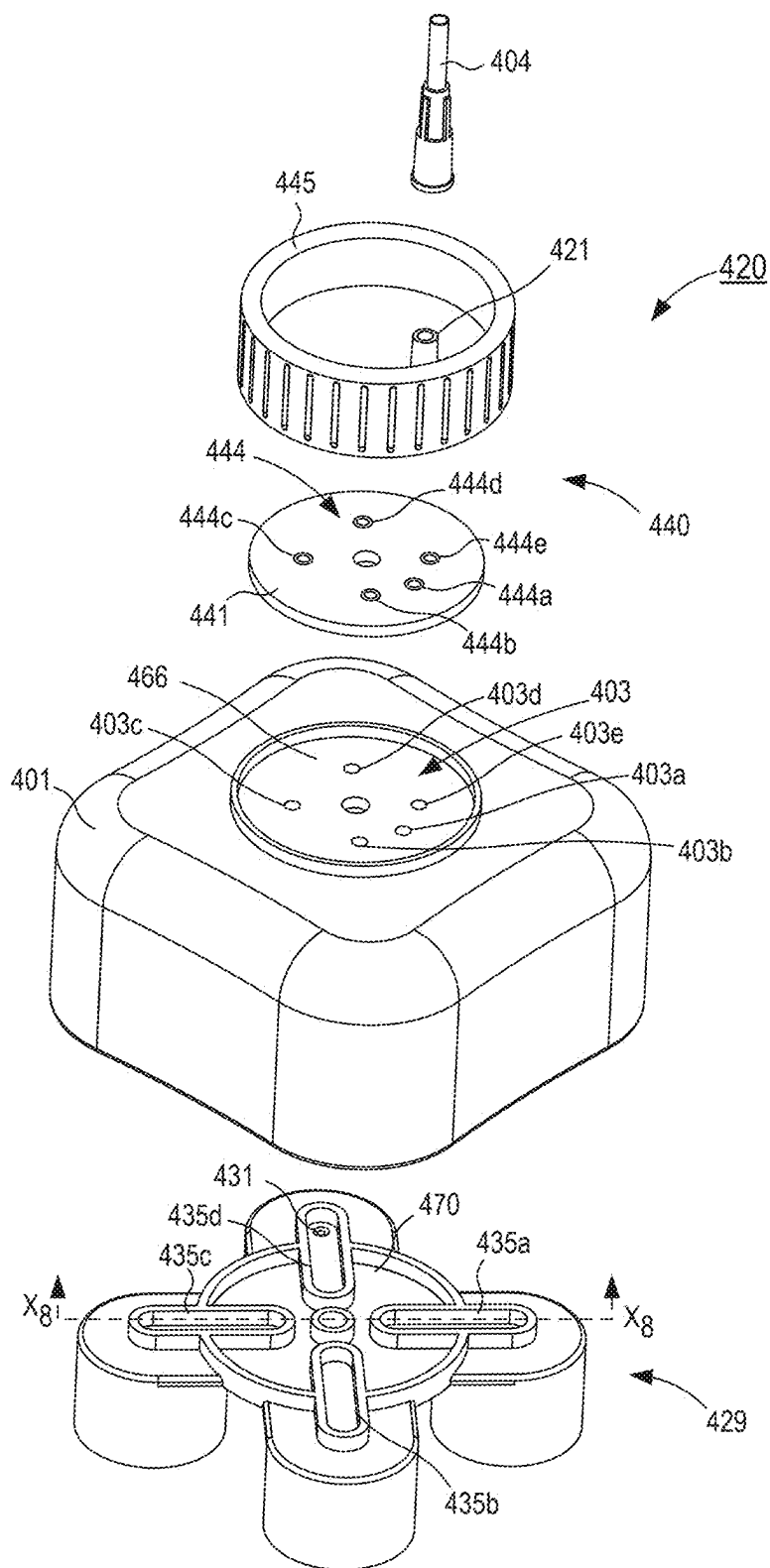


FIG. 17

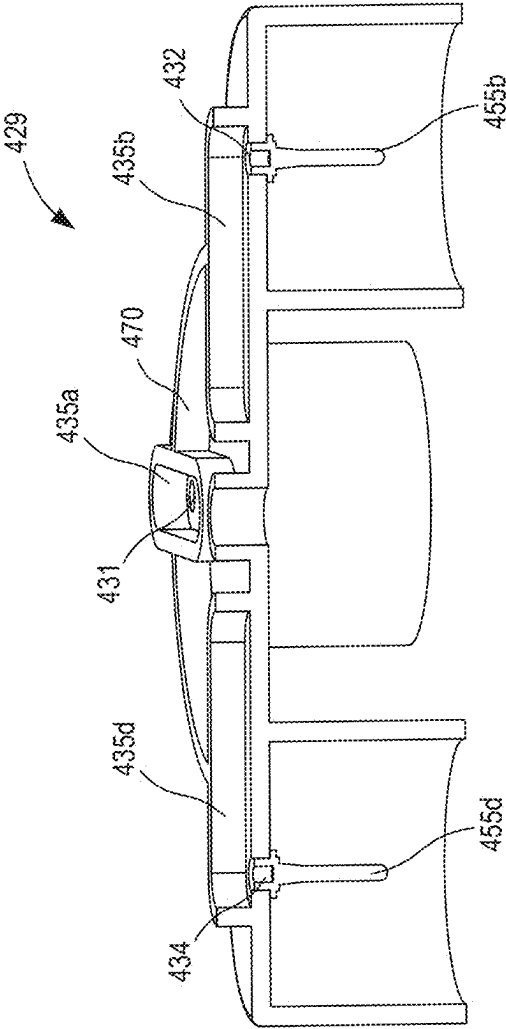


FIG. 18

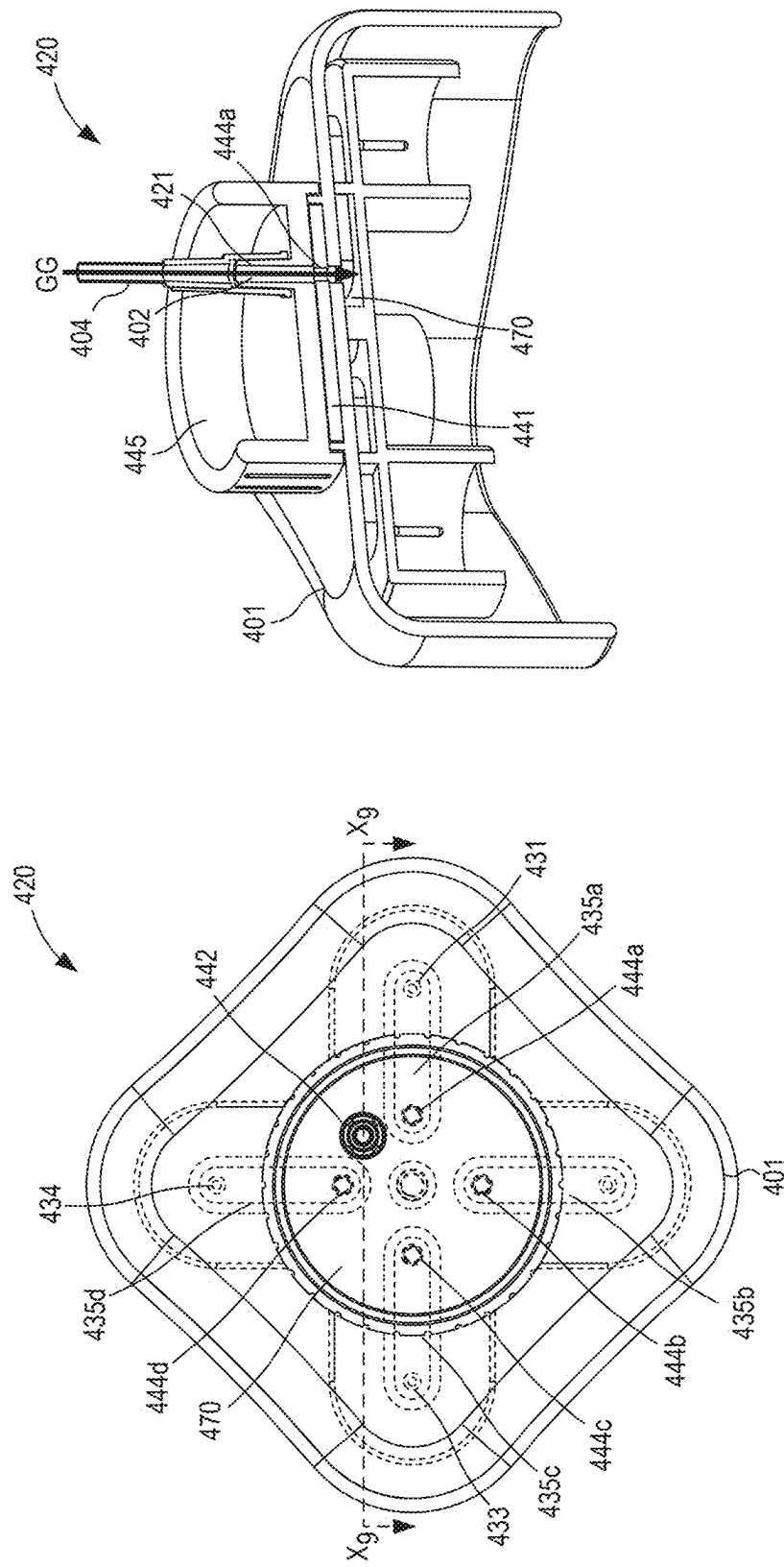


FIG. 20

FIG. 19

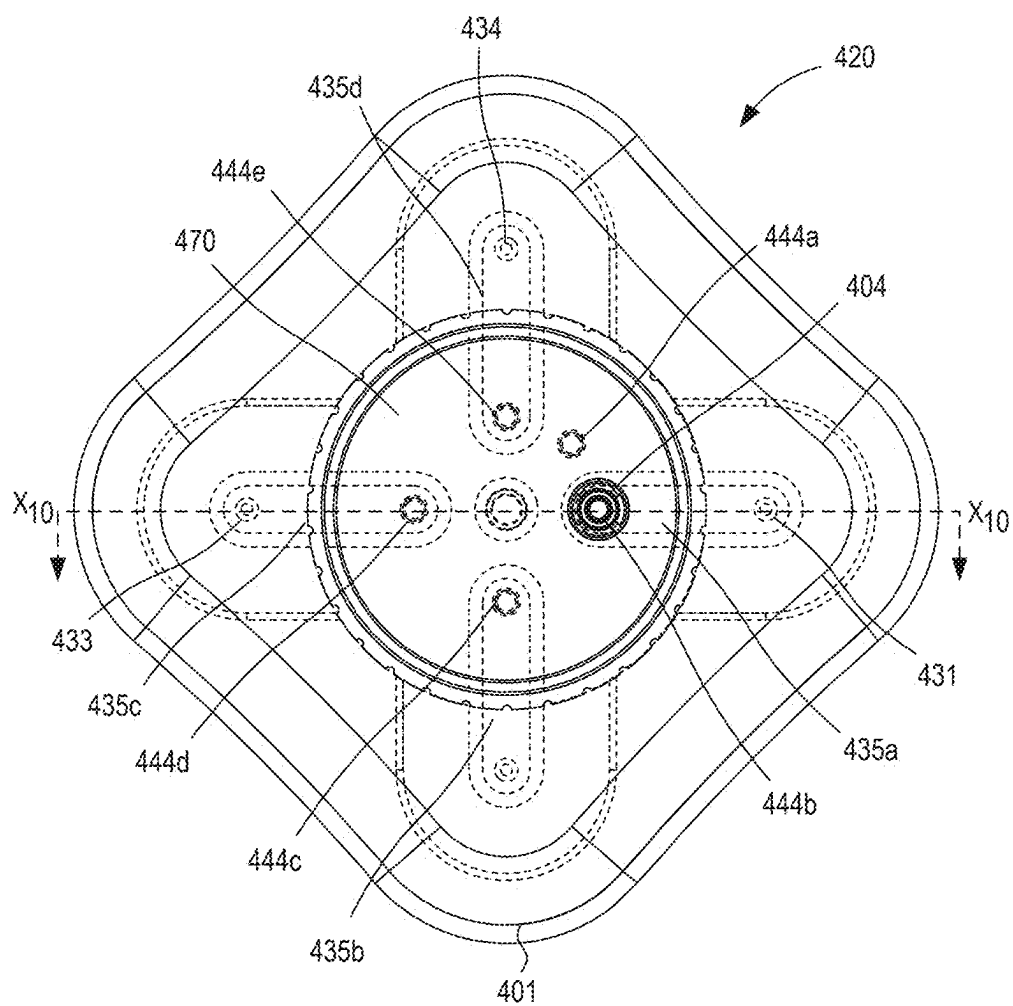


FIG. 21

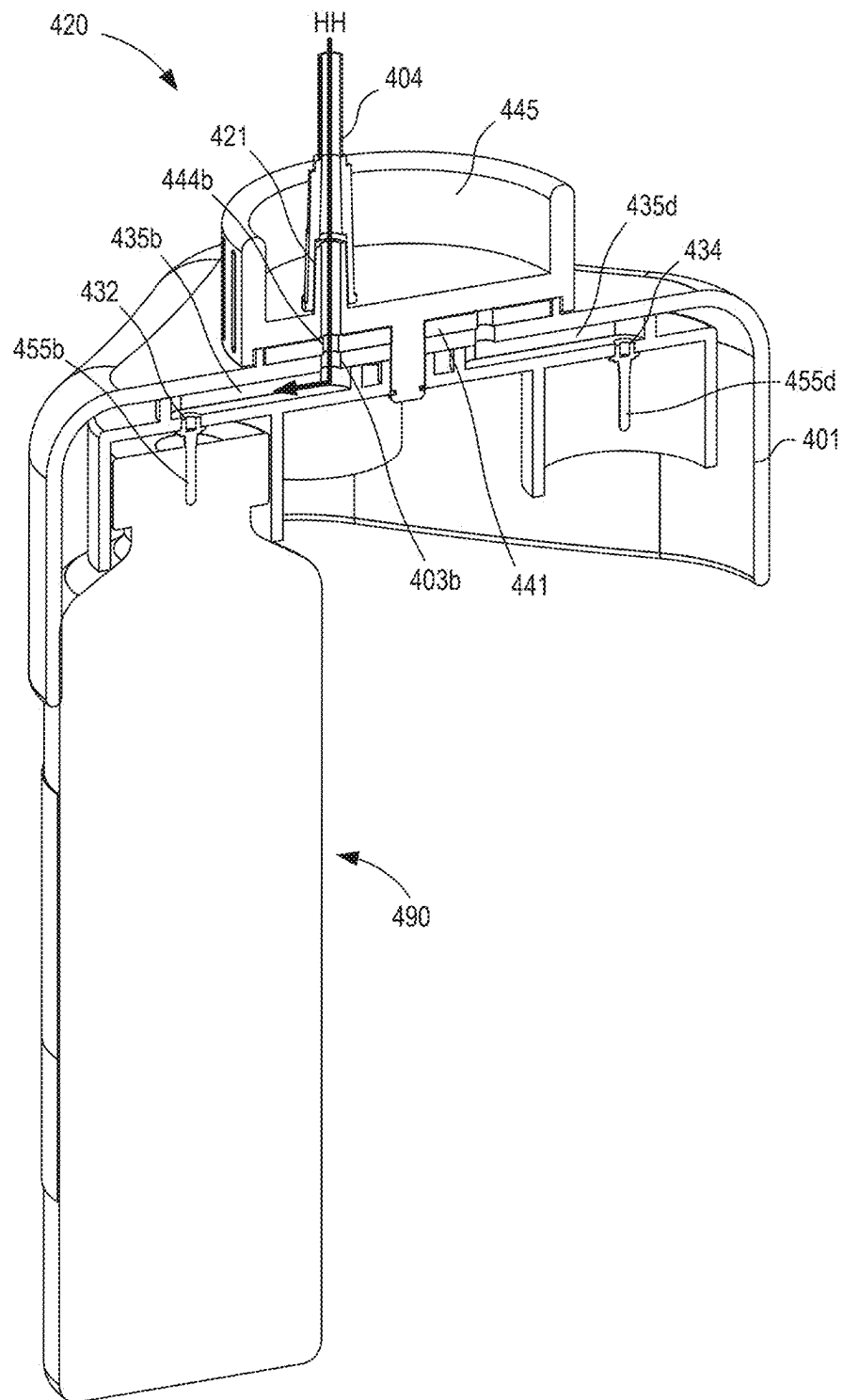


FIG. 22

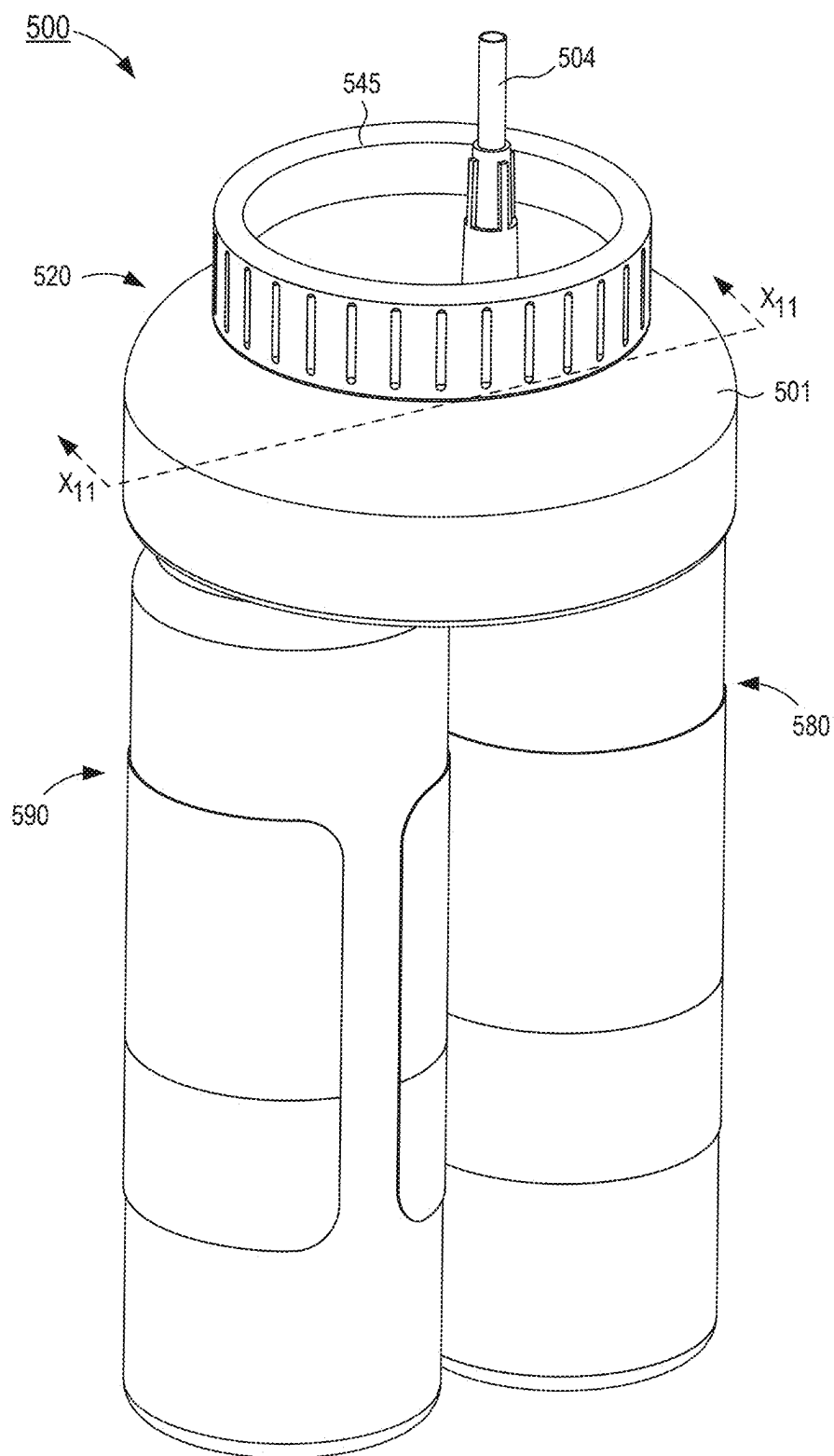


FIG. 23

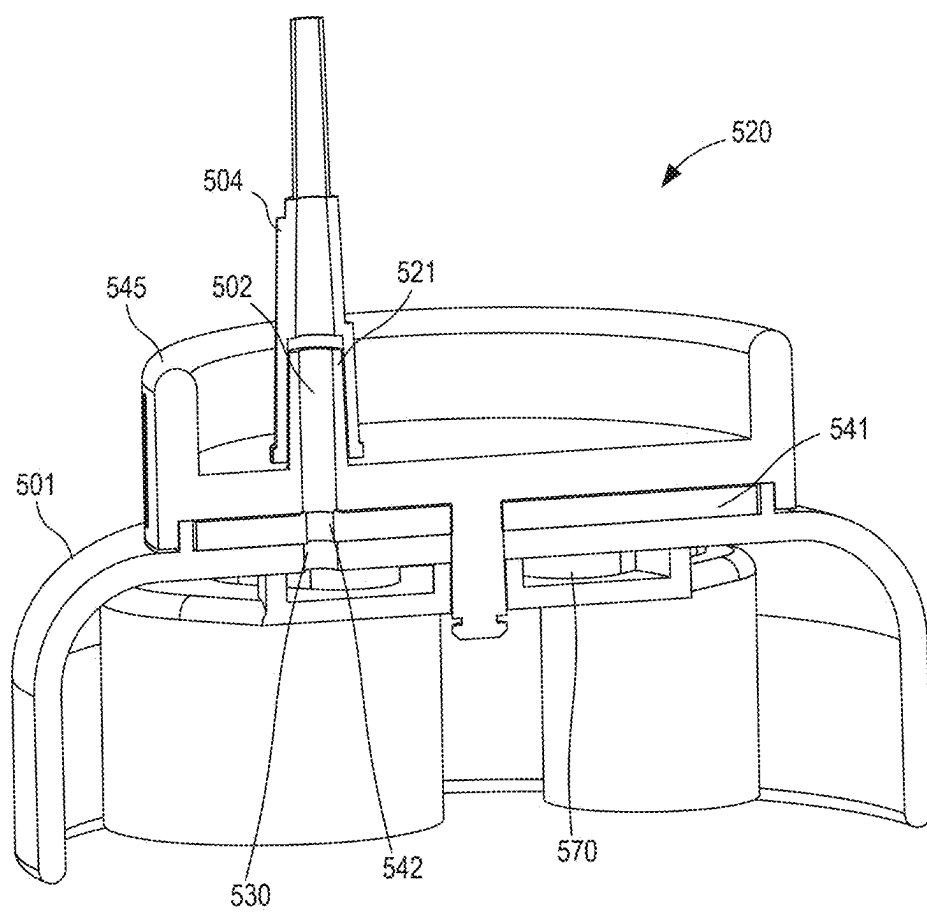


FIG. 24

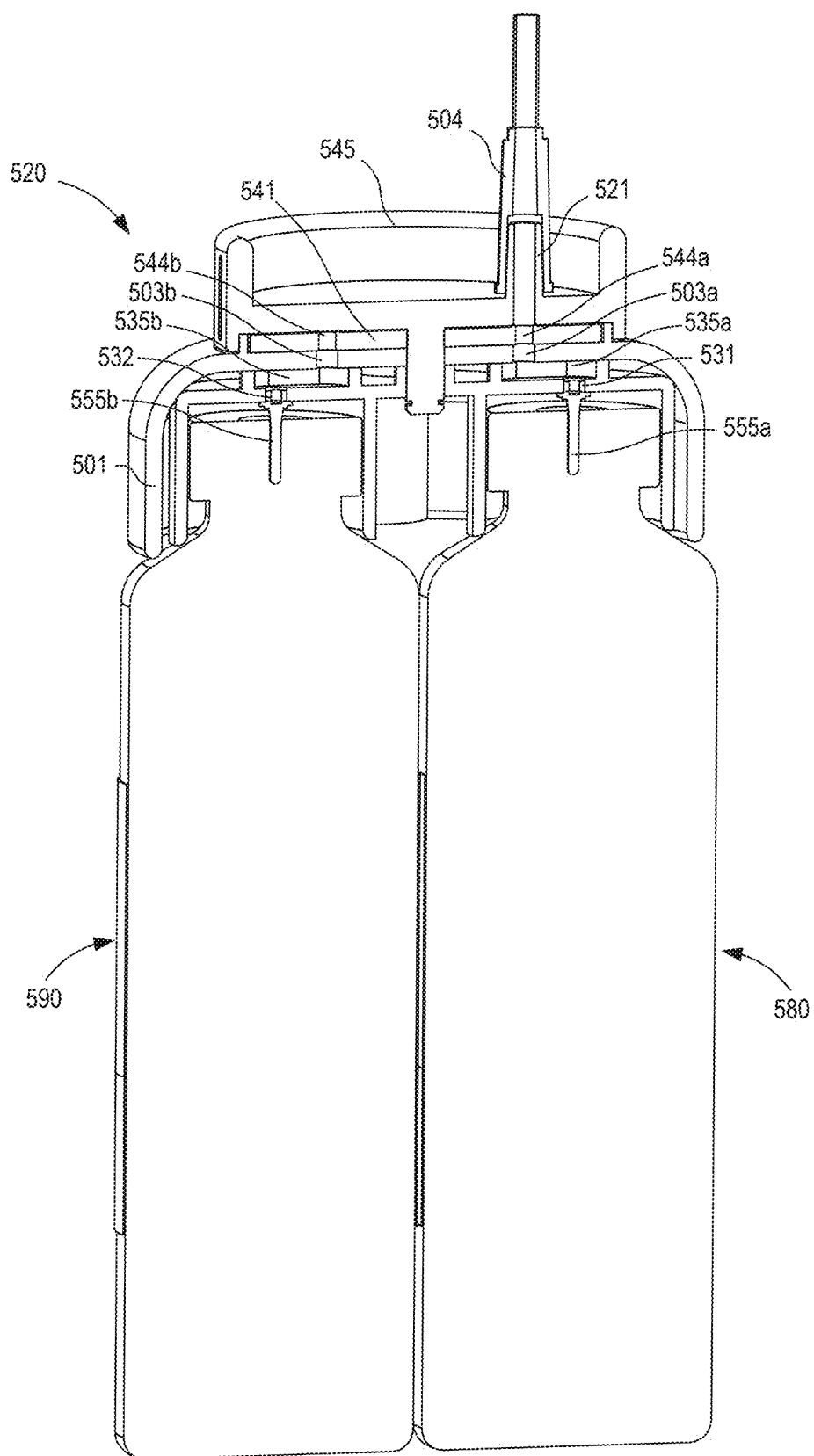


FIG. 25

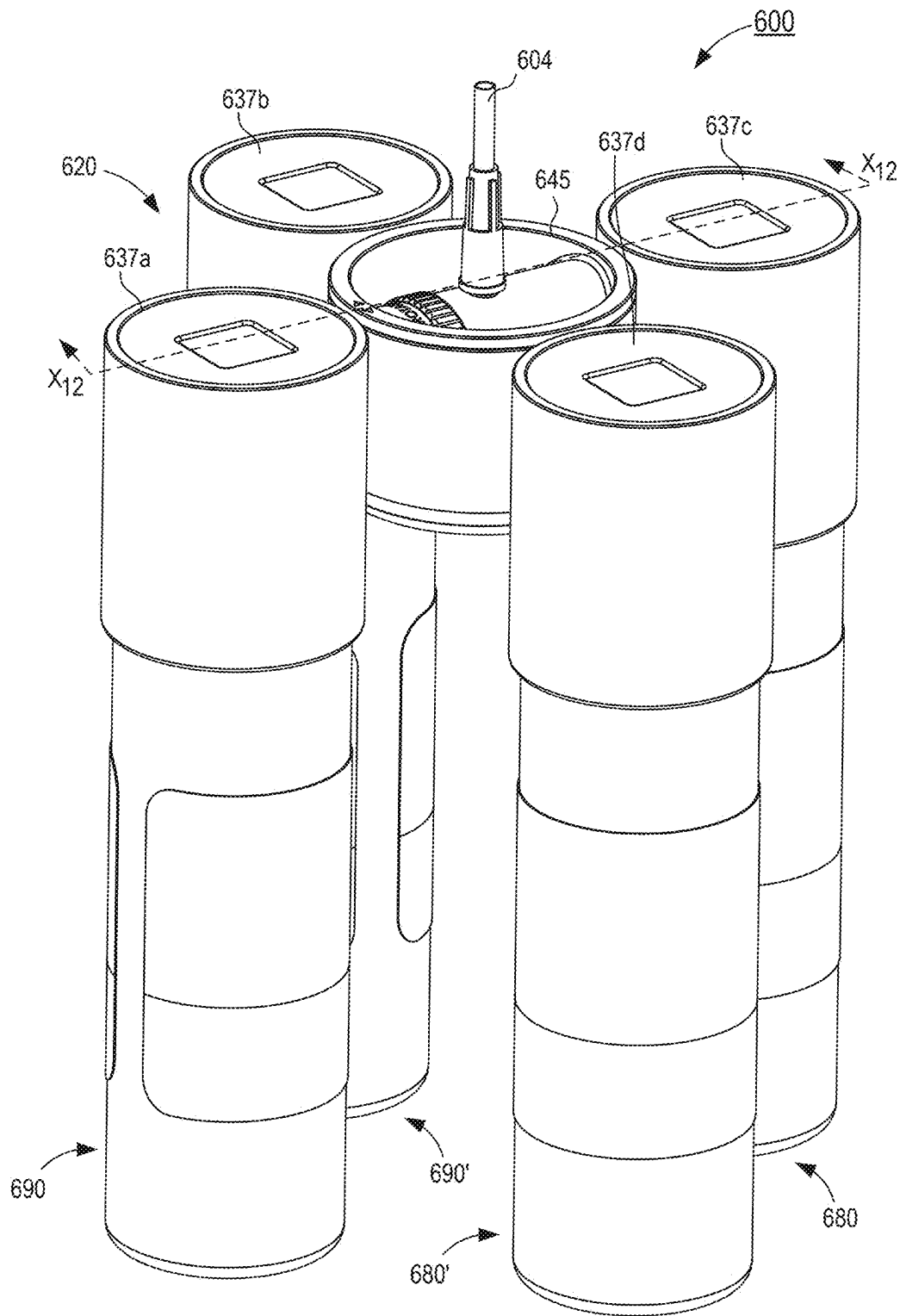


FIG. 26

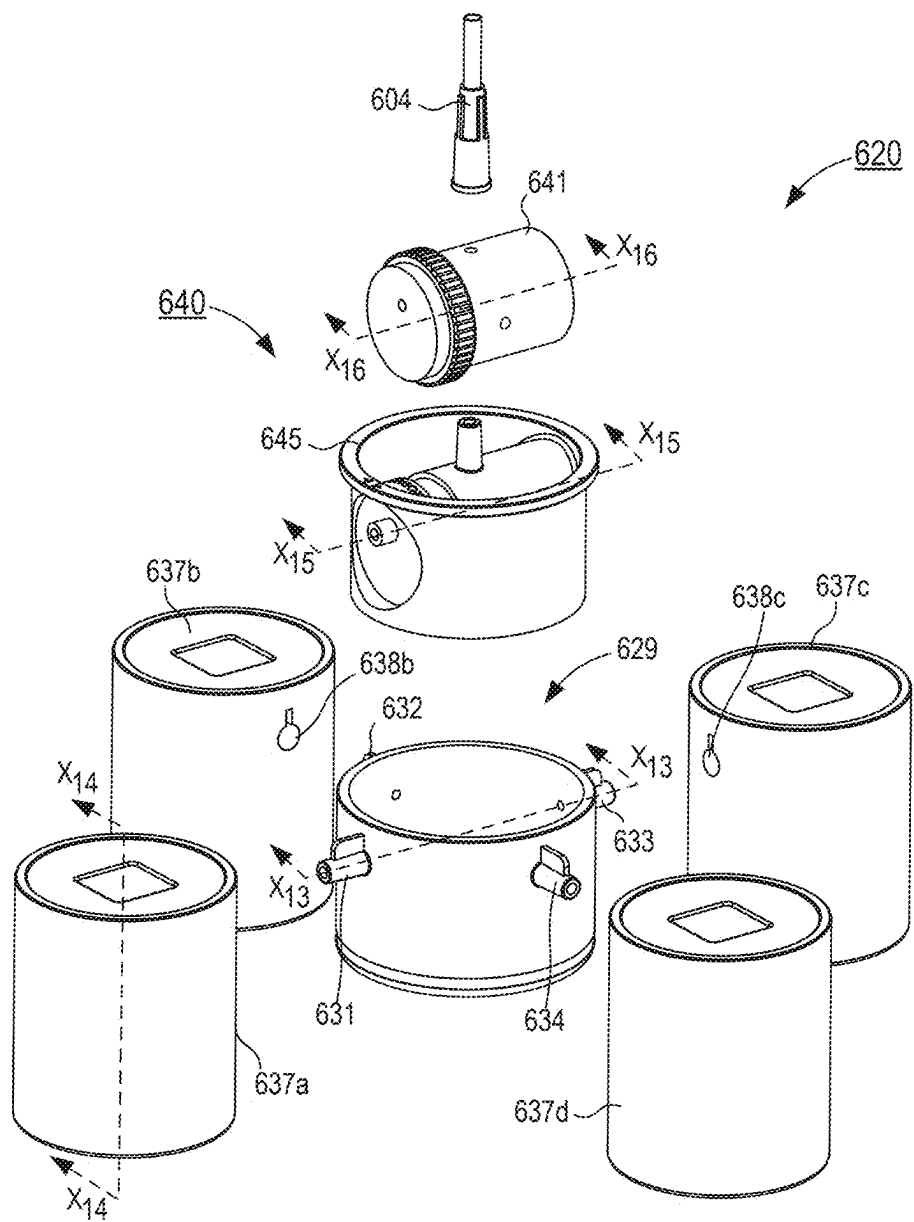
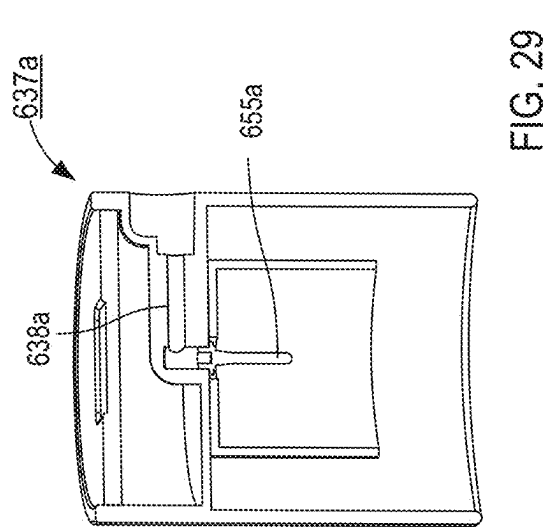
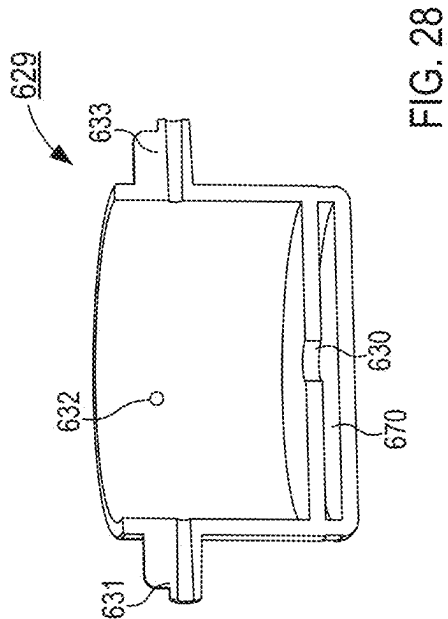
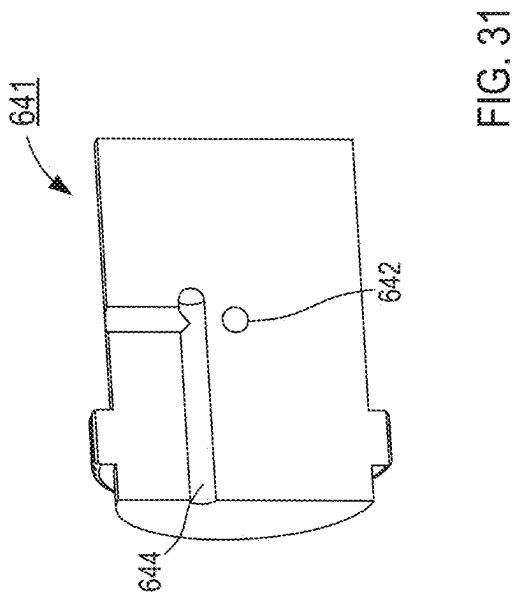
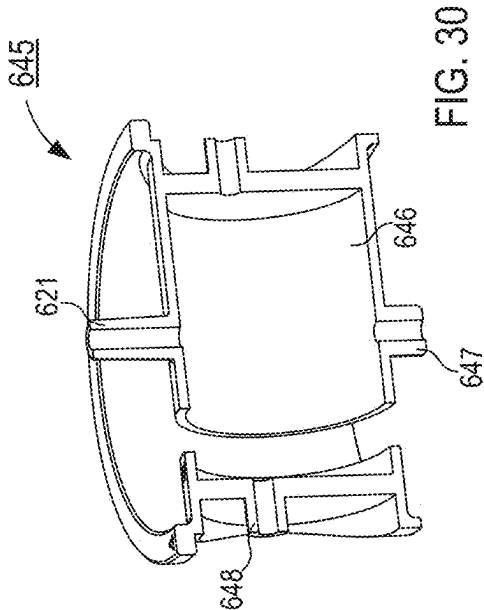


FIG. 27



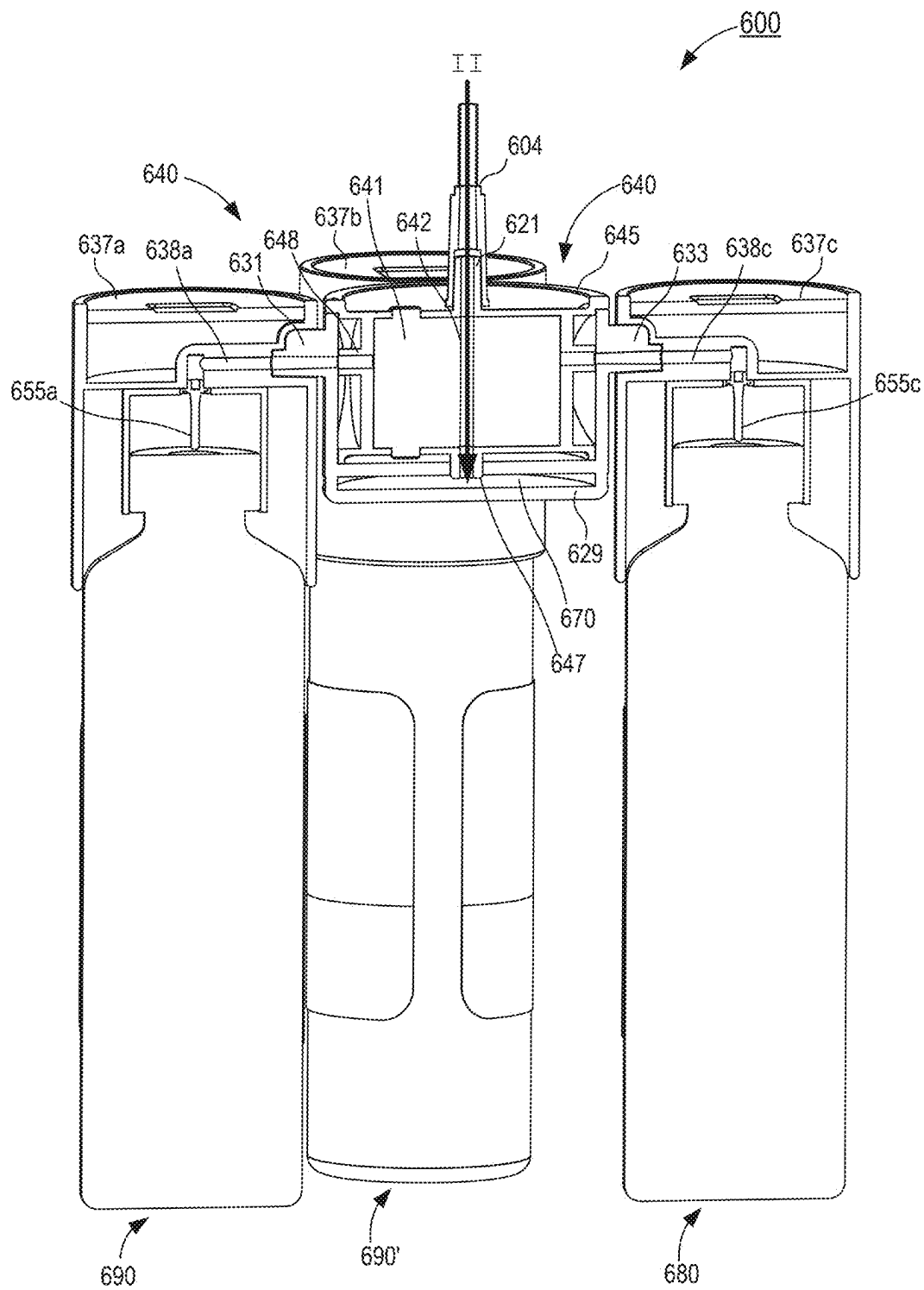


FIG. 32

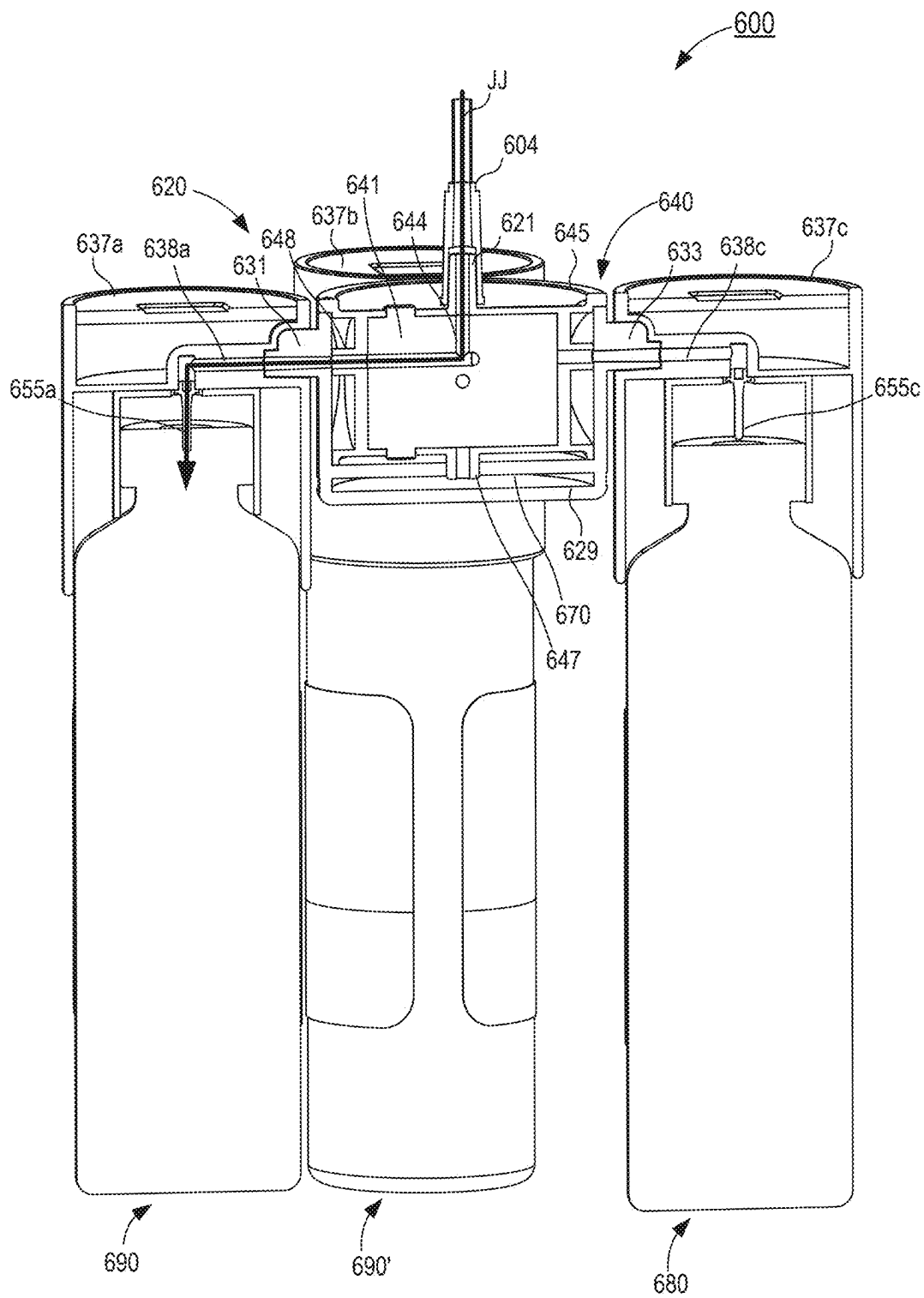


FIG. 33

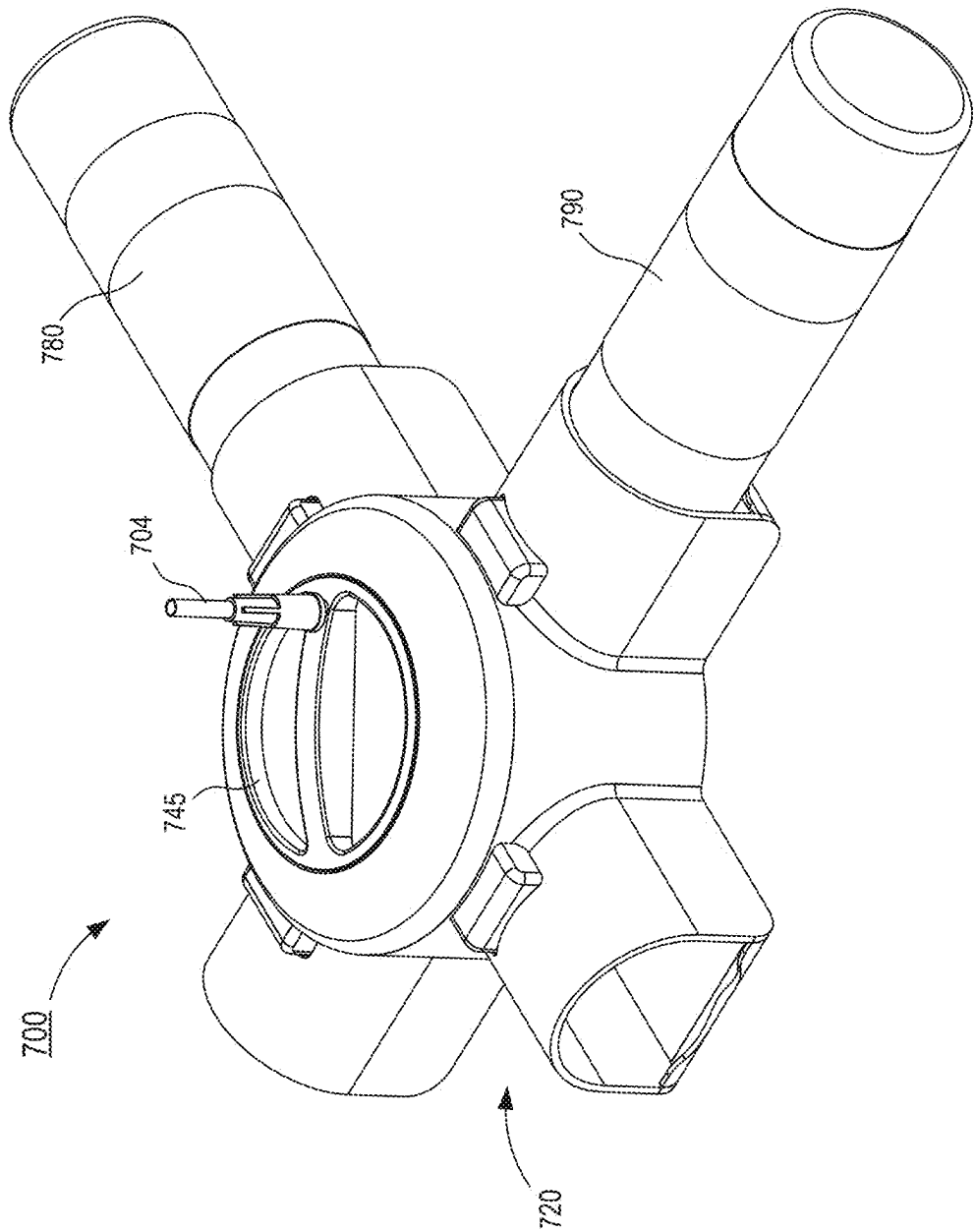


FIG. 34

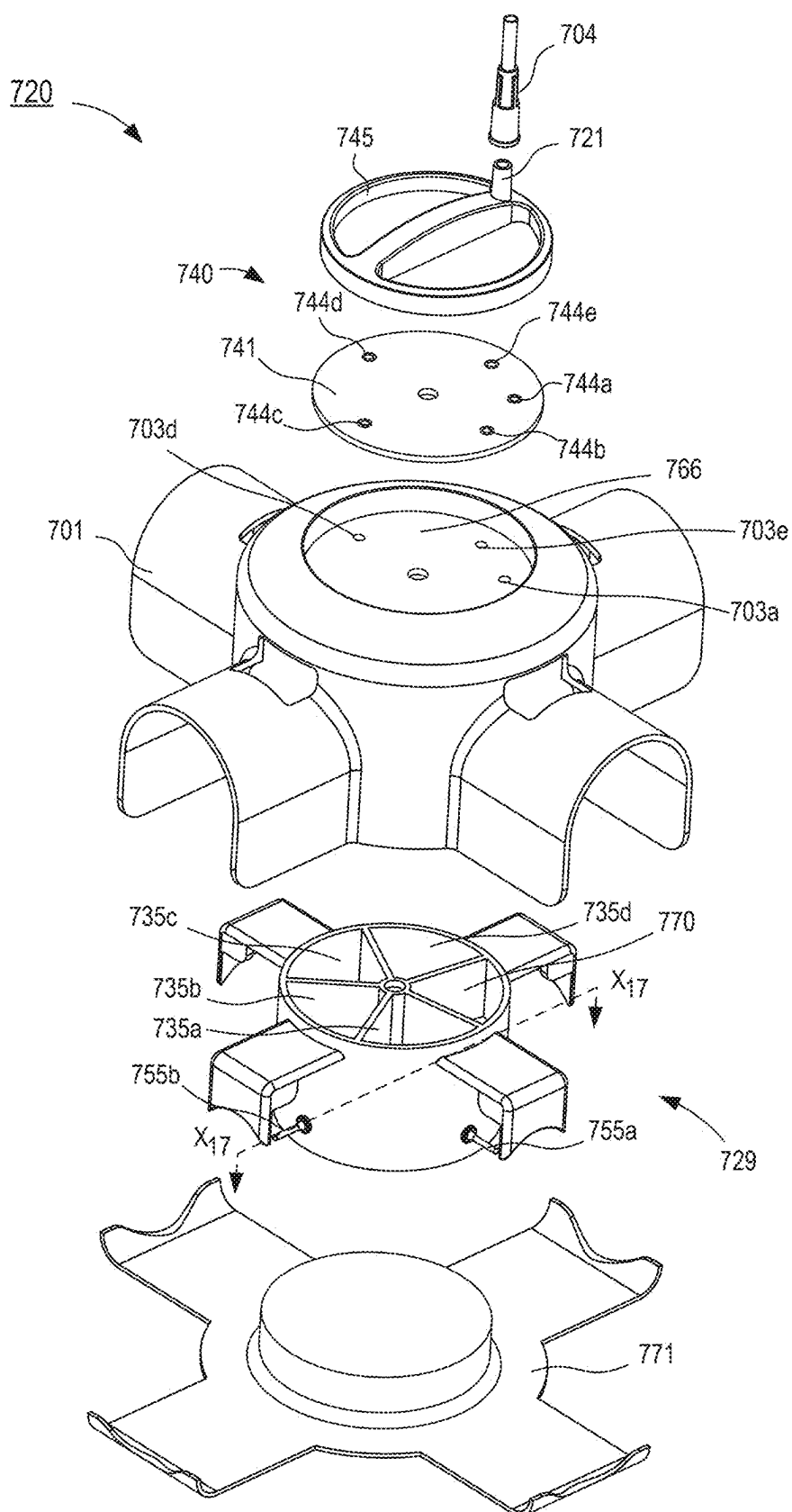


FIG. 35

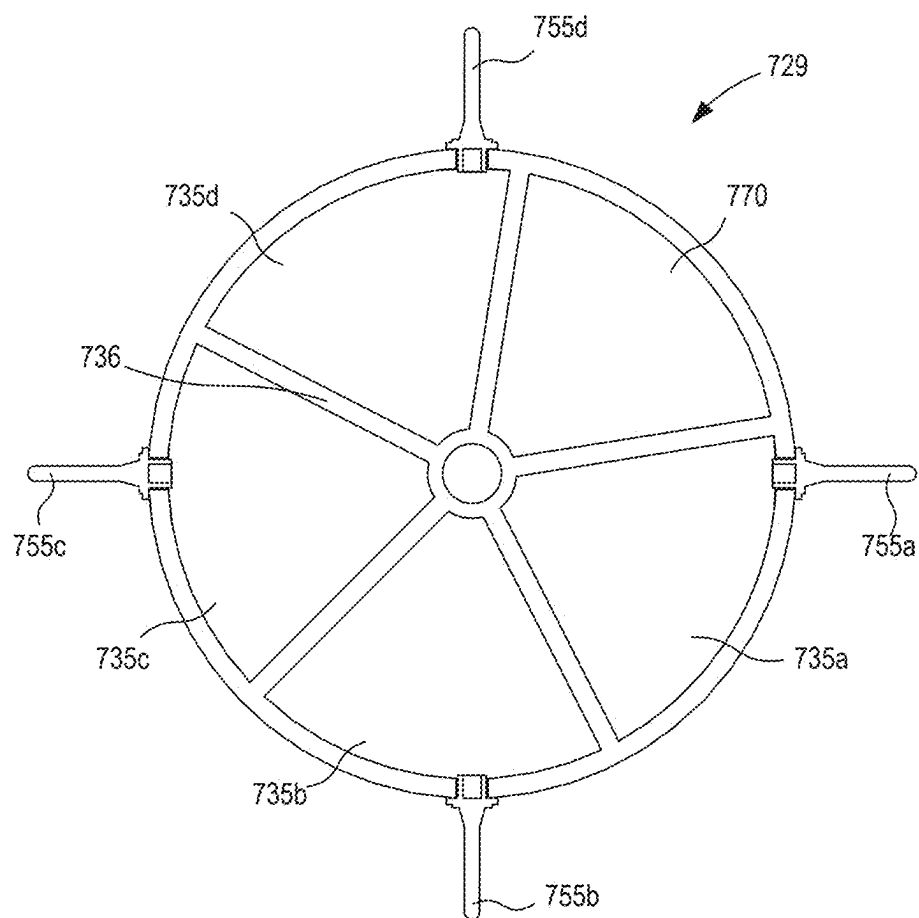


FIG. 36

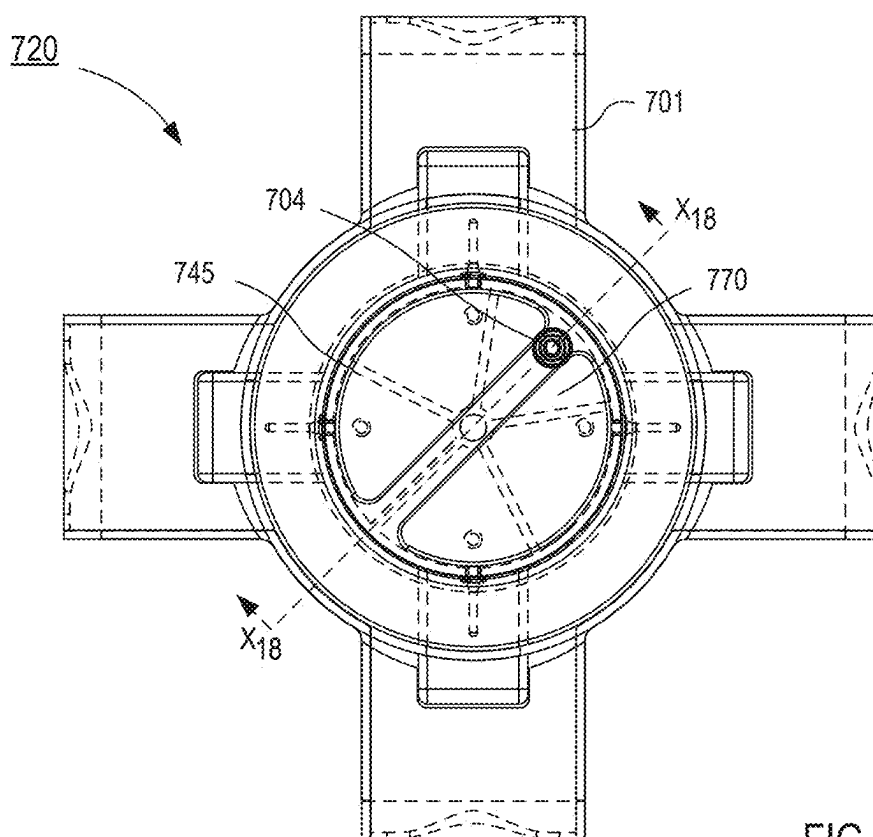


FIG. 37

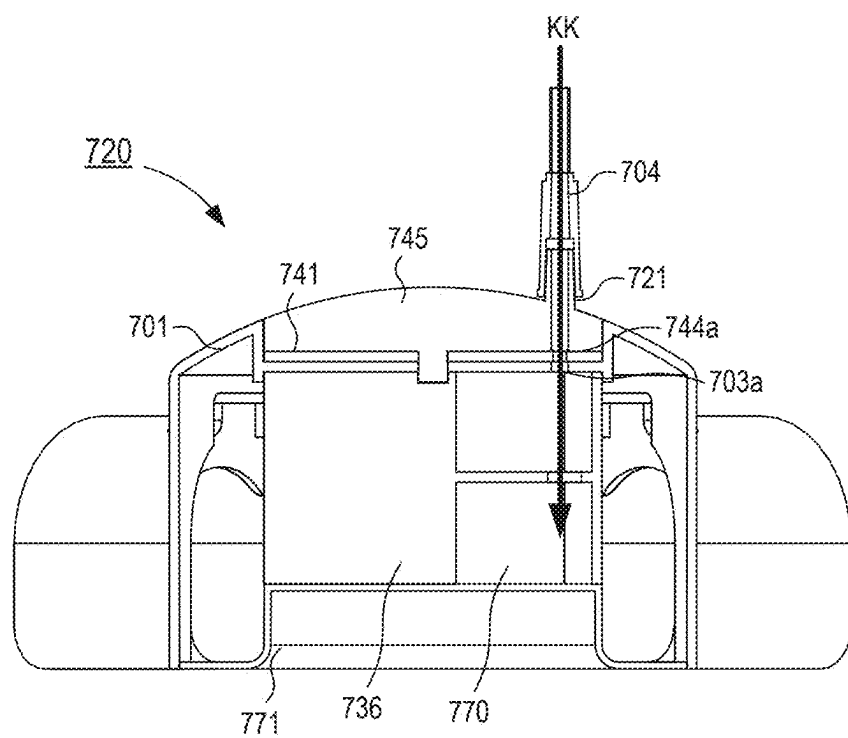


FIG. 38

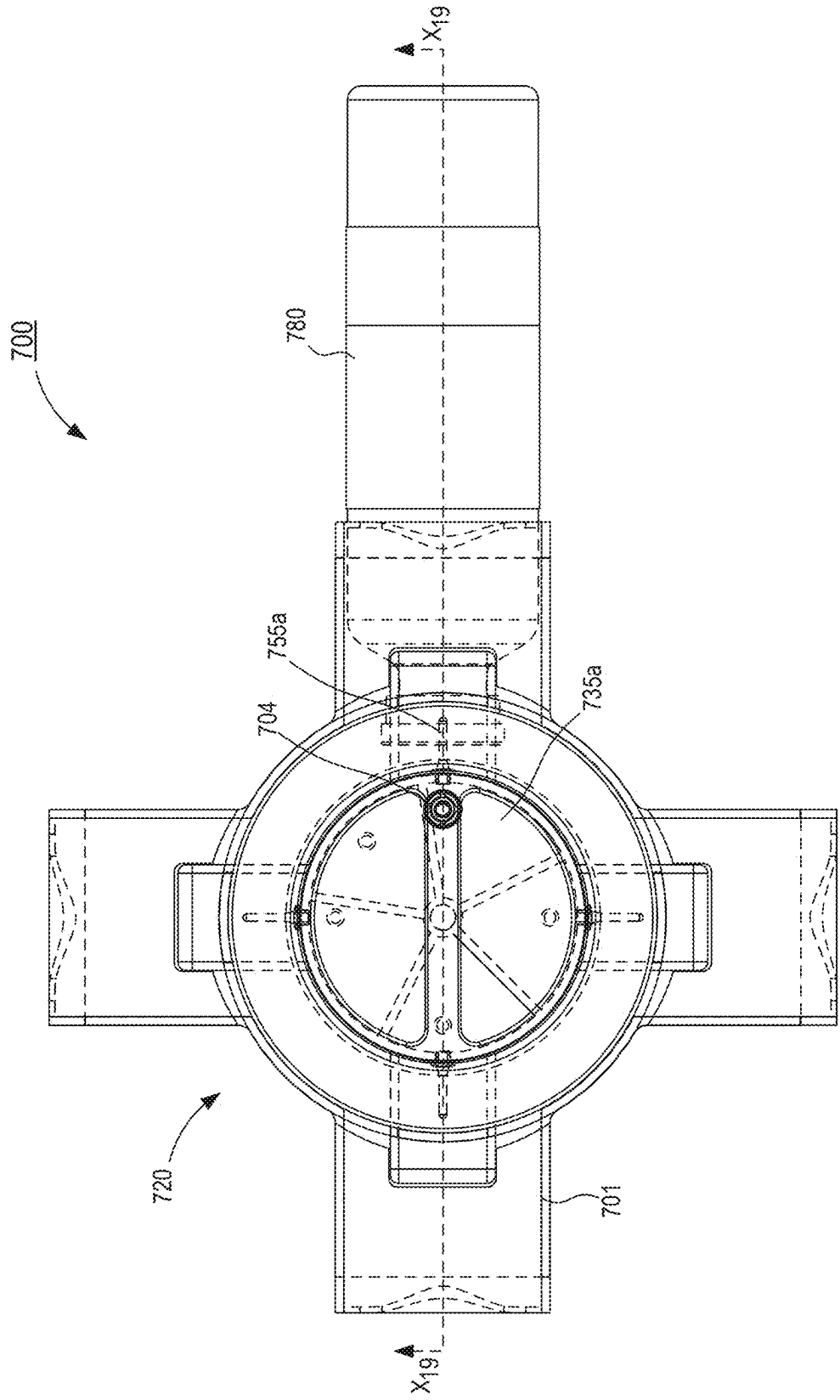


FIG. 39

Q
4
G
L

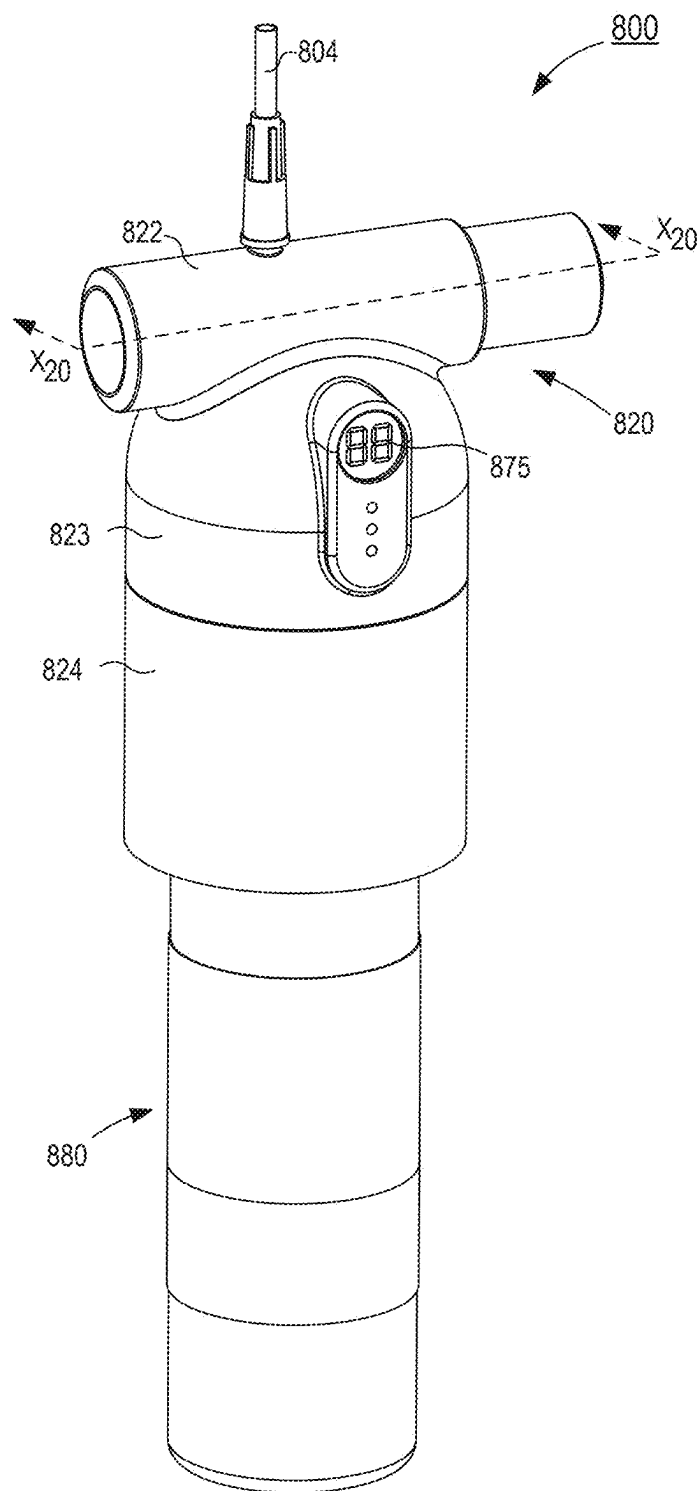


FIG. 41

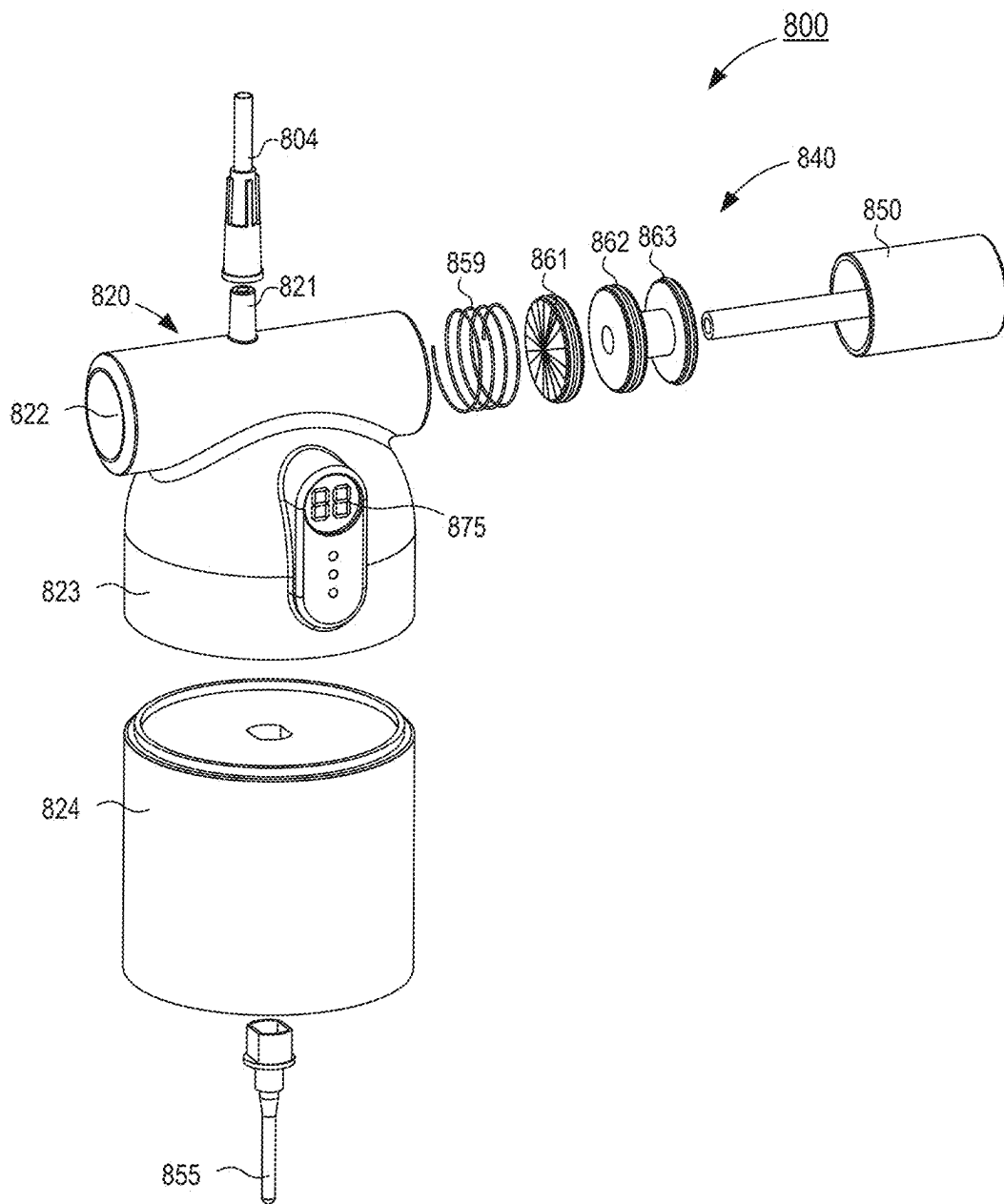


FIG. 42

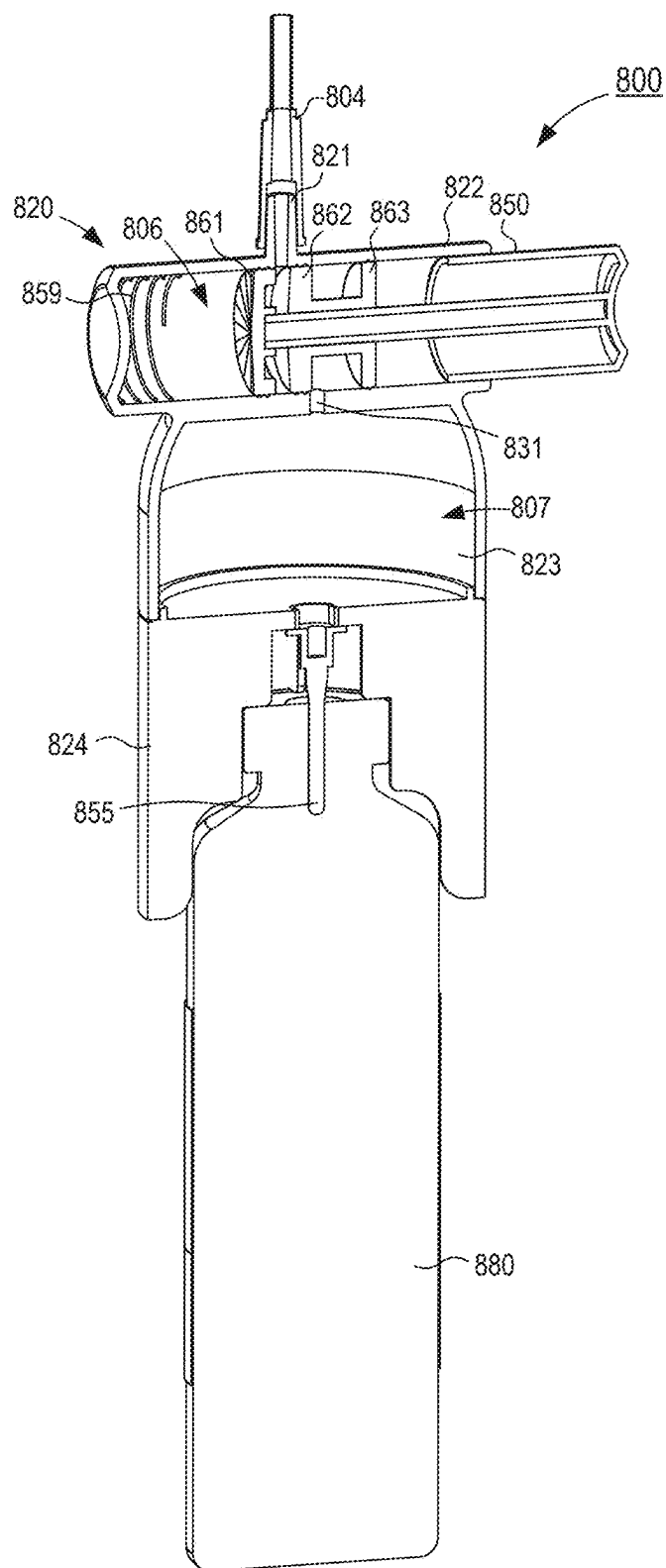


FIG. 43

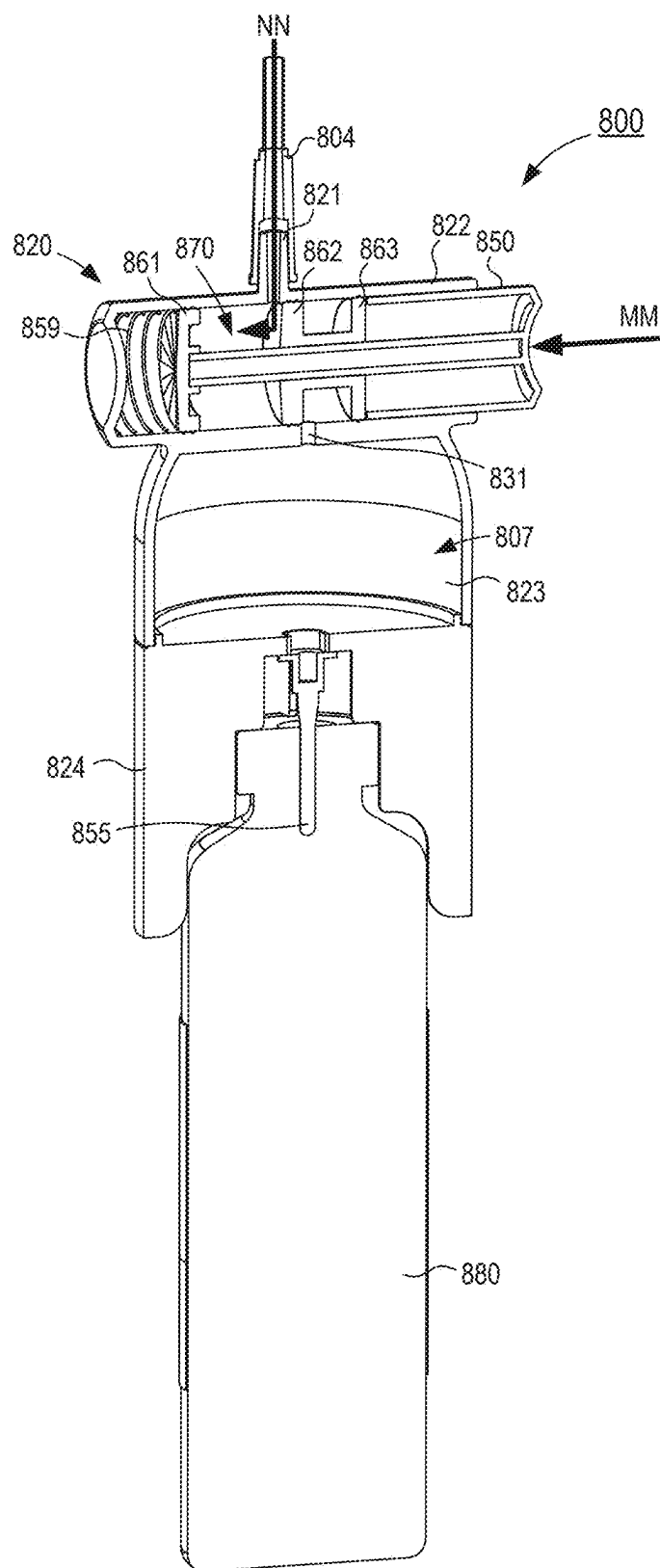


FIG. 44

FIG. 45

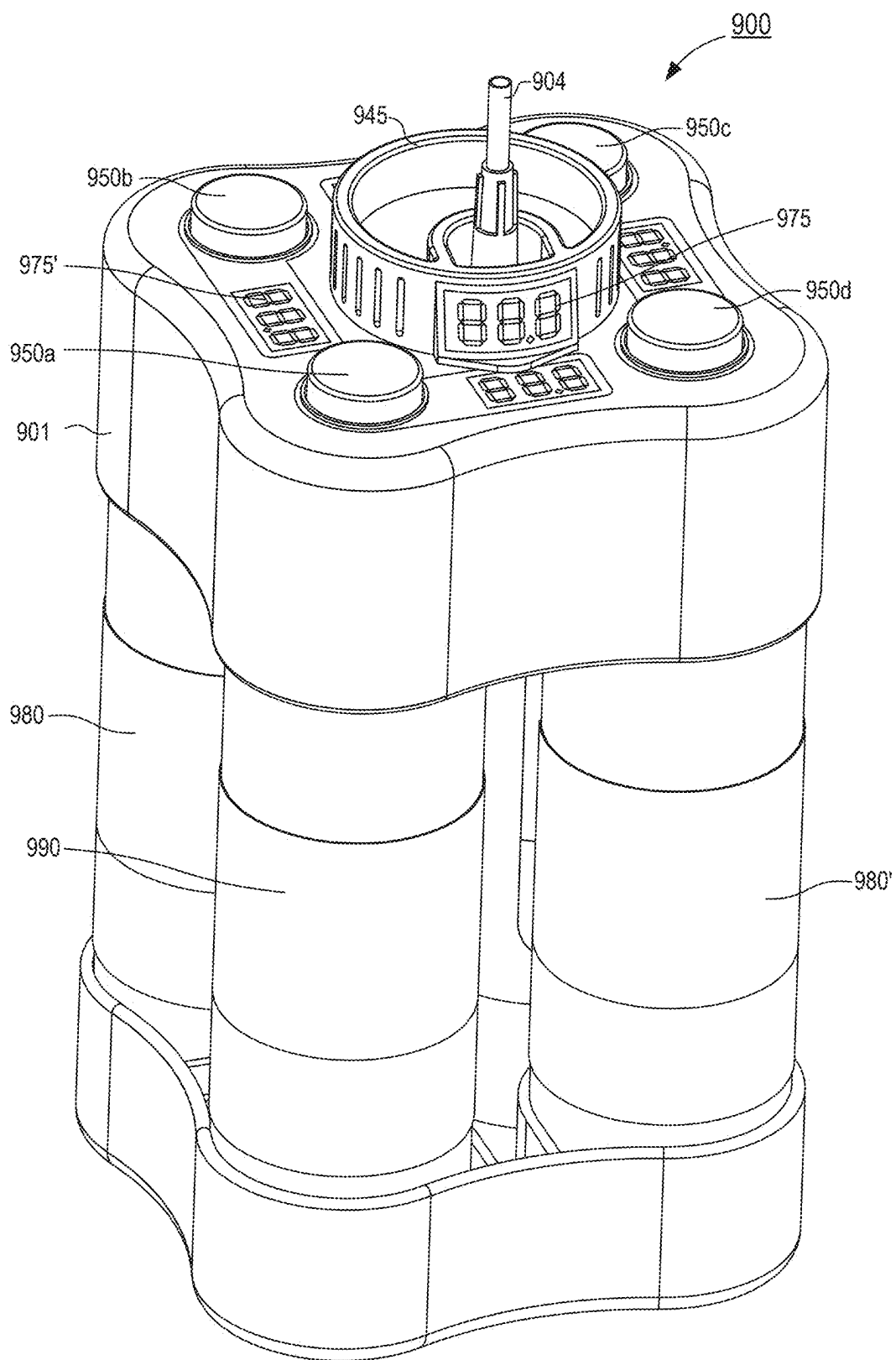


FIG. 46

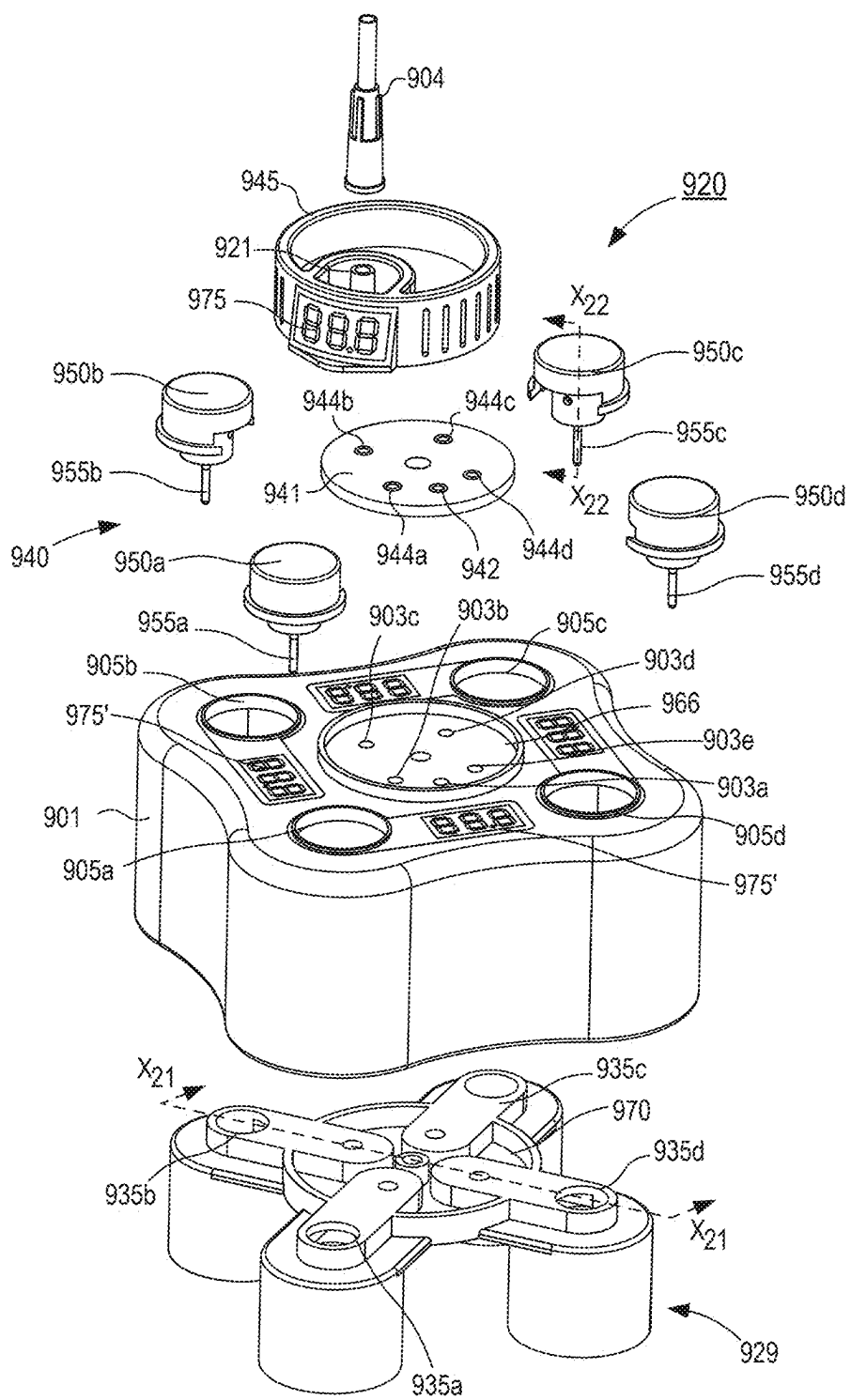


FIG. 47

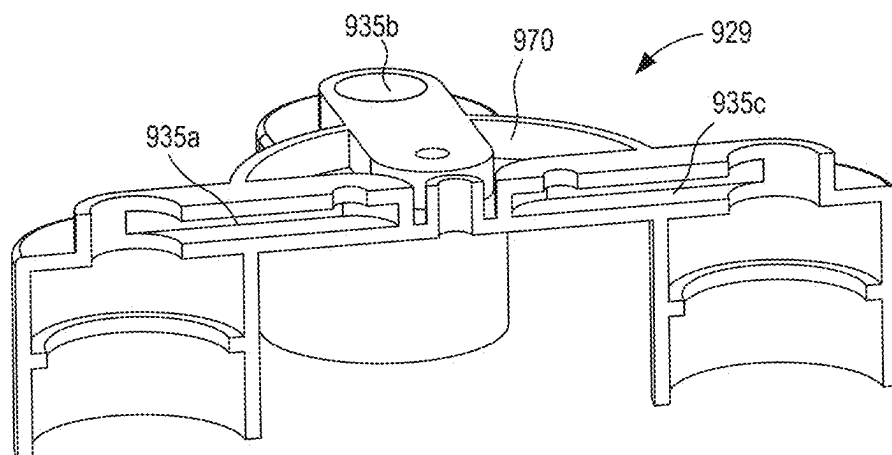


FIG. 48

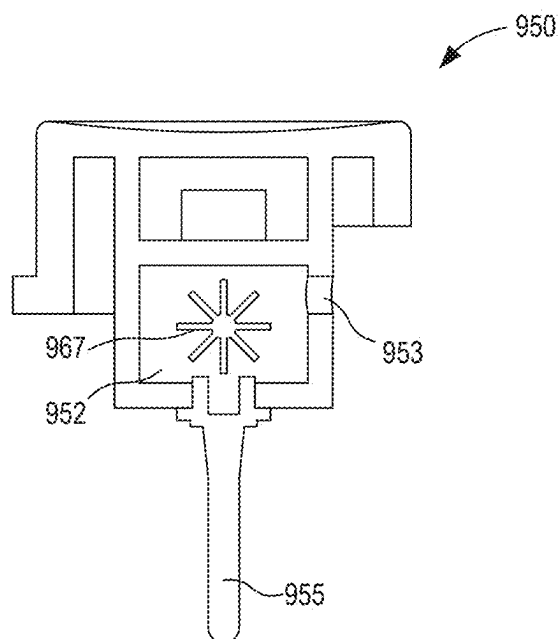


FIG. 49

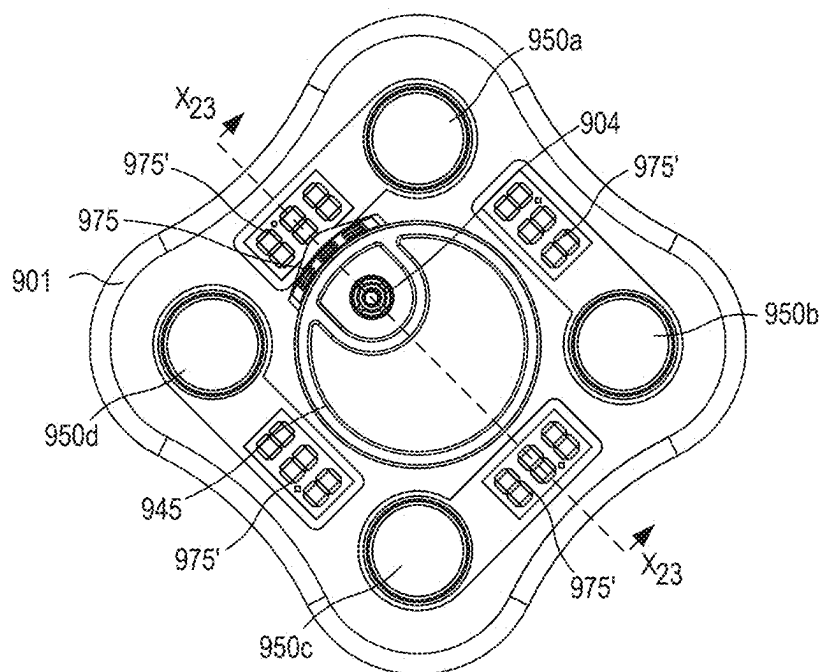


FIG. 50

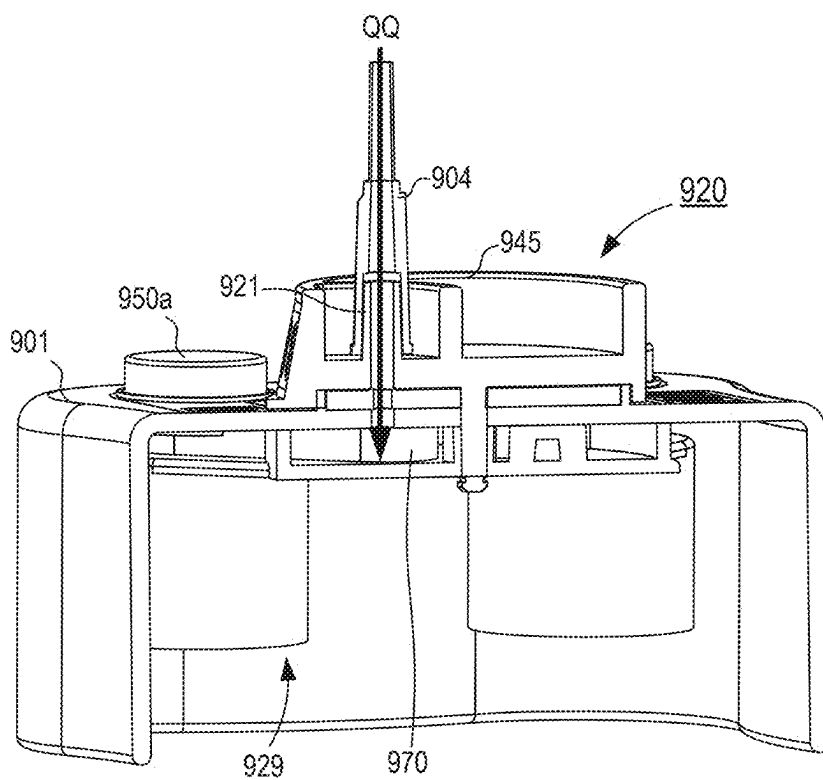


FIG. 51

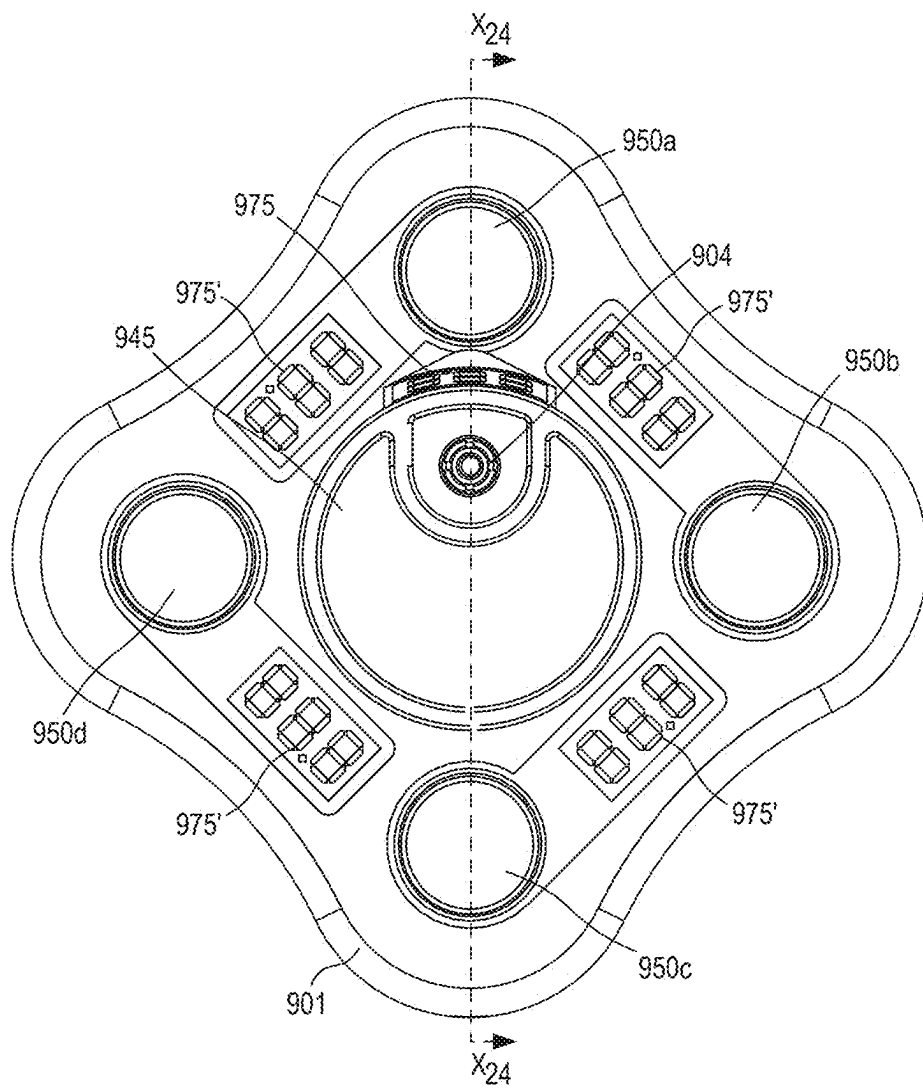


FIG. 52

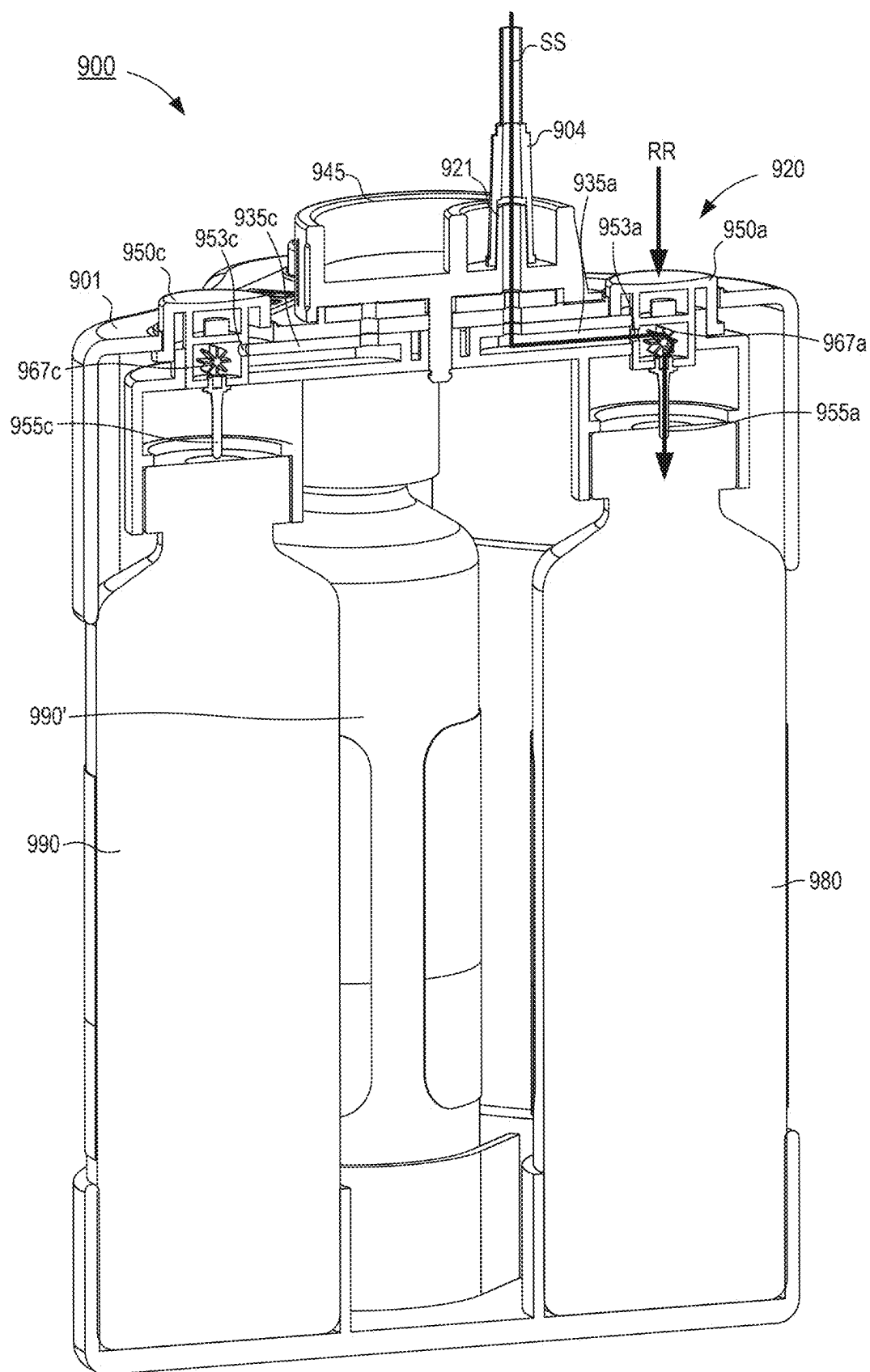


FIG. 53

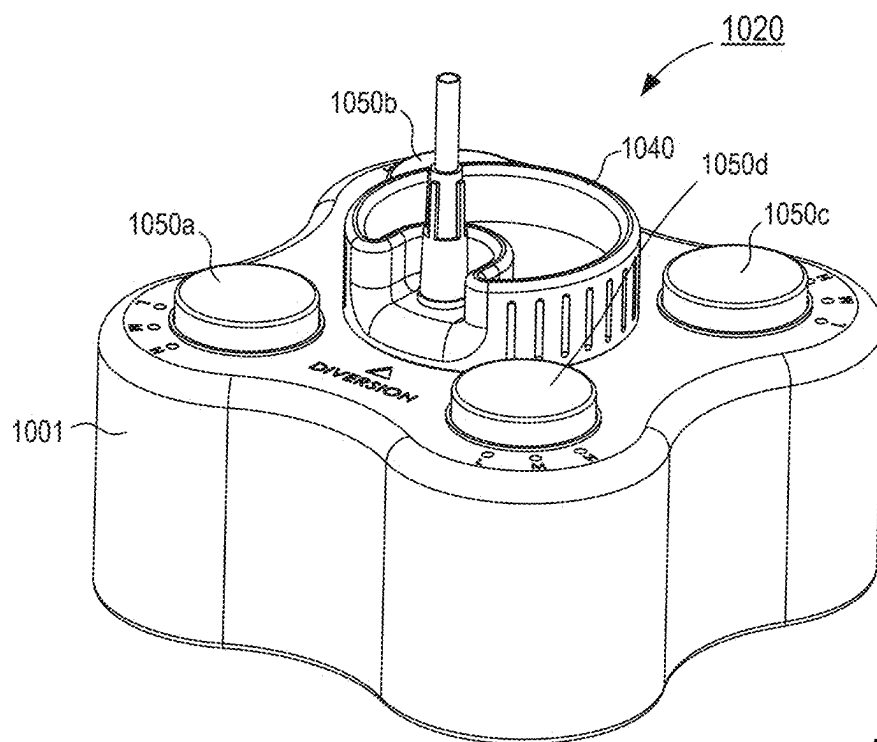


FIG. 54

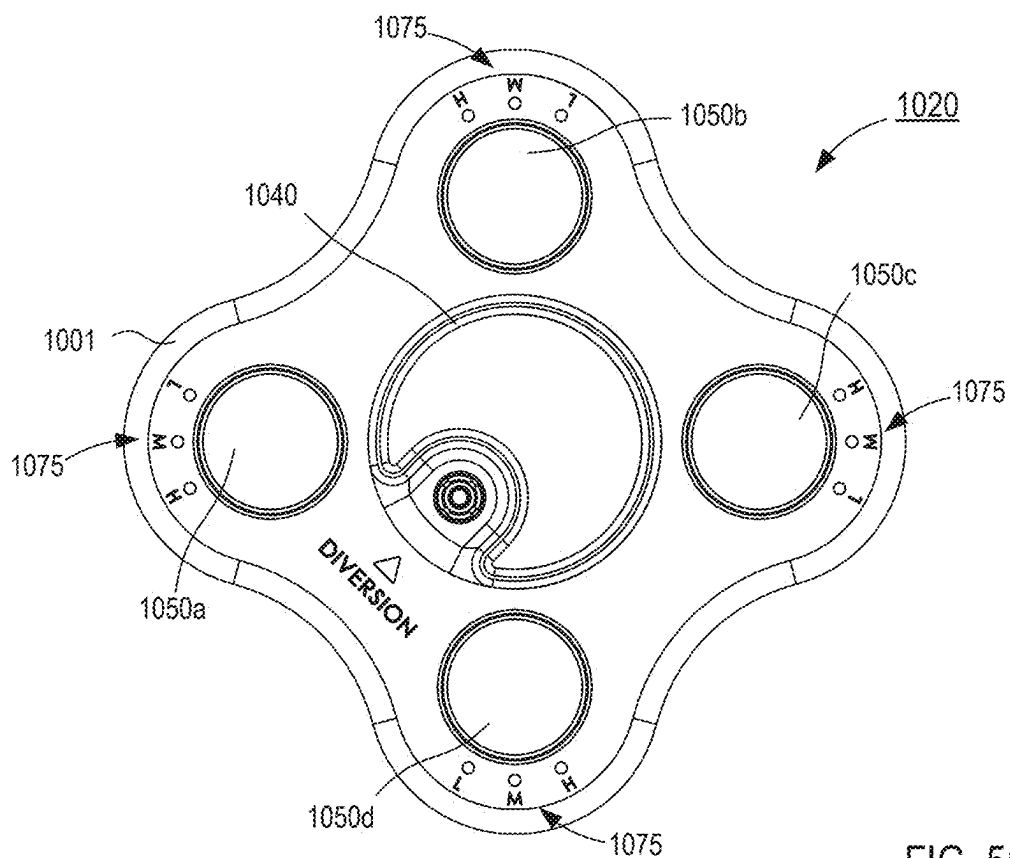


FIG. 55

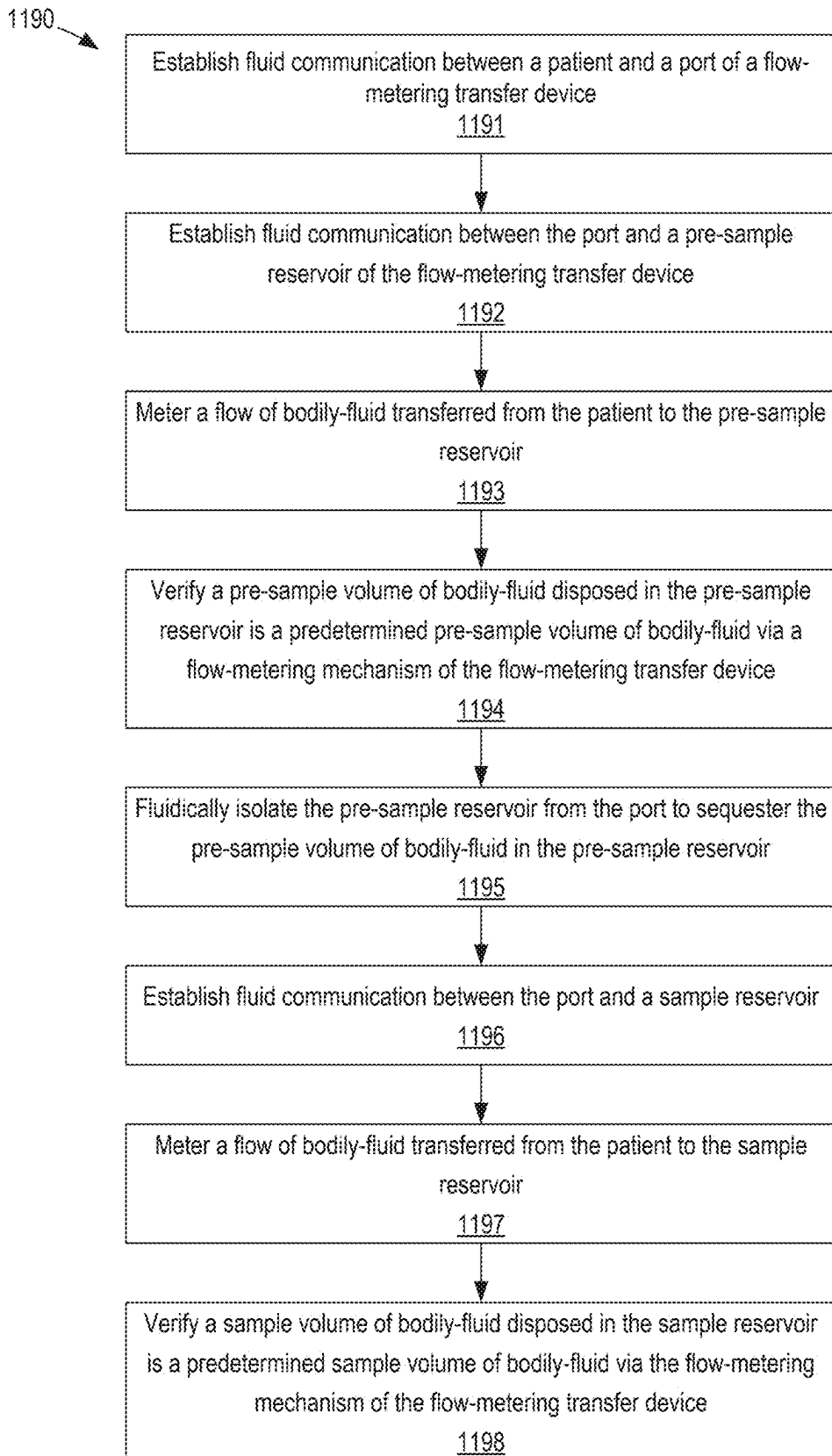


FIG. 56

STERILE BODILY-FLUID COLLECTION DEVICE AND METHODS

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] This application is a continuation of U.S. patent application Ser. No. 14/728,318, filed Jun. 2, 2015, entitled, “Sterile Bodily-Fluid Collection Device and Methods,” which is a continuation of International Patent Application Serial No. PCT/US2013/073080, filed Dec. 4, 2013, entitled, “Sterile Bodily-Fluid Collection Device and Methods,” and a continuation-in-part of U.S. patent application Ser. No. 14/096,826 (now U.S. Pat. No. 10,251,590), filed Dec. 4, 2013, entitled, “Sterile Bodily-Fluid Collection Device and Methods,” each of which claims priority to and the benefit of U.S. Provisional Application Ser. No. 61/733,199, filed Dec. 4, 2012, entitled, “Sterile Bodily-Fluid Collection Device and Methods,” the disclosures of which are incorporated herein by reference in their entireties.

BACKGROUND

[0002] Embodiments described herein relate generally to the parenteral procurement of bodily-fluid samples, and more particularly to devices and methods for parenterally-procuring bodily-fluid samples with reduced contamination from microbes or other contaminants exterior to the bodily-fluid source, such as dermally-residing microbes.

[0003] Health care practitioners routinely perform various types of microbial tests on patients using parenterally-obtained bodily-fluids. In some instances, patient samples (e.g., bodily-fluids) are tested for the presence of one or more potentially undesirable microbes, such as bacteria, fungi, or yeast (e.g., *Candida*). Microbial testing may include incubating patient samples in one or more sterile vessels containing culture media that is conducive to microbial growth, real-time diagnostics, and/or molecular PCR-based approaches. Generally, when such microbes are present in the patient sample, the microbes flourish over time in the culture medium. After a variable amount of time (e.g., a few hours to several days), organism growth can be detected by automated, continuous monitoring. Such automated monitoring can detect carbon dioxide produced by organism growth. The culture medium can then be tested for the presence of the microbes. The presence of microbes in the culture medium suggests the presence of the same microbes in the patient sample which, in turn, suggests the presence of the same microbes in the bodily-fluid of the patient from which the sample was obtained. Accordingly, when microbes are determined to be present in the culture medium, the patient may be prescribed one or more antibiotics or other treatments specifically designed to treat or otherwise remove the undesired microbes from the patient.

[0004] Patient samples, however, can become contaminated during procurement and/or can be otherwise susceptible to false positive results. One way in which contamination of a patient sample may occur is by the transfer of microbes from a bodily surface (e.g., dermally-residing microbes) dislodged during needle insertion into a patient and subsequently transferred to a culture medium with the patient sample. The bodily surface and/or other undesirable external microbes may be dislodged either directly or via dislodged tissue fragments, hair follicles, sweat glands and other skin adnexal structures. Another possible source of

contamination is from the person drawing the patient sample. For example, a doctor, phlebotomist, nurse, etc. can transfer contaminants from their body (e.g., finger, arms, etc.) to the patient sample and/or to the equipment containing the patient sample. Expanding further, equipment and/or devices used during a patient sample procurement process (e.g., patient to needle, needle/tubing to sample vessels, etc.) often include multiple fluidic interfaces that can each introduce points of potential contamination. The use of such equipment and/or devices typically includes manual intervention to connect and/or fluidically couple various interfaces. Since these interfaces are not preassembled and sterilized as a single fluidically coupled system, external contaminants can be introduced to the patient sample via the user (e.g., doctor, phlebotomist, etc.) and/or other sources (e.g. ambient air, contaminants on surfaces of tables and counters in patient room, microbes transferred from linens or clothing, etc.). In some instances, such contaminants may thrive in a culture medium and eventually yield a positive microbial test result, thereby falsely indicating the presence of such microbes in vivo.

[0005] In some instances, false positive results and/or false negative results can be attributed to a specific volume of the patient sample. For example, overfilling of volume-sensitive blood culture bottles can lead to false positive results as noted in the instructions for use and/or warning labeling from manufacturers of such culture bottles, as well as associated automated continuous monitoring microbial detection systems. On the other hand, as another example, insufficient patient sample volume within a culture medium can result in false negative results. By way of example, in a study performed by the Mayo Clinic entitled, “Optimized Pathogen Detection with 30- Compared to 20-Milliliter Blood Culture Draws,” published in the December 2011 issue of Journal of Clinical Microbiology, a patient sample volume of 20 milliliters (mL) can result in detection of about 80% of bacteremias present in a patient sample, a patient sample volume of 40 mL can result in detection of about 88% of the bacteremias, and a patient sample volume of 60 mL can result in detection of about 99% of the bacteremias.

[0006] Such inaccurate results as a result of contamination, insufficient patient sample volume, and/or the like are a concern when attempting to diagnose or treat a suspected illness or condition. For example, false negative results from microbial tests may result in a misdiagnosis and/or delayed treatment of a patient illness which, in some cases, could result in the death of the patient. Conversely, false positive results from microbial tests may result in the patient being unnecessarily subjected to one or more anti-microbial therapies, which may cause serious side effects to the patient including, for example, death, as well as produce an unnecessary burden and expense to the health care system due to extended length of patient stay and/or other complications associated with erroneous treatments. Additionally, the use of diagnostic imaging equipment attributable to these false positive results is also a concern from both a cost as well as patient safety perspective as unnecessary exposure to concentrated radiation associated with a variety of imaging procedures (e.g., CT scans) has many known adverse impacts on long-term patient health.

[0007] As such, a need exists for sterile “all-in-one” bodily-fluid collection devices and methods that reduce microbial contamination in bodily-fluid test samples by, for example, minimizing exposure of the patient sample and/or

fluidic interfaces to ambient non-sterile conditions and/or other sources of external contamination. Additionally, a need exists for such bodily-fluid collection devices to include a means for accurately metering, measuring, and/or otherwise assessing and confirming a volume of bodily-fluid transferred from a patient to a sample reservoir or culture medium that can be visually, tactically, or otherwise communicated to a healthcare practitioner procuring the patient sample in substantially real-time (e.g. at the patient bedside).

SUMMARY

[0008] Devices for parenterally-procuring bodily-fluid samples with reduced contamination from microbes exterior to the bodily-fluid source, such as dermally-residing microbes and/or other undesirable external contaminants, are described herein. In some embodiments, an apparatus for obtaining a bodily fluid sample from a patient includes a pre-sample reservoir, a diversion mechanism, and a flow metering mechanism. The pre-sample reservoir is configured to receive a first volume of bodily-fluid withdrawn from the patient. The diversion mechanism includes an inlet port, a first outlet port, and a second outlet port, and defines a first fluid flow path and a second fluid flow path. The inlet port can be coupled to a lumen-defining device for receiving bodily-fluids from the patient. The first outlet port and the second outlet port are configured to fluidically couple the pre-sample reservoir and a sample reservoir, respectively, to the diversion mechanism. The first fluid flow path is configured to place the first outlet port in fluid communication with the inlet port and a second fluid flow path configured to place the second outlet port in fluid communication with the inlet port. The flow metering mechanism is in fluid communication with the first fluid flow path and the second fluid flow path. The flow metering mechanism is configured to meter a flow of the first volume of bodily-fluid through the first fluid flow path into the pre-sample reservoir and to meter a flow of a second volume of bodily-fluid through the second fluid flow path into the sample reservoir. The flow metering mechanism is configured to display a volumetric indicator associated with the first volume and the second volume.

BRIEF DESCRIPTION OF THE DRAWINGS

[0009] FIG. 1 is a schematic illustration of a bodily-fluid collection device according to an embodiment.

[0010] FIG. 2 is a perspective view of the bodily-fluid collection device according to an embodiment.

[0011] FIG. 3 is an exploded perspective view of the bodily-fluid collection device of FIG. 2.

[0012] FIG. 4 is a cross-sectional side view of a housing included in the bodily-fluid collection device of FIG. 2, taken along the line X₁-X₁ in FIG. 3.

[0013] FIG. 5 is a cross-sectional view of a movable member included in the bodily-fluid collection device of FIG. 2, taken along the line X₂-X₂ in FIG. 3.

[0014] FIGS. 6 and 7 are cross-sectional views of a flow controller included in the bodily-fluid collection device of FIG. 2, taken along the line X₃-X₃ and X₄-X₄ in FIG. 3, respectively.

[0015] FIG. 8 is a top view of the bodily-fluid collection device of FIG. 2 in a first configuration.

[0016] FIG. 9 is a cross-sectional view of the bodily-fluid collection device of FIG. 2 in the first configuration, taken along the line X₅-X₅ in FIG. 8.

[0017] FIG. 10 is a top view of the bodily-fluid collection device of FIG. 2 in a second configuration.

[0018] FIG. 11 is a cross-sectional view of the bodily-fluid collection device of FIG. 2 in the second configuration, taken along the line X₆-X₆ in FIG. 10.

[0019] FIGS. 12 and 13 are cross-sectional views of the bodily-fluid collection device of FIG. 2, in a third configuration and a fourth configuration, respectively, taken along the line X₆-X₆ in FIG. 10.

[0020] FIG. 14 is a perspective view of a bodily-fluid collection device according to an embodiment.

[0021] FIG. 15 is a cross-sectional side view of the bodily-fluid collection device of FIG. 14, taken along the line X₇-X₇.

[0022] FIG. 16 is a perspective view of a bodily-fluid collection device according to an embodiment.

[0023] FIG. 17 is an exploded perspective view of a diversion mechanism included in the bodily-fluid collection device of FIG. 16.

[0024] FIG. 18 is a cross-sectional side view of a distribution member included in the bodily-fluid collection device of FIG. 16, taken along the line X₈-X₈ in FIG. 16.

[0025] FIG. 19 is a top view of the bodily-fluid collection device of FIG. 16 in a first configuration.

[0026] FIG. 20 is a cross-sectional view of a portion the bodily-fluid collection device of FIG. 16 in the first configuration, taken along the line X₉-X₉ in FIG. 19.

[0027] FIG. 21 is a top view of the bodily-fluid collection device of FIG. 16 in a second configuration.

[0028] FIG. 22 is a cross-sectional view of the bodily-fluid collection device of FIG. 16 in the second configuration, taken along the line X₁₀-X₁₀ in FIG. 21.

[0029] FIG. 23 is a perspective view of a bodily-fluid collection device according to an embodiment.

[0030] FIG. 24 is a cross-sectional view of a portion of the bodily-fluid collection device of FIG. 23 taken along the line X₁₁-X₁₁, in a first configuration.

[0031] FIG. 25 is a cross-sectional view of the bodily-fluid collection device of FIG. 23 taken along the line X₁₁-X₁₁, in a second configuration.

[0032] FIG. 26 is a perspective view of a bodily-fluid collection device according to an embodiment.

[0033] FIG. 27 is an exploded perspective view of a diversion mechanism included in the bodily-fluid collection device of FIG. 26.

[0034] FIG. 28 is a cross-sectional view of a distribution member included in the bodily-fluid collection device of FIG. 26 taken along the line X₁₃-X₁₃ in FIG. 27.

[0035] FIG. 29 is a cross-sectional view of a coupling member included in the bodily-fluid collection device of FIG. 26 taken along the line X₁₄-X₁₄ in FIG. 27.

[0036] FIG. 30 is a cross-sectional view of a dial included in the bodily-fluid collection device of FIG. 26 taken along the line X₁₅-X₁₅ in FIG. 27.

[0037] FIG. 31 is a cross-sectional view of a valve included in the bodily-fluid collection device of FIG. 26 taken along the line X₁₆-X₁₆ in FIG. 27.

[0038] FIG. 32 is a cross-sectional view of the bodily-fluid collection device of FIG. 26 in a first configuration, taken along the line X₁₂-X₁₂.

[0039] FIG. 33 is a cross-sectional view of the bodily-fluid collection device of FIG. 26 in a second configuration, taken along the line X_{12} - X_{12} .

[0040] FIG. 34 is a perspective view of a bodily-fluid collection device according to an embodiment.

[0041] FIG. 35 is an exploded perspective view of a diversion mechanism included in the bodily-fluid collection device of FIG. 34.

[0042] FIG. 36 is a cross-sectional view of a distribution member included in the diversion mechanism of FIG. 35 taken along the line X_{17} - X_{17} .

[0043] FIG. 37 is a top view of a portion of the bodily-fluid collection device of FIG. 34 in a first configuration.

[0044] FIG. 38 is a cross-sectional view of the portion the bodily-fluid collection device of FIG. 37 in the first configuration, taken along the line X_{18} - X_{18} .

[0045] FIG. 39 is a top view of the bodily-fluid collection device of FIG. 33 in a second configuration.

[0046] FIG. 40 is a cross-sectional view of the bodily-fluid collection device of FIG. 33 in the second configuration, taken along the line X_{19} - X_{19} in FIG. 39.

[0047] FIG. 41 is a perspective view of a bodily-fluid collection device in a first configuration according to an embodiment.

[0048] FIG. 42 is an exploded perspective view of a portion of the bodily-fluid collection device of FIG. 41.

[0049] FIG. 43 is a cross-sectional view of the bodily-fluid collection device of FIG. 41 in a first configuration, taken along the line X_{20} - X_{20} .

[0050] FIG. 44 is a cross-sectional view of the bodily-fluid collection device of FIG. 41 in a second configuration, taken along the line X_{20} - X_{20} .

[0051] FIG. 45 is a cross-sectional view of the bodily-fluid collection device of FIG. 41 in a third configuration, taken along the line X_{20} - X_{20} .

[0052] FIG. 46 is a perspective view of a bodily-fluid collection device in a first configuration according to an embodiment.

[0053] FIG. 47 is an exploded perspective view of a portion of the bodily-fluid collection device of FIG. 45.

[0054] FIG. 48 is a cross-sectional side view of a distribution member included in the bodily-fluid collection device of FIG. 46, taken along the line X_{21} - X_{21} in FIG. 47.

[0055] FIG. 49 is a cross-sectional view of a movable member included in the bodily-fluid collection device of FIG. 46, taken along the line X_{22} - X_{22} in FIG. 47.

[0056] FIG. 50 is a top view of the bodily-fluid collection device of FIG. 46 in a first configuration.

[0057] FIG. 51 is a cross-sectional view of a portion the bodily-fluid collection device of FIG. 46 in the first configuration, taken along the line X_{23} - X_{23} in FIG. 50.

[0058] FIG. 52 is a top view of the bodily-fluid collection device of FIG. 46 in a second configuration.

[0059] FIG. 53 is a cross-sectional view of the bodily-fluid collection device of FIG. 46 in the second configuration, taken along the line X_{24} - X_{24} in FIG. 52.

[0060] FIG. 54 is a perspective view of a bodily-fluid collection device according to an embodiment.

[0061] FIG. 55 is a top view of the bodily-fluid collection device of FIG. 54.

[0062] FIG. 56 is a flowchart illustrating a method of obtaining a bodily-fluid sample with reduced contamination using a collection device according to an embodiment.

DETAILED DESCRIPTION

[0063] Devices for parenterally-procuring bodily-fluid samples with reduced contamination from microbes exterior to the bodily-fluid source, such as dermally-residing microbes and/or other undesirable external contaminants, are described herein. In some embodiments, an apparatus for obtaining a bodily fluid sample from a patient includes a pre-sample reservoir, a diversion mechanism, and a flow metering mechanism. The pre-sample reservoir is configured to receive a first volume of bodily-fluid withdrawn from the patient. The diversion mechanism includes an inlet port, a first outlet port, and a second outlet port, and defines a first fluid flow path and a second fluid flow path. The inlet port can be coupled to a lumen-defining device for receiving bodily-fluids from the patient. The first outlet port and the second outlet port are configured to fluidically couple the pre-sample reservoir and a sample reservoir, respectively, to the diversion mechanism. The first fluid flow path is configured to place the first outlet port in fluid communication with the inlet port and a second fluid flow path configured to place the second outlet port in fluid communication with the inlet port. The flow metering mechanism is in fluid communication with the first fluid flow path and the second fluid flow path. The flow metering mechanism is configured to meter a flow of the first volume of bodily-fluid through the first fluid flow path into the pre-sample reservoir and to meter a flow of a second volume of bodily-fluid through the second fluid flow path into the sample reservoir. The flow metering mechanism is configured to display a volumetric indicator associated with the first volume and the second volume.

[0064] In some embodiments, an apparatus for obtaining a bodily-fluid sample from a patient includes a pre-sample reservoir, a diversion mechanism, a flow controller, and a movable member. The pre-sample reservoir is configured to receive a first volume of bodily-fluid withdrawn from the patient. The diversion mechanism includes an inlet port, a first outlet port, and a second outlet port. The inlet port is coupleable to a lumen-defining device for receiving bodily-fluids from the patient. The first outlet port fluidically couples the pre-sample reservoir to the diversion mechanism and the second outlet port fluidically couples a sample reservoir to the diversion mechanism. The flow controller is at least partially disposed within the diversion mechanism and can be moved between a first configuration, in which the flow controller defines at least a portion of a fluid flow path between the inlet port and the first outlet port, and a second configuration, in which the flow controller defines at least a portion of a fluid flow path between the inlet port and the second outlet port. The movable member movably coupled to the diversion mechanism and movable through the second outlet port between a first configuration, in which the sample reservoir is fluidically isolated from the fluid flow path between the inlet port and the second outlet port, and a second configuration, in which the sample reservoir is in fluid communication with the fluid flow path between the inlet port and the second outlet port. The sample reservoir is configured to receive a second volume of bodily-fluid withdrawn from the patient when the flow controller is in its second configuration and the movable member is in its second configuration.

[0065] In some embodiments, an apparatus for obtaining a bodily-fluid sample from a patient includes a pre-sample reservoir, a diversion mechanism, and a flow controller. The

pre-sample reservoir is configured to receive a first volume of bodily-fluid withdrawn from the patient. The diversion mechanism includes a housing and a distribution member. The housing defines a first aperture in fluid communication with the pre-sample reservoir and a second aperture. The distribution member is at least partially disposed within the housing and defines a fluid flow channel in fluid communication with the second aperture. The distribution member includes a coupling portion that is in fluid communication with the flow channel and is configured to be physically and fluidically coupled to a sample reservoir. The flow controller includes an inlet port couplable to a lumen-defining device for receiving bodily-fluids from the patient. The flow controller is rotatably coupled to the diversion mechanism and movable between a first configuration, in which the inlet port is in fluid communication with the first aperture, and a second configuration, in which the inlet port is in fluid communication with the second aperture.

[0066] In some embodiments, a method of using a flow-metering transfer device having a diversion mechanism with an inlet port configured to be selectively placed in fluid communication with a pre-sample reservoir and a sample reservoir, and a flow-metering mechanism configured to meter a flow of bodily-fluid from the patient to the pre-sample reservoir and to the sample reservoir includes establishing fluid communication between the patient and the inlet port of the flow-metering transfer device. Fluid communication is then established between the port and the pre-sample reservoir. A flow of bodily-fluid transferred from the patient to the pre-sample reservoir is metered. The method includes verifying a pre-sample volume of bodily-fluid disposed in the pre-sample reservoir is a first pre-sample volume of bodily-fluid via the flow-metering mechanism of the flow-metering transfer device. With the pre-sample volume disposed in the pre-sample reservoir, the pre-sample reservoir is fluidically isolated from the port to sequester the pre-sample volume of bodily-fluid in the pre-sample reservoir. With the pre-sample reservoir fluidically isolated, the method includes establishing fluid communication between the port and the sample reservoir. A flow of bodily-fluid transferred from the patient to the sample reservoir is metered. The method includes verifying a sample volume of bodily-fluid disposed in the sample reservoir is a first sample volume of bodily-fluid via the flow-metering mechanism of the flow-metering transfer device.

[0067] In some embodiments, an apparatus includes a diversion mechanism and a flow controller. The diversion mechanism can define an inlet port, a first outlet port, a second outlet port, and a third outlet port. The first outlet port is fluidically coupled to a pre-sample reservoir, the second outlet port is fluidically coupled to a first sample reservoir, and the third outlet port is fluidically coupled to a second sample reservoir, and so forth. All of the fluid reservoirs can be fluidically isolated from each other. The flow controller includes various fluidic channels that can allow fluidic movement in specified directions and can be configured to be operably coupled to the diversion mechanism. In use, when the diversion mechanism is at a first configuration, the flow controller can allow a flow of bodily-fluid to enter the pre-sample reservoir. The diversion mechanism can be moved to a second configuration, where the flow controller can allow a flow of bodily-fluid to enter the first sample reservoir. Additionally, the diversion mechanism

can then be moved to a third configuration, whereby the flow controller can allow a flow of bodily-fluid to enter the second sample reservoir.

[0068] In some embodiments, a bodily-fluid collection device can be configured to selectively divert a first, predetermined volume of a bodily-fluid to a pre-sample reservoir before permitting the flow of a second volume of the bodily-fluid into a first sample reservoir and/or a third volume of the bodily-fluid into a second sample reservoir. In this manner, the second and/or third volumes of bodily-fluid can be used for diagnostic or other testing, while the first volume of bodily-fluid, which may contain microbes from a bodily surface or other source external to the patient from which the sample is procured, is isolated. In some embodiments, the bodily-fluid collection device can include additional sample reservoirs (e.g., 3, 4, 5, 6 or more) depending on the analysis and/or testing protocols to be performed.

[0069] In some embodiments, a bodily-fluid collection device can include flow metering to ensure the proper volume of bodily-fluid is collected from a patient and/or transferred into a specific pre-sample and/or sample reservoir. The bodily-fluid collection device can be configured to automatically divert and/or control the fluid flow after metered volumes of bodily-fluid are collected. For example, after a first metered pre-sample volume is collected, a diversion mechanism can be configured to divert the bodily-fluid flow to a first sample reservoir and then after a first metered sample volume is collected, the diversion mechanism can be configured to divert the bodily-fluid flow to a second sample reservoir and so on. In some embodiments, the bodily-fluid collection device can include a metered volume display such as, for example, a liquid crystal display (LCD), to provide a visual indication to the user of how much bodily-fluid has been collected into each specific, individual sample reservoir. In some embodiments, multiple displays can be provided to allow for customized pre-sample and/or sample volume collection.

[0070] As used in this specification and the appended claims, the singular forms “a,” “an” and “the” include plural referents unless the context clearly dictates otherwise. Thus, for example, the term “a member” is intended to mean a single member or a combination of members, “a material” is intended to mean one or more materials, or a combination thereof.

[0071] As used herein, “bodily-fluid” can include any fluid obtained from a body of a patient, including, but not limited to, blood, cerebrospinal fluid, urine, bile, lymph, saliva, synovial fluid, serous fluid, pleural fluid, amniotic fluid, and the like, or any combination thereof.

[0072] As used herein, the terms “first, predetermined amount,” “first amount,” and “first volume” describe an amount of bodily-fluid configured to be received or contained by a first reservoir or a pre-sample reservoir. While the terms “first amount” and “first volume” do not explicitly describe a predetermined amount, it should be understood that the first amount is the first, predetermined amount unless explicitly described differently.

[0073] As used herein, the terms “second amount” and “second volume” describe an amount of bodily-fluid configured to be received or contained by a second reservoir or sample reservoir. The second amount can be any suitable amount of bodily-fluid and need not be predetermined. Conversely, when explicitly described as such, the second

amount received and contained by the second reservoir or sample reservoir can be a second, predetermined amount.

[0074] As used herein, the term “set” can refer to multiple features or a singular feature with multiple parts. For example, when referring to set of walls, the set of walls can be considered as one wall with distinct portions, or the set of walls can be considered as multiple walls. Similarly stated, a monolithically constructed item can include a set of walls. Such a set of walls can include, for example, multiple portions that are in discontinuous from each other. A set of walls can also be fabricated from multiple items that are produced separately and are later joined together (e.g., via a weld, an adhesive or any suitable method).

[0075] As used herein, the terms “proximal” and “distal” refer to the direction closer to and away from, respectively, a user who would place the device into contact with a patient. Thus, for example, the end of a device first touching the body of the patient would be the distal end, while the opposite end of the device (e.g., the end of the device being manipulated by the user) would be the proximal end of the device.

[0076] As used herein, the terms “about,” “approximately,” and “substantially” when used in connection with a numerical value is intended to convey that the value so defined is nominally the value stated. Said another way, the terms about, approximately, and substantially when used in connection with a numerical value generally include the value stated plus or minus a given tolerance. For example, in some instances, a suitable tolerance can be plus or minus 10% of the value stated; thus, about 0.5 would include 0.45 and 0.55, about 10 would include 9 to 11, about 1000 would include 900 to 1100. In other instances, a suitable tolerance can be plus or minus an acceptable percentage of the last significant figure in the value stated. For example, a suitable tolerance can be plus or minus 10% of the last significant figure; thus, about 10.1 would include 10.09 and 10.11, approximately 25 would include 24.5 and 25.5. Such variance can result from manufacturing tolerances or other practical considerations (such as, for example, tolerances associated with a measuring instrument, acceptable human error, or the like).

[0077] When describing a relationship between a predetermined volume of bodily-fluid and a collected volume of bodily-fluid it is to be understood that the values include a suitable tolerance such as those described above. For example, when stating that a collected volume of bodily-fluid is substantially equal to a predetermined volume of bodily-fluid, the collected volume and the predetermined volume are nominally equal within a suitable tolerance. In some instances, the tolerances can be determined by the intended use of the collected volume of bodily-fluid. For example, in some instances, an assay of a blood culture can be about 99% accurate when the collected volume of blood is within 1.0% to 5.0% of the manufacturer’s (or evidence-based best practices) recommended volume. By way of an example, a manufacturer’s recommended volume for an assay of a bodily-fluid can be 10 milliliters (mL) per sample collection bottle, with a total of four or six collection bottles used (i.e., an aggregate volume of 40 ml to 60 ml) plus or minus 5% for about 99% confidence. Thus, a collected volume of 10.5 mL would provide results with over about 99% confidence, while a collected volume of 11 mL would provide results with less than about 99% confidence. In other instances, a suitable tolerance can be 0.1%, 0.5%, 1.0%,

2.0%, 3.0%, 4.0%, 5.0%, 6.0%, 7.0%, 8.0%, 9.0%, 10.0%, or any fraction of a percent therebetween. In still other instances, a tolerance can be greater than 10.0%. Thus, any of the embodiments described herein can include and/or can be used in conjunction with any suitable flow-metering mechanism and/or device that is configured to meter a flow and/or otherwise measure a volume of bodily-fluid within a suitable tolerance. Moreover, the flow-metering mechanism and/or device can be arranged such as to minimize or eliminate tolerance stacking that can result from a combination of inaccurate measurement, human error, and/or the like.

[0078] FIG. 1 is a schematic illustration of a portion of a bodily-fluid collection device 100, according to an embodiment. Generally, the bodily-fluid collection device 100 (also referred to herein as “fluid collection device” or “collection device”) is configured to permit the withdrawal of bodily-fluid from a patient such that a first portion or volume of the withdrawn fluid is diverted away from a second and/or third portion or volume of the withdrawn fluid that is to be used as a biological sample, such as for testing for the purpose of medical diagnosis and/or treatment. In other words, the collection device 100 is configured to transfer a first, predetermined volume of a bodily-fluid to a pre-sample collection reservoir and a second and third volume (or, in some embodiments, a fourth, fifth and so on) of bodily-fluid to one or more sample collection reservoirs fluidically isolated from the pre-sample collection reservoir, as described in more detail herein.

[0079] The collection device 100 includes a diversion mechanism 120, a flow controller 140, a pre-sample reservoir 170, a first sample reservoir 180, and a second sample reservoir 190, different than the first sample reservoir 180. The diversion mechanism 120 includes an inlet port 121 and at least two outlet ports, such as a first outlet port 125, and a second outlet port 126 as shown in FIG. 1. In some embodiments, the diversion mechanism 120 can include a set of outlet ports equal to a total number of pre-sample reservoirs and sample reservoirs. For example, the diversion mechanism 120 can include five outlet ports when the collection device 100 has one pre-sample reservoir and four sample reservoirs. In some embodiments, the diversion mechanism 120 can be operatively coupled to an actuator (not shown in FIG. 1) which can facilitate the movement of the diversion mechanism 120 between multiple configurations. The inlet port 121 is configured to be fluidically coupled to a medical device defining a pathway X for withdrawing and/or conveying the bodily-fluid from the patient to the collection device 100. For example, the inlet port 121 can be fluidically coupled to a needle or other lumen-defining device (e.g., flexible sterile tubing). In this manner, the diversion mechanism 120 can receive the bodily-fluid from the patient via the needle or any other lumen-defining device.

[0080] The first outlet port 125 of the diversion mechanism 120 can be fluidically coupled to the pre-sample reservoir 170. In some embodiments, the pre-sample reservoir 170 is monolithically formed with the first outlet port 125 and/or a portion of the diversion mechanism 120. In other embodiments, the pre-sample reservoir 170 can be mechanically and/or fluidically coupled to the diversion mechanism 120 via an adhesive, a resistance fit, a mechanical fastener, any number of mating recesses, a threaded coupling, and/or any other suitable coupling or combination

thereof. Similarly stated, the pre-sample reservoir **170** can be physically (e.g., mechanically) coupled to the diversion mechanism **120** such that an interior volume defined by the pre-sample reservoir **170** is in fluid communication with the first outlet port **125** of the diversion mechanism **120**. In still other embodiments, the pre-sample reservoir **170** can be operably coupled to the first outlet port **125** of the diversion mechanism **120** via an intervening structure (not shown in FIG. 1), such as flexible sterile tubing. More particularly, the intervening structure can define a lumen configured to place the pre-sample reservoir **170** in fluid communication with the first outlet port **125**.

[0081] The pre-sample reservoir **170** is configured to receive and contain the first, predetermined volume of the bodily-fluid. In some embodiments, the pre-sample reservoir **170** is configured to contain the first volume of the bodily-fluid such that the first volume is fluidically isolated from a second and/or third volume of the bodily-fluid (which can be the same or different than the first volume of bodily-fluid) that is subsequently withdrawn from the patient. The pre-sample reservoir **170** can be any suitable reservoir for containing a bodily-fluid, such as a pre-sample reservoir described in detail in U.S. Pat. No. 8,197,420 entitled, “Systems and Methods for Parenterally Procuring Bodily-Fluid Samples with Reduced Contamination,” issued Jun. 12, 2012 (referred to henceforth as the “420 patent”), the disclosure of which is incorporated herein by reference in its entirety.

[0082] In some embodiments, the second outlet port **126** of the diversion mechanism **120** is configured to be fluidically coupled to a lumen-defining device that can be coupled to the first sample reservoir **180** and the second sample reservoir **190**. Optionally, in other embodiments, the second outlet port **126** of the diversion mechanism **120** can be coupled to the first sample reservoir **180** and the diversion mechanism **120** can have a third outlet port (not shown) coupled to the second sample reservoir **190**. In some embodiments, the first sample reservoir **180** can be monolithically formed with the second outlet port **126** and/or a portion of the diversion mechanism **120**. In other embodiments, the first sample reservoir **180** can be mechanically coupled to the second outlet port **126** or operably coupled to the second outlet port **126** via an intervening structure, such as described above with reference to the pre-sample reservoir **170**. The first sample reservoir **180** is configured to receive and contain the second volume of the bodily-fluid. For example, the second volume of bodily-fluid can be an amount withdrawn from the patient subsequent to withdrawal of the first pre-sample volume. In some embodiments, the first sample reservoir **180** is configured to contain the second volume of the bodily-fluid in such a manner that the second volume is fluidically isolated from the first volume of the pre-sample bodily-fluid.

[0083] The first sample reservoir **180** and the second sample reservoir **190** can be any suitable sterile reservoir for containing a bodily-fluid including, for example, a sample reservoir as described in the ‘420 patent incorporated by reference above. In some embodiments, the second volume can be any suitable volume of bodily-fluid and need not be predetermined. In other embodiments, the transfer of the bodily-fluid to the first sample reservoir **180** and/or the second sample reservoir **190** can be metered or the like such that the second volume is a second predetermined volume.

[0084] The second sample reservoir **190** can be any suitable sample reservoir. In some embodiments, the second sample reservoir **190** can be substantially similar to the first sample reservoir **180** described above. The second sample reservoir **190** can be fluidically coupled to the second output port **126** as described above. The fluidic coupling of the second outlet port **126** to the second sample reservoir **190** can be substantially similar to the fluidic coupling of the second outlet port **126** to the first sample reservoir **180**, as described in detail above. Therefore, such portions are not described in further detail herein and should be considered substantially similar unless explicitly described differently. Furthermore, additional outlet ports of the diversion mechanism **120** and sample reservoirs (not shown in FIG. 1) can be substantially similar to the second outlet port **126** and the first sample reservoir **180**.

[0085] In some embodiments, the pre-sample reservoir **170**, the first sample reservoir **180**, and the second sample reservoir **190** can be coupled to (or formed with) the diversion mechanism **120** in a similar manner. In other embodiments, the pre-sample reservoir **170**, the first sample reservoir **180**, and the second sample reservoir **190** need not be similarly coupled to the diversion mechanism **120**. For example, in some embodiments, the pre-sample reservoir **170** can be monolithically formed with the diversion mechanism **120** (e.g., the first outlet port **124**) and the first sample reservoir **180** and/or the second sample reservoir **190** can be operably coupled to the diversion mechanism **120** (e.g., the second outlet port **126**) via an intervening structure, such as a flexible sterile tubing or any combination thereof.

[0086] In some embodiments, the collection device **100** can further include an actuator (not shown in FIG. 1) and a flow controller **140** that defines a first fluid flow path **142**, a second fluid flow path **144**, and optionally additional fluid flow paths (not shown in FIG. 1). In some embodiments, the actuator can be included in or otherwise operably coupled to the diversion mechanism **120**. In this manner, the actuator can be configured to control fluid movement within the flow controller **140** (e.g., between different configurations). For example, the actuator can be movable between a first position corresponding to a first configuration of the flow controller **140**, a second position, different than the first position, corresponding to a second configuration of the flow controller **140**, and so on. In some embodiments, the actuator can be configured for uni-directional movement. For example, the actuator can be moved from its first position to its second position, but cannot be moved from its second position to its first position. Similarly, the actuator can be moved from its second position to a third position, but cannot be moved from its third position back to its second position. In this manner, the flow controller **140** is prevented from being moved into its second or third configuration before its first configuration, thus requiring that the first amount of the bodily-fluid be directed to the pre-sample reservoir **170** and not the sample reservoirs **180** and/or **190** which is designed to contain the second and/or third volume of the withdrawn fluid that is to be used as a biological sample, such as for testing for the purpose of medical diagnosis and/or treatment.

[0087] The flow controller **140** is configured such that when in the first configuration, the first fluid flow path **142** fluidically couples the inlet port **121** to the first outlet port **125**, and when in the second configuration, the second fluid flow path **144** fluidically couples the inlet port **121** to the

second outlet port **126**. In some embodiments, an actuator as described above can be configured to move the flow controller **140** in a translational motion between the first configuration, and the second configuration, and optionally a third or fourth configuration. For example, in some embodiments, the flow controller **140** can be in the first configuration when the flow controller **140** is in a distal position relative to the collection device **100**. In such embodiments, the actuator can be actuated to move the flow controller **140** in the proximal direction to a proximal position relative to the collection device **100**, thereby placing the flow controller **130** in the second configuration. In other embodiments, the actuator can also be actuated to move the flow controller **140** in a rotational motion between the first configuration and the second configuration or optionally a third or fourth configuration.

[0088] Accordingly, when the flow controller **140** is in the first configuration, the second outlet port **126** (and optionally additional outlet ports coupled to sample reservoirs) is fluidically isolated from the inlet port **121**. Similarly, when the flow controller **140** is in the second configuration, the first outlet port **125** is fluidically isolated from the inlet port **121**. And optionally, if the flow controller **140** is in a third configuration (not shown in FIG. 1), the first outlet port **125** and the second outlet port **126** are fluidically isolated from the inlet port **121**. In this manner, the flow controller **140** can direct, or divert the first amount of the bodily-fluid to the pre-sample fluid reservoir **170** via the first outlet port **125** when the flow controller **140** is in the first configuration and can direct, or divert the second amount of the bodily-fluid to the first sample fluid reservoir **180** via the second outlet port **126** when the flow controller **140** is in the second configuration.

[0089] In some embodiments, at least a portion of the actuator can be operably coupled to the pre-sample fluid reservoir **170**. In this manner, the actuator (or at least the portion of the actuator) can be configured to introduce or otherwise facilitate the development of a vacuum within the “pre-sample” fluid reservoir **170**, thereby initiating flow of the bodily-fluid through the collection device **100** and into the pre-sample fluid reservoir **170** when the diversion mechanism **120** is in its first configuration. The actuator can include any suitable mechanism for actuating the flow of bodily-fluid into the collection device **100**, such as, for example, a rotating disc, a plunger, a slide, a dial, a button, a handle, a lever, and/or any other suitable mechanism or combination thereof. Examples of suitable actuators are described in more detail herein with reference to specific embodiments.

[0090] In some embodiments, the diversion mechanism **120** can be configured such that the first amount of bodily-fluid need to be conveyed to the pre-sample fluid reservoir **170** before the diversion mechanism **120** will permit the flow of the second amount of bodily-fluid to be conveyed through the diversion mechanism **120** to the first sample fluid reservoir **180** and/or to the second sample fluid reservoir **190**. In this manner, the diversion mechanism **120** can be characterized as requiring compliance by a health care practitioner regarding the collection of the first, predetermined amount (e.g., a pre-sample) prior to a collection of the second and/or third amount (e.g., a sample) of bodily-fluid. Similarly stated, the diversion mechanism **120** can be configured to prevent a health care practitioner from collecting the second amount, or the sample, of bodily-fluid into the

first sample fluid reservoir **180** without first diverting the first amount, or pre-sample, of bodily-fluid to the pre-sample reservoir **170**. In this manner, the health care practitioner is prevented from including (whether intentionally or unintentionally) the first amount of bodily-fluid, which is more likely to contain bodily surface microbes and/or other undesirable external contaminants, in the bodily-fluid sample to be used for analysis. In other embodiments, the fluid collection device **100** need not include a forced-compliance feature or component.

[0091] In some embodiments, the diversion mechanism **120** can have a fourth configuration (not shown in FIG. 1), different than the first, second, and third configurations. When in the fourth configuration, the diversion mechanism **120** can fluidically isolate the inlet port **121** from the first outlet port **125**, the second outlet port **126**, and optionally a third outlet port simultaneously. Therefore, when the diversion mechanism **120** is in its fourth configuration, flow of bodily-fluid from the inlet port **121** to the pre-sample fluid reservoir **170**, the first sample fluid reservoir **180**, and the second sample fluid reservoir **190** is prevented. In use, for example, the diversion mechanism **120** can be actuated (e.g., manually or automatically) to place the diversion mechanism **120** in the first configuration such that a bodily-fluid can flow from the inlet port **121** to the pre-sample fluid reservoir **170**, then moved to the second configuration such that the bodily-fluid can flow from the inlet port **121** to the first sample fluid reservoir **180**, and optionally moved to the third configuration such that the bodily-fluid can flow from the inlet port **121** to the second sample fluid reservoir **190**, then moved to the fourth configuration to stop the flow of bodily-fluid into and/or through the diversion mechanism **120**. In this manner, the device is effectively “locked” and self-contained in the fourth configuration such that any residual bodily-fluid in the device **100** is prevented from being communicated and/or otherwise exposing health care practitioner and/or patient to potential dangerous fluids. This optional safety feature can prevent potential exposure to bodily-fluid samples that can be infected with pathogens such as HIV, Hepatitis C, etc.

[0092] In some embodiments, one or more portions of the collection device **100** are disposed within a housing (not shown in FIG. 1). For example, in some embodiments, at least a portion of one or more of the diversion mechanism **120**, the first pre-sample reservoir **170**, and the sample reservoirs **180** and **190** can be disposed within the housing. In such an embodiment, at least a portion of the diversion mechanism **120** is accessible through the housing to allow the user to actuate the flow controller **140** to control the flow of bodily-fluid from the patient (e.g., a vein) to the collection device **100**. Examples of suitable housings are described in more detail herein with reference to specific embodiments.

[0093] In some embodiments, the collection device **100** can optionally include one or more flow metering devices that can meter a flow of bodily-fluid through the collection device. For example, a flow metering device can be in fluid communication with the first fluid flow path **142** and/or the second fluid flow path **144** to meter a flow of bodily-fluid therethrough. In other embodiments, a flow metering device can be in fluid communication with and/or otherwise disposed in the first port **125** and/or the second port **126**. The flow metering device can include an indicator or the like (e.g., a dial, a display, color, a haptic output device, an electrical signal output device such as a wireless radio

signal, Bluetooth radio signal, etc.) that can be configured to provide an indication to a user that is associated with a predetermined volume being transferred to the pre-sample reservoir **170**, the first sample reservoir **180**, and/or the second sample reservoir **190**. In some embodiments, the flow metering device can be operably coupled to, for example, an actuator or the like such as those described above. In such embodiments, the flow metering device can be operable in actuating the actuator to move the flow controller **140** between its first configuration and its second configuration based on a desired volume of bodily-fluid having flown through the flow metering device. Thus, the flow metering device can be used to ensure a desired volume of bodily-fluid is transferred to the pre-sample reservoir **170**, the first sample reservoir **180**, and/or the second sample reservoir **190**, which in turn, can prevent insufficient, inaccurate and/or false results in, for example, microbial testing to the patient sample or the like.

[0094] Referring now to FIGS. 2-13, a collection device **200** includes a diversion mechanism **220**, a flow controller **240**, a pre-sample reservoir **270**, a first sample reservoir **280**, and a second sample reservoir **290**, different than the first sample reservoir **280**. As further described herein, the collection device **200** can be moved between a first, a second, and a third configuration to deliver a flow of a bodily-fluid that is substantially free from microbes exterior the body, such as, for example, dermally residing microbes and/or other undesirable external contaminants. The collection device **200** can be any suitable shape, size, or configuration. For example, while shown in FIGS. 2-13 with the sample reservoirs **280** and/or **290** as being oriented vertically with respect to the housing **201**, the collection device **200** can have the sample reservoirs **280** and/or **290** oriented in a plane with respect to the housing **201**, or conically disposed with respect to the housing **201**, and/or so forth.

[0095] The diversion mechanism **220** includes a housing **201** and movable members **250** and **250'**. As shown in FIGS. 2-4, the housing **201** is coupled to the pre-sample reservoir **270**, the first sample reservoir **280**, and the second sample reservoir **290**. The housing **201** includes an inlet port **221**, a first outlet port **230**, a second outlet port **231**, a third outlet port **232**, and defines an inner flow channel **235** that can define a fluid flow path for collecting bodily-fluids from the patient. The inlet port **221** can be selectively placed in fluid communication with the inner flow channel **235**. More specifically, the inlet port **221** defines an inlet lumen **202** that can be placed in fluid communication with the inner flow channel **235**. In this manner, the inlet port **221** extends from a portion of the housing **201** such that the inner flow channel **235** can be placed in fluid communication with a volume substantially outside the housing **201**, via the inlet lumen **202**. The inlet port **221** can be fluidically coupled to a medical device (not shown) that defines a fluid flow pathway for withdrawing and/or conveying bodily-fluid from a patient to the collection device **200**. For example, the inlet port **221** can be fluidically coupled to a needle or other lumen-defining device (e.g., flexible sterile tubing) either directly or indirectly via an adapter **204**. Similarly stated, the inlet lumen **202** defined by the inlet port **221** is placed in fluid communication with a lumen defined by a lumen-defining device, when the lumen-defining device is coupled to the inlet port **221**. Expanding further, when the lumen-defining device is disposed within a portion of a body of the patient (within a vein or the spinal cavity of a patient, for

example), the inner flow channel **235** of the housing **201** can be placed in fluid communication with the portion of the body of the patient.

[0096] The inner flow channel **235** defined by the housing **201** is a central lumen that extends along a length of the housing **201** and that can be placed in fluid communication with the bodily-fluid of the patient following venipuncture (other method employed to gain access to bodily-fluid) as described herein. The inner flow channel **235** forms a fluid flow pathway for transferring bodily-fluid between the inlet port **221** and the first outlet port **230**, the second outlet port **231**, and the third outlet port **232**. More specifically, when the inner flow channel **235** is placed in fluid communication with the patient (e.g., via the medical device coupled to the inlet port **221**), the first outlet port **230**, the second outlet port **231**, and the third outlet port **232** can be selectively placed in fluid communication with the inner flow channel **235** to allow bodily-fluid to flow into at least one of the pre-sample reservoir **270**, the first sample reservoir **280**, or the second sample reservoir **290**. In some embodiments, the bodily-fluid is prevented from flowing to the second outlet port **231** and the third outlet port **232** prior to a predetermined volume of bodily-fluid being collected in the pre-sample reservoir **270**. In some embodiments, the second outlet port **231** and the third outlet port **232** can be placed in fluid communication with the inner flow channel **235** simultaneously. In some embodiments, the second outlet port **231** and the third outlet port **232** can be placed in fluid communication with the inner flow channel **235** sequentially.

[0097] The movable members **250**, **250'** are configured to be actuated (e.g., moved) by the user from a first position and a second position relative to the housing **201** to direct fluid flow into the first sample reservoir **280** and the second sample reservoir **290**. The movable members **250** and **250'** are substantially the same and therefore are described with reference to a single movable member **250**. As shown in FIG. 5, the movable member **250** includes a boss **251** that defines an inner cavity **252**, an inlet port **253**, a first outlet port **254**, and a piercing member **255** that defines a lumen **256** fluidically coupled to the inner cavity **252**. The inlet port **253** and the outlet port **254** extend through the walls of the boss **251** that defines the inner chamber **252** of the movable member **250**. The movable member **250** is configured to be mounted on a support **257** of the housing **201** (see FIG. 4) such that the boss **251** is disposed within a bore **258** (see FIG. 4) and at least a portion of the movable member **250** is received in an annular chamber **260**. Optionally, a bias member **259** (e.g., a spring) can be disposed in the annular chamber **260** to return the movable member **250** back to its first position after being actuated by the user. In some embodiments, the movable member **250**, the annular chamber **260**, the bore **258** or the boss **251** can include mechanical locking features configured to hold the movable member **250** in the second position (e.g., a depressed position) after being actuated by the user.

[0098] As described herein, in the first configuration, the movable member **250** is disposed in a manner such that the movable member **250** is spaced apart from the inner flow channel **235**. In such a configuration, no fluid flow path can be established between a part of the body of a patient (e.g., a vein, spinal cavity, etc.) and the sample reservoirs **280** and/or **290**. Said another way, when in the movable member **250** is in its first configuration, the first sample reservoir **280** and the second sample reservoir **290** are fluidically isolated

from the inner flow channel 235 defined by the housing 201. The movable member 250 can be actuated by the user to move the movable member 250 from the first configuration to the second configuration and into alignment with the inner flow channel 235. The force exerted by the user can be sufficient to deform (e.g., compress) the bias member 259, thereby allowing the piercing member 255 to be inserted into the sample reservoir 280 and/or 290. In the second configuration, the inlet port 253 and the outlet port 254 are substantially aligned with the inner flow channel 235 placing the inner cavity 252 in fluid communication with the inner flow channel 235. Thus, with the movable member 250 in the second configuration, a fluid flow pathway is established between the inner flow channel 235, the inner cavity 252, the lumen 256 of the piercing member 255, and the sample reservoir 280. Said another way, in such a configuration, bodily-fluid can flow from the patient (e.g., a vein, spinal cavity, etc.), through the diversion mechanism 220, and into the first sample reservoir 280 and/or the second sample reservoir 290 as described in greater detail herein.

[0099] The pre-sample reservoir 270 can be any suitable reservoir for containing a bodily-fluid such as, for example, single use disposable collection tubes, vacuum based collection tubes, and/or the like. The pre-sample reservoir 270 is configured to be fluidically coupled to the first outlet port 230 of the collection device 200 (either directly or via an intervening structure such as sterile flexible tubing) in any suitable manner. For example, in some embodiments, a portion of the pre-sample reservoir 270 can form a friction fit within a portion of the first outlet port 230. In other embodiments, the pre-sample reservoir 270 can be coupled to the first outlet port 230 via a threaded coupling, an adhesive, a snap fit, a mechanical fastener and/or any other suitable coupling method. In some embodiments, the pre-sample reservoir 270 can be monolithically formed with the housing 201. The pre-sample reservoir 270 can be configured to maintain negative pressure conditions (vacuum conditions) inside (the pre-sample reservoir 270) that can allow drawing of bodily-fluid from the inlet port 221 to the pre-sample reservoir 270 through outlet port 230 via vacuum suction. The pre-sample reservoir 270 is configured to contain the first amount of the bodily-fluid, where the first amount of bodily-fluid can be a predetermined or undetermined amount, such that the first amount of bodily-fluid is fluidically isolated from a second and/or third amount of the bodily-fluid that is subsequently withdrawn from the patient.

[0100] The sample reservoirs 280 and/or 290 can be any suitable reservoirs for containing a bodily-fluid, including, for example, single use disposable collection tubes, vacuum based collection tubes, a sample reservoir as described in the '420 patent incorporated by reference above, and/or the like. In some embodiments, sample reservoirs 280 and/or 290 can be substantially similar to or the same as known sample containers such as, for example, a Vacutainer®, or the like. The sample reservoir 280 and 290 include a sample container 282 and 292, respectively, and a vacuum seal 284 and 294, respectively. The vacuum seal 284 or 294 maintains negative pressure conditions (vacuum conditions) inside the sample container 282 or 292, respectively, that can allow drawing of bodily-fluid from the inner flow channel 235 to the sample container 282 or 292, respectively via vacuum suction. The sample reservoirs 280 and/or 290 can be configured to be fluidically coupled to the second outlet port 231 and third outlet port 232, respectively, of the collection

device 200 (either directly or via an intervening structure such as sterile flexible tubing) in any suitable manner. The sample reservoirs 280 and/or 290 can be moved relative to the outlet ports 231 and/or 232 to place the sample reservoirs 280 and/or 290 in fluid communication with the outlet ports 231 and/or 232. The sample reservoirs 280 and 290 can be configured to contain a second or third amount of the bodily-fluid. The second or third amount of bodily-fluid can be a predetermined or undetermined amount, such that the second or third amount of bodily-fluid is fluidically isolated from the first amount of the bodily-fluid that is withdrawn from the patient. In some configurations, the sample reservoirs 280 and/or 290 can be coupled to the collection device 200 by being monolithically formed with the housing 201 in a manner similar to the pre-sample reservoir 270, thus, they are not described in detail herein. In some instances, the sample reservoirs 280 and/or 290 can be transparent such that the user can have visual feedback to confirm bodily-fluid flow into the sample reservoirs 280 and/or 290.

[0101] In some embodiments, the sample reservoirs 280 and 290 and the diversion mechanism 220 (and/or the portions of the collection device 200 other than the sample reservoirs 280 and 290) are independently formed (e.g., not monolithically formed) and coupled together during, for example, a manufacturing process. In some instances, the sample reservoirs 280 and 290 can be coupled to the diversion mechanism 220 in a substantially sterile or hermetic environment (e.g., an environment filled with ethylene oxide or the like). Thus, the interface between the sample reservoirs 280 and 290 and the diversion mechanism 220 is substantially sterilized prior to use. Moreover, the collection device 200 can be shipped and/or stored in a pre-assembled manner such as to maintain the substantially sterile interface between the sample reservoirs 280 and 290 and the diversion mechanism 220.

[0102] As shown in FIGS. 6 and 7, the flow controller 240 includes a first member 241 and a second member 245. The first member 241 is configured to be disposed in a recess 266 of the housing 201 (see e.g., FIG. 4), and can be made of any number of materials that are biocompatible such as, for example, titanium, graphite, pyrolytic carbon, polyester, polycarbonate, polyurethane, elastomeric material and/or the like. In some embodiments, the second member 245 serves as an actuator to move the first member 241 from a first configuration to a second configuration. More specifically, when the first member 241 is disposed in the recess 266, the second member 245 can be moved between a first position and a second position to move the flow controller 240 between the first and second configuration. In some embodiments, the housing 201 can selectively limit movement of the second member 245 from its first position to its second position. In some embodiments, the housing 201 can be configured to prevent movement of the second member 245 once it has been moved to the second position. Said another way, the housing 201 can include a locking mechanism that prevents the second member 245 from being moved from the second position back to the first position. The second member 245 and/or the housing 201 can also include mechanical detents and/or other indicators that provide visual or tactile feedback to ensure precise positioning of the second member 245.

[0103] The first member 241 can include multiple channels for directing fluid flow following a venipuncture (and/or other method of accessing a patient's bodily-fluid). For

example, as shown in FIGS. 6 and 7, the first member 241 includes a first flow channel 242 and a second flow channel 244. When the second member 245 is in the first position (see e.g., FIGS. 8 and 9), the flow controller 240 is placed in the first configuration and the first flow channel 242 establishes fluid communication between the inlet port 221 and the first outlet port 230 while fluidically isolating the inlet port 221 from the inner flow channel 235. When the second member 245 is in the second position (see e.g., FIGS. 10-13), the flow controller 240 is placed in the second configuration and the second flow channel 244 establishes fluid communication between the inlet port 221 and the inner flow channel 235 while fluidically isolating the inlet port 221 from the first outlet port 230. Additional second member 245 positions corresponding to additional first member 241 flow channels and/or flow controller 240 configurations can be included to further direct/isolate fluid flow between the patient and the collection device 200. For example, the second member 245 can have a third position corresponding to a third configuration of the flow controller 240 that substantially prevents fluid flow between the patient and the collection device 200 altogether. Said another way, in some embodiments, the dial can be moved to a third position after all bodily-fluid samples are taken from the patient to substantially seal the samples in the collection device 200 from the external environment.

[0104] In operation, the collection device 200 can be used to collect bodily-fluids (e.g., blood) from a patient with reduced contamination from dermally-residing microbes and/or other undesirable external contaminants. For example, the inlet port 221 of the collection device 200 is fluidically coupled to a needle or other lumen-defining device (e.g., flexible sterile tubing) via the adapter 204. Following venipuncture (or other bodily-fluid access method), the second member 245 is rotated until it reaches the first position as shown in FIGS. 8 and 9. Alternatively, the second member 245 can be pre-set in the first position and the collection device 200 can be otherwise sealed to preserve the vacuum in the pre-sample reservoir 270 and the sterility of the collection device 200. For example, the inlet port 221 and/or the adapter 204 can include a valve that is opened when the collection device 200 is coupled to the needle or other lumen-defining device.

[0105] As described above, when the second member 245 is in the first position, the flow controller 240 is placed in the first configuration and the first flow channel 242 of the first member 241 establishes fluid communication between the inlet port 221 and the first outlet port 230 while fluidically isolating the inlet port 221 from the inner flow channel 235. Additionally, the first and second sample reservoirs 280 and 290 are fluidically isolated from the inlet port 221 in the first configuration and a fluid flow path is defined between a portion of the body of a patient (e.g. a vein) and the pre-sample reservoir 270 as indicated by the arrow AA in FIG. 9. As described above, fluid reservoirs used in the collection device 200 such as the pre-sample reservoir 270, and the sample reservoirs 280 and 290 can be configured to define a negative pressure (i.e., a pressure less than the fluid pressure of the portion of the body that the collection device 200 is being used to withdraw bodily-fluid from) so that once fluid communication is established between a portion of the body of the patient (e.g., a vein) and the pre-sample reservoir 270, the negative pressure within the pre-sample reservoir 270 is such that the pressure differential between

the pre-sample reservoir 270 and the portion of the body of the patient draws the bodily-fluid into the pre-sample reservoir 270. In this first configuration, the flow controller 240 also fluidically isolates the pre-sample reservoir 270 from the inner flow channel 235. Thus, a first amount (predetermined or undetermined) of bodily-fluid can be received into the pre-sample reservoir 270 immediately after venipuncture (for example) and isolated from subsequent samples. In this manner, the collection device 200 can be used to prevent the first amount of bodily-fluid, which is most likely to contain bodily surface microbes and/or other undesirable external contaminants, from contaminating subsequent amounts of the bodily-fluid samples that are collected and used for diagnostic or other testing that can be impacted by the contaminants.

[0106] Following collection of the volume of bodily-fluid pre-sample in the pre-sample reservoir 270, the second member 245 can be rotated until it reaches the second position as shown in FIGS. 10 and 11. When the second member 245 is in the second position, the flow controller 240 is placed in the second configuration and the second flow channel 244 of the first member 241 establishes fluid communication between the inlet port 221 and the inner flow channel 235, while fluidically isolating the first outlet port 230 (i.e., the pre-sample reservoir 270) from the inlet port 221. Said another way, in the second configuration, the flow controller 240 establishes a fluid flow path between a portion of the body of a patient (e.g. a vein) and the inner flow channel 235 via the second flow channel 244 as indicated by arrow BB in FIG. 11.

[0107] With the flow controller 240 in the second configuration, the movable members 250 and/or 250' can be actuated (i.e., depressed) from the first position to the second position by the user to establish fluid communication between a part of the body of a patient (e.g., a vein) and the first sample reservoir 280 and/or the second sample reservoir 290. More specifically, the movable member 250 is moved from its first position to its second configuration to pass the piercing member 255 through the outlet port 231 in such a manner that the piercing member 255 can puncture the vacuum seal 284 of the first sample reservoir 280 to be disposed inside the sample container 282, as indicated by the arrow CC in FIG. 12. While in the second position, the inlet port 253 and the outlet port 254 of the movable member 250 are substantially aligned with, and in fluid communication with, the inner flow channel 235, which allows the bodily-fluid to flow from the inner flow channel 235, into the inner cavity 252 of the movable member 250, and out the lumen 256 of the piercing member 255 into the first sample reservoir 280. The pressure differential between the sample reservoir 280 (e.g., vacuum or negative pressure) and the inner flow channel 235 draws the bodily-fluid into the sample reservoir 280. Said another way, in the second configuration, the movable member 250 establishes a fluid flow path between the inner flow channel 235 and the first sample reservoir 280 as indicated by the arrow DD in FIG. 12. Once a desired volume of bodily-fluid (e.g., the second amount) is collected in the first sample reservoir 280, the user can release the movable member 250 allowing the bias member 259 to move the button 250 back to its first position. With the movable member 250 back in its first position, the piercing member 255 is removed from the first sample

reservoir **280** and the seal **284** (e.g., a self sealing septum) fluidically isolates the first sample reservoir **280** from the inner flow channel **235**.

[0108] In a similar manner, while the flow controller **240** is in the second configuration, the movable member **250'** can be actuated (depressed) from its first position to its second position by the user, as indicated by the arrow EE in FIG. 13. In this manner, fluid communication is established between a part of the body of a patient (e.g., a vein) and the second sample reservoir **290** (via the outlet port **232**) in a manner similar to that of the movable member **250** and first sample reservoir **280** described above. Said another way, in the second configuration, the movable member **250'** establishes a fluid flow path between the inner flow channel **235** and the second sample reservoir **290** as indicated by the arrow FF in FIG. 13. Once a desired volume of bodily-fluid (e.g., the third amount) is collected in the second sample reservoir **290**, the user can release the movable member **250'** allowing the bias member **259'** to move the button **250'** back to its first position. Although shown and described as being a sequential process, the order of fill and/or sequencing is not necessarily required (i.e., sample reservoir **280** does not necessarily have to be filled before sample reservoir **290**, etc.). Said another way, once the flow controller **240** is moved to the second configuration, the first sample reservoir **280** and the second sample reservoir **290** (and any additional sample reservoirs) can be filled in any order, at the same time (e.g., simultaneously), and/or at overlapping time intervals. For example, the user can begin to fill the first sample reservoir **280** and then after the first sample reservoir **280** is partially filled, the user can depress the movable member **250'** to begin filling the second sample reservoir **290** while the first sample reservoir **280** is finished filling. Additionally, adjustments in the volume of the bodily-fluid collected in the sample reservoirs **280** and/or **290** can be made possible by actuating (inserting) the movable members **250** and/or **250'** repeatedly. As described above, the second member **245** can have a third position corresponding to a third configuration of the flow controller **240** that can substantially prevent fluid flow between the patient and the collection device **200** altogether to substantially seal the samples in the collection device **200** from the external environment.

[0109] Although not shown in FIGS. 2-13, the collection device **200** can include a flow metering device or the like that can be configured to meter a volume of bodily-fluid that is transferred to the pre-sample reservoir **270**, the first sample reservoir **280**, and/or the second sample reservoir **290**. For example, in some embodiments, the first member **241** of the flow controller **240** can include a flow metering device that is in fluid communication with the first flow channel **242** and the second flow channel **244**. In other embodiments, a flow metering device can be disposed within the inner cavity **252** of the movable members **250** and/or **250'**. Thus, a volume of bodily-fluid sample transferred to and disposed in the first sample reservoir **280** and the second sample reservoir **290** can be metered and/or controlled such that the volume of bodily-fluid sample disposed in each sample reservoir **280** and **290** is a predetermined volume such as, for example, 10 mL, 20 mL, 30 mL, etc.

[0110] Although the collection device **200** is shown and described as including a first sample reservoir **280** and a second sample reservoir **290**, in other embodiments, a collection device can include any number of pre-sample

and/or sample reservoirs. For example, FIGS. 14 and 15 illustrate a collection device **300** according to an embodiment. As shown, certain aspects of the collection device **300** can be substantially similar to corresponding aspects of the collection device **200** described above with reference to FIGS. 2-13. Thus, similar aspects are not described in further detail herein.

[0111] As shown in FIGS. 14 and 15, the collection device **300** includes a diversion mechanism **320**, a flow controller **340**, a pre-sample reservoir **370**, a first sample reservoir **380**, a second sample reservoir **380'**, a third sample reservoir **390**, and a fourth sample reservoir **390'**. The pre-sample reservoir **370** can be substantially similar to the pre-sample reservoir **270** described in detail above. In some embodiments, the sample reservoirs **380**, **380'**, **390**, and **390'** can be substantially similar to the sample reservoirs **280** and **290** described in detail above. In some embodiments, the sample reservoirs **380**, **380'**, **390**, and **390'** can have substantially the same shape and size and can include, for example substantially the same culture medium. In other embodiments, the sample reservoirs **380**, **380'**, **390**, and **390'** can have substantially the same shape and size and can include one of an aerobic culture medium or an anaerobic culture medium. For example in some embodiment, the first sample reservoir **380** and the third sample reservoir **390** can include an aerobic culture medium, while the second sample reservoir **380'** and the fourth sample reservoir **390'** can include an anaerobic culture medium. In other embodiments, the sample reservoirs **380**, **380'**, **390**, and **390'** can each include an aerobic or an anaerobic culture medium in any arrangement or combination.

[0112] The diversion mechanism **320** includes a housing **301** and a set of movable members **350**, **350'**, **350''**, and **350'''**. The movable members **350**, **350'**, **350''**, and **350'''** are, for example, substantially similar to the movable member **250** described above with reference to FIG. 5. Thus, the movable members **350**, **350'**, **350''**, and **350'''** can be moved between a first position and a second position relative to the housing **301** to be placed in fluid communication with the sample reservoirs **380**, **380'**, **390**, and **390'**, respectively. The housing **301** includes and/or defines an inlet port **321**, a first outlet port **330** configured to be placed in fluid communication with the pre-sample reservoir **370**, a second outlet port **331** configured to be placed in fluid communication with the first sample reservoir **380**, a third outlet port **332** configured to be placed in fluid communication with the second sample reservoir **380'**, a fourth outlet port **333** configured to be placed in fluid communication with the third sample reservoir **390**, and a fifth outlet port **334** configured to be placed in fluid communication with the fourth sample reservoir **390'**. Moreover, the housing **301** defines an inner flow channel **335** that can be selectively placed in fluid communication with the inlet port **321** and the outlet ports **331**, **332**, **333**, and **334** in a similar manner as described above with reference to the inner flow channel **235** of the housing **201**.

[0113] The flow controller **340** is, for example, substantially similar to the flow controller **240** described above with reference to FIGS. 6-13. Thus, the flow controller **340** can be rotated between a first configuration and a second configuration to selectively define a portion of a fluid flow path between the patient and the pre-sample reservoir **370** or the sample reservoirs **380**, **380'**, **390**, and **390'**. In this manner, a user can manipulate the collection device **300** in a similar

manner as described above with reference to the collection device 200 in FIGS. 8-13. Thus, a first volume of bodily-fluid can be transferred to and disposed in the pre-sample reservoir 370 and subsequent volumes of bodily-fluid can be transferred to and disposed in the sample reservoirs 380, 380', 390, and 390'.

[0114] FIGS. 16-22 illustrate a collection device 400 according to an embodiment. The collection device 400 includes a diversion mechanism 420, a flow controller 440, and sample reservoirs 480, 480', 490 and 490'. As further described herein, the collection device 400 can be moved between a first, a second, a third, a fourth, and a fifth configuration to deliver a flow of a bodily-fluid that is substantially free from microbes exterior the body, such as, for example, dermally residing microbes and/or other undesirable external contaminants. The collection device 400 can be any suitable shape, size, or configuration. For example, while shown in FIGS. 16-22 with the sample reservoirs 480, 480', 490 and 490' oriented vertically with respect to the housing 401, the collection device 400 can have the sample reservoirs 480, 480', 490 and 490' oriented in any suitable plane with respect to the housing 401, or conically disposed with respect to the housing 401, and/or so forth.

[0115] The sample reservoirs 480, 480', 490 and 490' are substantially similar or the same in form and function to the sample reservoirs 280 and/or 290 of the collection device 200 and thus, are not described in detail herein. As discussed above, the sample reservoirs 480, 480', 490 and 490' maintain negative pressure conditions (vacuum conditions) that can allow drawing of bodily-fluid from a patient to the sample reservoirs 480, 480', 490 and 490' via suction. In some embodiments, sample reservoirs 480 and 480' can be aerobic culture bottles and sample reservoirs 490 and 490' can be anaerobic culture bottles and the collection device 400 can be used to collect multiple aerobic and multiple anaerobic blood culture samples from a single venipuncture. As described in further detail herein, the sample reservoirs 480, 480', 490 and 490' can each be placed in fluid communication with at least a portion of the diversion mechanism 420 to receive a volume of a bodily-fluid sample. The volume of the bodily-fluid samples can be a predetermined or undetermined amount. Moreover, once a desired volume of bodily-fluid is disposed in the sample reservoirs 480, 480', 490, 490', each sample reservoir 480, 480', 490, and 490' can be fluidically isolated from at least a portion of the diversion mechanism 420, as described in further detail herein.

[0116] The diversion mechanism 420 includes a housing 401 and a distribution member 429. The housing 401 of the diversion mechanism 420 is physically and fluidically coupled to the distribution member 429, and provides and/or defines a set of fluid flow pathways for collecting bodily-fluids from the patient. The housing 401 defines a recess 466 and a set of outlet apertures 403. The recess 466 is configured to receive a seal member 441 included in the flow controller 440, as described in further detail herein. The set of outlet apertures 403 includes a first outlet aperture 403a, a second outlet aperture 403b, a third outlet aperture 403c, a fourth outlet aperture 403d, and a fifth outlet aperture 403e that are each configured to define a different fluid flow path in fluid communication with different portions of the distribution member 429. More specifically, the distribution member 429 defines and/or forms at least a portion of a pre-sample reservoir 470 in fluid communication with the first outlet aperture 403a, and a first flow channel 435a in

fluid communication with the second outlet aperture 403b, second flow channel 435b in fluid communication with the third outlet aperture 403c, a third flow channel 435c in fluid communication with the fourth outlet aperture 403d, and a fourth flow channel 435 in fluid communication with the fifth outlet aperture 403e.

[0117] As shown in FIGS. 17 and 18, the distribution member 429 defines a chamber or volume that defines at least a portion of the pre-sample reservoir 470. The pre-sample reservoir 470 is configured to contain bodily-fluids such as, for example, blood, plasma, urine, and/or the like. The first outlet aperture 403a of the housing 401 can be substantially aligned with an open portion of the pre-sample reservoir 470 to allow the pre-sample reservoir 470 to receive a flow of bodily-fluid from the patient. For example, the pre-sample reservoir 470 can receive and contain a first amount or volume of the bodily-fluid, where the first amount of bodily-fluid can be a predetermined or undetermined amount. Moreover, the arrangement of the diversion mechanism 420 can be such that the pre-sample reservoir 470 is maintained in fluidic isolation from the flow channels 435a, 435b, 435c, and 435d and/or subsequent volumes of bodily-fluid withdrawn from the patient, as described in further detail herein. While the pre-sample reservoirs 270 and 370 are described above as maintaining a negative pressure, the pre-sample reservoir 470 does not maintain negative pressure conditions (vacuum conditions), and hence other mechanisms such as, for example, gravitational pull can be used to draw the bodily-fluid into the pre-sample reservoir 470.

[0118] The flow channels 435a-435d extend radially from a center of the distribution member 429 and are arranged such that each flow channel 435a, 435b, 435c, and 435d is fluidically isolated from the pre-sample reservoir 470 and the other flow channels. In this manner, the flow channels 435a, 435b, 435c, and 435d can direct and/or otherwise define a fluid flow path between a first end portion that is substantially aligned with the outlet apertures 403b, 403c, 403d, and 403e, respectively, and a second end portion. As shown in FIGS. 17 and 18, the distribution member 429 defines a first outlet port 431 disposed at the second end portion of the first flow channel 435a, a second outlet port 432 disposed at the second end portion of the second flow channel 435b, a third outlet port 433 disposed at the second end portion of the third flow channel 435c, and a fourth outlet port 434 disposed at the second end portion of the fourth flow channel 435d. Moreover, the distribution member 429 includes a first piercing member 455a, a second piercing member 455b, a third piercing member 455c, and a fourth piercing member 455d that are physically and fluidically coupled to the first outlet port 431, the second outlet port 432, the third outlet port 433, and the fourth outlet port 434, respectively. As such, the piercing members 455a-455d can be used to puncture a vacuum seal of the sample reservoirs 480, 480', 490 and 490' which can initiate a flow of bodily-fluid, as described in further detail herein. Although not shown in FIGS. 17 and 18, the sample reservoirs 480, 480', 490 and 490' can be physically coupled to a portion of the distribution member 429 (either directly or via an intervening structure such as sterile flexible tubing) in any suitable manner that can allow the sample reservoirs 480, 480', 490, and 490' to be placed in fluid communication with the outlet ports 431, 432, 433, and 434, respectively.

[0119] The flow controller 440 includes a dial 445 and a seal member 441. The seal member 441 is disposed in the recess 466 of the housing 401 (see e.g., FIG. 20). More particularly, the flow controller 440 can be coupled to the housing 401 such that the seal member 441 is disposed between and in contact with a surface of the housing 401 defining the recess 466 and a surface of the dial 445. Moreover, the seal member 441 can have a size and a shape such that, when the flow controller 440 is coupled to the housing 401, the seal member 441 forms a substantially fluid tight seal with the surface of the dial 445 and the surface of the housing 401 that defines the recess 466 (see e.g., FIG. 20), as described in further detail herein. The seal member 441 can be made of any number of materials that are biocompatible such as, for example, silicone, polylactides, polyglycolides, polylactide-co-glycolides (PLGA), polyanhydrides, polyorthoesters, polyetheresters, polycaprolactones, polyesteramides, poly(butyric acid), poly(valeric acid), polyurethanes, nylons, polyesters, polycarbonates, polyacrylates, polymers of ethylene-vinyl acetates and other acyl substituted cellulose acetates, polystyrenes, polyvinyl chloride, polyvinyl fluoride, poly(vinyl imidazole), chlorosulphonate polyolefins, polyethylene oxide, and/or blends and copolymers thereof.

[0120] As shown in FIG. 17, the seal member 441 defines a set of apertures 444 that can direct a flow of bodily-fluid following a venipuncture (or other method of accessing bodily-fluid). For example, the set of apertures 444 defined by the seal member 441 includes a first aperture 444a, a second aperture 444b, a third aperture 444c, a fourth aperture 444d, and a fifth aperture 444e. The arrangement of the seal member 441 is such that when the seal member 441 is disposed in the recess 466, the first aperture 444a, the second aperture 444b, the third aperture 444c, the fourth aperture 444d, and the fifth aperture 444e are substantially aligned with the first outlet aperture 403a, the second outlet aperture 403b, the third outlet aperture 403c, the fourth outlet aperture 403d, and the fifth outlet aperture 403e of the housing 401, respectively.

[0121] The dial 445 of the flow controller 440 is rotatably coupled to the housing 401 and movable between a first position, a second position, a third position, a fourth position, and a fifth position relative to the housing 401. The dial 445 includes an inlet port 421 that defines a lumen 402. The inlet port 421 can be fluidically coupled to a medical device (not shown) that defines a fluid flow pathway for withdrawing and/or conveying bodily-fluid from a patient to the collection device 400. For example, the inlet port 421 can be fluidically coupled to a needle or other lumen-defining device (e.g., flexible sterile tubing) either directly or indirectly via an adapter 404. Similarly stated, the inlet lumen 402 defined by the inlet port 421 is placed in fluid communication with a lumen defined by a lumen-defining device, when the lumen-defining device is coupled to the inlet port 421. In this manner, the inlet port 421 can be configured to selectively place the pre-sample reservoir 470, the first sample reservoir 480, the second sample reservoir 480', the third sample reservoir 490, and the fourth sample reservoir 490' in fluid communication with the patient, as described in further detail herein.

[0122] As described above, the dial 445 is movable between the first, the second, the third, the fourth, and the fifth positions. When the dial 445 is in the first position, the flow controller 440 is placed in a first configuration and the

inlet port 421 can be substantially aligned with the first aperture 444a of the seal member 441 and the first outlet aperture 403a of the housing 401. In this manner, first aperture 444a of the seal member 441 establishes fluid communication between the inlet port 421 and the first outlet aperture 403a while fluidically isolating the inlet port 421 from the outlet apertures 403b, 403c, 403d, and 403e which in turn, fluidically isolates the inlet port 421 from the flow channels 435a-435d. With the first outlet port 403a aligned with an open portion of the pre-sample reservoir 470, the first aperture 444a and the first outlet aperture 403a establish fluid communication between the inlet port 421 and the pre-sample reservoir 470. When the dial 445 is rotated (or actuated) to the second position, the flow controller 440 is placed in a second configuration and the second outlet aperture 444b establishes fluid communication between the inlet port 421 and the second outlet aperture 403b while fluidically isolating the inlet port 421 from the outlet apertures 403a, 403c, 403d, and 403e. With the second outlet aperture 403b aligned with the first end portion of the first flow channel 435a, the second aperture 444b and the second outlet aperture 403b establish fluid communication between the inlet port 421 and the first flow channel 435a.

[0123] The collection device 400 works in a similar manner when the dial 445 is rotated to the third, fourth and fifth positions. Thus, when the inlet lumen 402 is placed in fluid communication with the patient (e.g., via the medical device coupled to the inlet port 421), the first outlet port 430, the second outlet port 431, the third outlet port 432, the fourth outlet port 433, and the fifth outlet port 434 can be selectively placed in fluid communication with the inlet lumen 402 to allow all the bodily-fluid to flow into at least one of the pre-sample reservoir 470, or one or more of the sample reservoirs 480, 480', 490 and 490'. In some embodiments, additional dial 445 positions corresponding to additional seal outlet apertures and/or flow controller 440 configurations can be included to further direct/isolate fluid flow between the patient and the collection device 400. For example, the dial 445 can have a sixth position corresponding to a sixth configuration of the flow controller 440 that substantially prevents fluid flow between the patient and the collection device 400 altogether. Said another way, in some embodiments, the dial 445 can be moved to a sixth position after all bodily-fluid samples are taken from the patient to substantially seal the samples in the collection device 400 from the external environment.

[0124] In some embodiments, the bodily-fluid is prevented from flowing to the outlet ports associated with the sample reservoirs (e.g., outlet ports 431-434) until after a predetermined volume of bodily-fluid is collected in the pre-sample reservoir 470. In some embodiments, the outlet ports associated with the sample reservoirs (e.g., outlet ports 431-434) can only be placed in fluid communication with the inlet lumen 402 sequentially (e.g., outlet port 431 must be in fluid communication with the inlet lumen 402 before outlet port 432, and so on). In some embodiments, the outlet ports associated with subsequent sample reservoirs (e.g., outlet ports 432-434) can only be placed in fluid communication with the inlet lumen 402 after a confirmed volume of bodily-fluid has been collected. In some embodiments, the outlet ports associated with the sample reservoirs (e.g., outlet ports 431-434) can be placed in fluid communication with the inlet lumen 402 in any random manner without any preference for order (e.g., outlet port 434 can be in fluid

communication with the inlet lumen 402 before outlet port 431, outlet port 432 can be in fluid communication with the inlet lumen 402 before outlet port 433, and so on).

[0125] In some embodiments, the housing 401 can selectively limit movement of the dial 445 from its first position to its second, third, fourth, and fifth positions. In some embodiments, the housing 401 can be configured to prevent movement of the dial 445 once it has been moved to the fifth position. Said another way, the housing 401 can include a locking mechanism that prevents the dial 445 from being moved from the fifth position back to the first position. The dial 445 and/or the housing 401 can also include mechanical detents and/or other indicators that provide visual or tactile feedback to ensure precise positioning of the dial 445 with respect to the outlet apertures 403a-403e of the housing 401.

[0126] In operation, the collection device 400 can be used to collect bodily-fluids (e.g., blood, plasma, urine, and/or the like) from a patient with reduced contamination. For example, the inlet port 421 of the collection device 400 can be fluidically coupled to a needle or other lumen-defining device (e.g., flexible sterile tubing). Following venipuncture (or other method of accessing bodily-fluid), the dial 445 is actuated (or rotated) until it reaches the first position, as shown in FIGS. 19 and 20. Alternatively, the dial 445 can be pre-set in the first position and the collection device 400 can be otherwise sealed to preserve the sterility of the collection device 400. For example, the inlet port 421 can include a valve that is opened when the collection device 400 is coupled to the needle or other lumen-defining device.

[0127] As described above, when the dial 445 is in the first position, the flow controller 440 is placed in the first configuration and the first aperture 444a of the seal member 441 establishes fluid communication between the inlet port 421 and the first outlet port 430 (contained within the housing 401) while fluidically isolating the inlet port 421 from the four flow channels 435a-435d. Additionally, the sample reservoirs 480, 480', 490 and 490' are fluidically isolated from the inlet port 421 in the first configuration and a fluid flow path is defined between a portion of the body of a patient (e.g. a vein) and the pre-sample reservoir 470 as indicated by the arrow GG in FIG. 20. In this first configuration, the bodily-fluid flows (e.g., by gravitation force, vacuum, etc.) from the portion of the body of the patient through the inlet lumen 402 of the inlet port 421, the first aperture 444a of the seal member 441, the first outlet port 430, and into the pre-sample reservoir 470. In the first configuration, the flow controller 440 also fluidically isolates the pre-sample reservoir 470 from the flow channels 435a-435d. Thus, a first amount (predetermined or undetermined) of bodily-fluid can be received into the pre-sample reservoir 470 immediately after venipuncture and isolated from subsequent samples. In this manner, the collection device 400 can be used to prevent the first amount of bodily-fluid, which is most likely to contain bodily surface microbes and/or other undesirable external contaminants, from contaminating subsequent amounts of the bodily-fluid samples that are collected and used for diagnostic or other testing that can be impacted by the contaminants.

[0128] Following collection of the bodily-fluid pre-sample in the pre-sample reservoir 470, the dial 445 can be actuated (or rotated) until it reaches the second position as shown in FIGS. 21 and 22. When the dial 445 is in the second position, the flow controller 440 is placed in the second configuration and the second aperture 444b of the seal

member 441 establishes fluid communication between the inlet port 421 and the flow channel 435a, while fluidically isolating the pre-sample reservoir 470 from the inlet port 421. Said another way, in the second configuration, the flow controller 440 establishes a fluid flow path between a portion of the body of a patient (e.g. a vein) and the flow channel 435a, as indicated by the arrow HH in FIG. 22. With the flow controller 440 in the second configuration, the sample reservoir 480 can be actuated by the user (e.g., pushed against the piercing member 455a) from a first configuration to a second configuration to establish fluid communication between a part of the body of a patient (e.g., a vein) and the first sample reservoir 480.

[0129] As described above, moving the sample reservoir 480 to the second configuration results in the piercing member 455a puncturing the vacuum seal of the sample reservoir 480 to be disposed inside the sample reservoir 480. In this second configuration, the part of the body of a patient (e.g., a vein) is exposed to vacuum suction force from the sample reservoir 480 due to the negative pressure conditions (vacuum) therein. The pressure differential between the sample reservoir 480 (e.g., vacuum or negative pressure) and the part of the body of the patient draws the bodily-fluid into the sample reservoir 480. The bodily-fluid flows from the part of the body of a patient through the inlet lumen 402 of the inlet port 421, the second aperture 444b of the seal member 441, the second outlet aperture 403b of the housing 401, and into the first flow channel 435a. The vacuum suction draws the flow of bodily-fluid through the first flow channel 435a into the sample reservoir 480 via the second outlet port 431 and the piercing member 455a. Said another way, in the second configuration, the flow controller 440 establishes a fluid flow path between the inlet port 421 and the sample reservoir 480. Once a desired volume of bodily-fluid (e.g., the second amount) is collected in the sample reservoir 480, the user can actuate (rotate) the flow controller 440 to the third position and/or move the sample reservoir 480 back to its first configuration to isolate the first sample reservoir 480 from the flow channel 435a. When the sample reservoir 480 is back in the first configuration, the piercing member 455a is removed from the sample reservoir 480 and the seal of the sample reservoir 480 (e.g., a self sealing septum) fluidically isolates the first sample reservoir 480 from the flow channel 435a. Filling the other sample reservoirs is done in a similar manner with the flow controller 440 being placed in the third, fourth and fifth configurations respectively.

[0130] Note that the order of fill and/or sequencing is not necessarily required (i.e., sample reservoir 480 does not necessarily have to be filled before sample reservoir 490, etc.). Said another way, the first sample reservoir 480 and the second sample reservoir 490 (and any additional sample reservoirs) can be filled in any order. For example, the user can begin to fill the first sample reservoir 480 and then after the first sample reservoir 480 is partially filled, the user can fill the second sample reservoir 490. Additionally, adjustments in the volume of the bodily-fluid collected in the sample reservoirs 480 and/or 490 can be made possible by repeated filling of the sample reservoirs 480 and/or 490. However, in other embodiments, the order of fill can be mechanically manipulated such that the second sample reservoir cannot be accessed until a specified amount of bodily-fluid is confirmed to have been placed into the first reservoir and so on. As described above, the dial 445 can

have a sixth position corresponding to a sixth configuration of the flow controller **440** that can substantially prevent fluid flow between the patient and the collection device **400** altogether to substantially seal the samples in the collection device **400** from the external environment.

[0131] Although the collection device **400** is shown and described above as including and/or otherwise coupling to a set of four sample reservoirs (e.g., the first sample reservoir **480**, the second sample reservoir **480'**, the third reservoir **490**, and the fourth reservoir **490'**), in other embodiments, a collection device can include and/or can be coupled to any suitable number of sample reservoirs. For example FIGS. **23-25** illustrate a collection device **500** according to an embodiment. As shown, certain aspects of the collection device **500** can be substantially similar to corresponding aspects of the collection device **500** described above with reference to FIGS. **16-22**. Thus, similar aspects are not described in further detail herein.

[0132] The collection device **500** includes a diversion mechanism **520**, a flow controller **540**, a first sample reservoir **580**, and a second sample reservoir **590**. The sample reservoirs **580** and **590** can be substantially similar to the sample reservoirs described in detail above. In some embodiments, the sample reservoirs **580** and **590** can have substantially the same shape and size and can include substantially the same culture medium. In other embodiments, the sample reservoirs **580** and **590** can have substantially the same shape and size and can include one of an aerobic culture medium or an anaerobic culture medium. In still other embodiments, the first sample reservoir **580** can have a first size that is substantially larger than a size of the second sample reservoir **590**.

[0133] As shown in FIGS. **24** and **25**, the diversion mechanism **520** includes a housing **501** and a distribution member **529**. The housing **501** of the diversion mechanism **520** is physically and fluidically coupled to the distribution member **529**, and provides and/or defines a set of fluid flow pathways for collecting bodily-fluids from the patient. As described above with reference to the housing **401**, the housing **501** can define a recess and a first outlet aperture **503a**, a second outlet aperture **503b**, and a third outlet aperture **503c**. The recess is configured to receive a seal member **541** included in the flow controller **540**, as described in detail above. The first outlet aperture **503a**, the second outlet aperture **503b**, and the third outlet aperture **503c** can be substantially similar in form and function as the first outlet aperture **403a**, the second outlet aperture **403b**, and the third outlet aperture **403c**, respectively, defined by the housing **401**. Similarly, the distribution member **529** defines a pre-sample reservoir **570**, a first flow channel **535a**, and a second flow channel **535b** that are substantially similar to the pre-sample reservoir **470**, the first flow channel **435a**, and the second flow channel **435b** included in the diversion member **429**. As such, the pre-sample reservoir **570** is in fluid communication with the first outlet aperture **503a**, the first flow channel **535a** is in fluid communication with the second outlet aperture **503b**, and the second flow channel **535b** is in fluid communication with the third outlet aperture **503c**, as described above with reference to the diversion mechanism **420**. As shown in FIG. **25**, the distribution member **529** defines a first outlet port **531** in fluid communication with the first flow channel **535a** and a first piercing member **555a**, and a second outlet port **532** in fluid communication with the second flow channel **535b** and a second

piercing member **555b**. As described above, the piercing members **555a** and **555b** can be used to puncture a vacuum seal of the sample reservoirs **580** and **590** which can initiate a flow of bodily-fluid, as described in further detail herein.

[0134] The flow controller **540** includes a dial **545** and a seal member **541**. The seal member **541** is disposed in the recess of the housing **501**, as described above. In this manner, when the flow controller **540** is coupled to the housing **501**, the seal member **541** forms a substantially fluid tight seal with a surface of the dial **545** and the surface of the housing **501** that defines the recess. As shown in FIGS. **24** and **25**, the seal member **541** defines a first aperture **544a**, a second aperture **544b**, and a third aperture **544c** that are substantially aligned with the first outlet aperture **503a**, the second outlet aperture **503b**, and the third outlet aperture **503c**, respectively, as described in detail above with reference to the seal member **441**.

[0135] The dial **545** of the flow controller **540** can be substantially similar in form and function as the dial **445**, while having a size that is suitable for coupling to the housing **501**. As such, the dial **545** can be rotatably coupled to the housing **501** and movable between a first position, a second position, and a third position relative to the housing **501**. The dial **545** includes an inlet port **521** that defines a lumen **502** and that can be fluidically coupled to a medical device (not shown) that defines a fluid flow pathway for withdrawing and/or conveying bodily-fluid from a patient to the collection device **500**. In this manner, the inlet port **521** can be configured to selectively place the pre-sample reservoir **570**, the first sample reservoir **580**, and the second sample reservoir **590**. More particularly, when the dial **545** is in the first position, the flow controller **540** is placed in a first configuration and the inlet port **521** is substantially aligned with the first aperture **544a** of the seal member **541** and the first outlet aperture **503a** of the housing **501**. In this manner, the first aperture **544a** of the seal member **541** establishes fluid communication between the inlet port **521** and the first outlet aperture **503a** and hence, places the inlet port **521** in fluid communication with the pre-sample reservoir **570**, as described in detail above with reference to the collection device **400**. Similarly, when the dial **545** is rotated (or actuated) to the second position, the flow controller **540** is placed in a second configuration and the second outlet aperture **544b** establishes fluid communication between the inlet port **521** and the second outlet aperture **503b** and hence, the first flow channel **535a**; and when the dial **545** is rotated to the third position, the flow controller **540** is placed in a third configuration and the third outlet aperture **544c** establishes fluid communication between the inlet port **521** and the third outlet aperture **503c** and hence, the second flow channel **535a**. In this manner, the collection device **500** can be used to transfer a first volume of a bodily-fluid to the pre-sample **570** and subsequently used to transfer a second volume and a third volume of the bodily-fluid to the first sample reservoir **580** and the second sample reservoir **590**, respectively, as described in detail above with reference to the collection device **400**.

[0136] FIGS. **26-33** illustrate a collection device **600** according to an embodiment. The collection device **600** includes a diversion mechanism **620**, a flow controller **640**, and sample reservoirs **680**, **680'**, **690** and **690'**. As further described herein, the collection device **600** can be moved between a first, a second, a third, a fourth, and a fifth configuration to deliver a flow of a bodily-fluid that is

substantially free from microbes exterior the body, such as, for example, dermally residing microbes and/or other undesirable external contaminants. The collection device 600 can be any suitable shape, size, or configuration. For example, aspects and/or portions of the collection device 600 can be substantially similar in form and/or function as corresponding aspects and/or portions of any of the collection devices 100, 200, 300, 400, and/or 500 described above. Thus, such similar aspects and/or portions are not described in further detail herein. By way of example, in some embodiments, the sample reservoirs 680, 680', 690, and 690' of the collection device 600 can be substantially similar and/or the same in form and function as the sample reservoirs 480, 480', 490, and 490', respectively, included in the collection device 400 of FIGS. 16-22.

[0137] The diversion mechanism 620 includes a distribution member 629 and a set of coupling members 637a, 637b, 637c, and 637d (see e.g., FIG. 27). The distribution member 629 is in fluid communication with the coupling members 637a, 637b, 637c, and 637d and is configured to provide and/or define a set of fluid flow pathways for collecting bodily-fluids from the patient. As shown in FIGS. 27 and 28, the distribution member 629 defines and/or forms a first outlet port 630 in fluid communication with a pre-sample reservoir 670, a second outlet port 631 in fluid communication with the first coupling member 637a, a third outlet port 632 in fluid communication with the second coupling member 637b, a fourth outlet port 633 in fluid communication with the third coupling member 637c, and a fifth outlet port 634 in fluid communication with the fourth coupling member 637d.

[0138] As shown in FIG. 28, the distribution member 629 defines a chamber or volume that defines at least a portion of the pre-sample reservoir 670. The pre-sample reservoir 670 is configured to contain bodily-fluids such as, for example, blood, plasma, urine, and/or the like. For example, the pre-sample reservoir 670 can receive and contain a first amount or volume of the bodily-fluid from the patient, where the first amount of bodily-fluid can be a predetermined or undetermined amount. Moreover, the arrangement of the diversion mechanism 620 and the flow controller 640 can be such that the pre-sample reservoir 670 is maintained in fluidic isolation from the coupling portions 637a, 637b, 637c, and 637d and/or subsequent volumes of bodily-fluid withdrawn from the patient, as described in further detail herein. In this manner, the outlet ports 631, 632, 633, and 634 can direct and/or otherwise define a fluid flow path between the flow controller 640 and the coupling members 637a, 637b, 637c, and 637d, respectively, as described in further detail herein. In some embodiments, the arrangement of the first outlet port 630 and the pre-sample reservoir 670 can be substantially similar in form and function as the pre-sample reservoirs 470 and/or 570. Thus, the pre-sample reservoir 670 is not described in further detail herein.

[0139] As shown in FIG. 29, the first coupling member 637a defines a flow channel 638a that is fluidically coupled to a piercing member 655a. As described above, the coupling member 637a can be physically and fluidically coupled to the distribution member 629. For example, the flow channel 638a can receive a portion of the second outlet port 631 of the distribution member 629 to physically and fluidically couple the coupling member 637a thereto. In some embodiments, a surface of the second outlet port 631 can form a substantially fluid tight seal with an inner surface

of the coupling portion 637a defining the flow channel 638a (e.g., a friction fit that can form a substantially hermetic seal). The piercing member 655a of the coupling portion 637a can be substantially similar in form and function as the piercing member 455a included in the collection device 400 of FIGS. 16-22. Thus, the piercing member 655a is not described in further detail herein. The second coupling member 637b, the third coupling member 637c, and the fourth coupling member 637d are similarly arranged. As such, the second coupling member 637b, the third coupling member 637c, and the fourth coupling member 637d each include a piercing member 655b, 655c, and 655d, respectively, and each define a flow channel 638b, 638c, and 638d, respectively. As described in further detail herein, the first coupling member 637a, the second coupling member 637b, the third coupling member 637c, and the fourth coupling member 637d can be used to selectively place the diversion mechanism 620 in fluid communication with the first sample reservoir 680, the second sample reservoir 680', the third sample reservoir 690, and the fourth sample reservoir 690', respectively.

[0140] As shown in FIGS. 30 and 31, the flow controller 640 includes a dial 645 and a seal member 641. The dial 645 of the flow controller 640 is rotatably disposed within the distribution member 629 (see e.g., FIGS. 32 and 33) and is movable between a first position, a second position, a third position, and a fourth position. The dial 645 includes an inlet port 621, a first outlet port 647, and a second outlet port 648 that are each in fluid communication with an inner volume 646 (see e.g., FIG. 30). The inner volume 646 is configured to receive a portion of the seal member 641, as described in further detail herein. The inlet port 621 can be fluidically coupled to a medical device (not shown) that defines a fluid flow pathway for withdrawing and/or conveying bodily-fluid from a patient to the collection device 600. For example, the inlet port 621 can be fluidically coupled to a needle or other lumen-defining device (e.g., flexible sterile tubing) either directly or indirectly via an adapter 604 (see e.g., FIGS. 26 and 27). The first outlet port 647 is in fluid communication with the pre-sample reservoir 670. For example, the first outlet port 647 can be rotatably disposed in the first outlet port 630 of the distribution member 629. The second outlet port 648 can be selectively placed in fluid communication with the second outlet port 631, the third outlet port 632, the fourth outlet port 633, and the fifth outlet port 634 when the dial 645 is in its first position, second position, third position, and fourth position, respectively. In this manner, the inner volume 646 of the dial 645 can be selectively placed in fluid communication with the pre-sample reservoir 670, the first sample reservoir 680, the second sample reservoir 680', the third sample reservoir 690, and the fourth sample reservoir 690', as described in further detail herein.

[0141] At least a portion of the seal member 641 of the flow controller 640 is rotatably disposed in the inner volume 646 of the dial 645 and movable between a first position and a second position. Moreover, the seal member 641 can have a size and a shape such that an outer surface of the seal member 641 forms a substantially fluid tight seal with an inner surface of the dial 645 that defines at least a portion of the inner volume 646. As shown in FIG. 31, the seal member 641 defines a first flow channel 642 and a second flow channel 644. When the seal member 641 is in its first position within the inner volume 646, the first flow channel

642 establishes fluid communication between the inlet port **621** and the first outlet port **647** while fluidically isolating the inlet port **621** from the second outlet port **648**. Similarly, when the seal member **641** is in its second position within the inner volume **646**, the second flow channel **644** establishes fluid communication between the inlet port **621** and the second outlet port **648** while fluidically isolating the inlet port **621** from the first outlet port **647**. The collection device **600** works in a similar manner when the dial **645** is rotated to the second, third, and fourth positions within the distribution member **629**. Thus, when the inlet port **621** is placed in fluid communication with the patient (e.g., via the medical device coupled to the inlet port **621** and/or the adapter **604**), the first outlet port **630**, the second outlet port **631**, the third outlet port **632**, the fourth outlet port **633**, and the fifth outlet port **634** of the distribution member **629** can be selectively placed in fluid communication with the inlet port **621** to allow the bodily-fluid to flow into the pre-sample reservoir **670**, the first sample reservoir **680**, the second sample reservoir **680'**, the third sample reservoir **690**, and the fourth sample reservoir **690'**, respectively.

[0142] In operation, the collection device **600** can be used to collect bodily-fluids (e.g., blood, plasma, urine, and/or the like) from a patient with reduced contamination. For example, the inlet port **621** of the collection device **600** can be fluidically coupled to a needle or other lumen-defining device (e.g., flexible sterile tubing). Following venipuncture (or other method of accessing bodily-fluid), the seal member **641** can be actuated (or rotated) until in its first position, as shown in FIG. 32. Alternatively, the seal member **641** can be pre-set in the first position and the collection device **600** can be otherwise sealed to preserve the sterility of the collection device **600**. When the seal member **641** is in its first position, the flow controller **640** establishes fluid communication between the inlet port **621** and the first outlet port **630** of the distribution member **629** while fluidically isolating the inlet port **621** from the coupling members **637a**, **637b**, **637c**, and **637d**. Thus, as indicated by the arrow II in FIG. 32, bodily-fluid can be transferred from the patient, through the inlet port **621**, the first flow channel **642**, the first outlet port **647** of the dial **645**, and the first outlet port **630** of the distribution member **629**, and into the pre-sample reservoir **670** in a similar manner as described above with reference to the collection device **400**.

[0143] Following collection of the bodily-fluid pre-sample in the pre-sample reservoir **670**, the seal member **641** can be actuated (e.g., rotated) from its first position to its second position relative to the dial **645**. Similarly, the dial **645** can be actuated (or rotated) until it reaches the second position relative to the distribution member **629**, as shown in FIG. 33. When the seal member **641** and the dial **645** are in the second position, the flow controller **640** is placed in a second configuration and the second flow channel **644** of the seal member **641** establishes fluid communication between the inlet port **621** and the flow channel **638a** of the first coupling member **637a**, while fluidically isolating the pre-sample reservoir **670** from the inlet port **621**. With the flow controller **640** in the second configuration, the sample reservoir **680** can be actuated by the user (e.g., pushed against the piercing member **655a**) from a first configuration to a second configuration to establish fluid communication between a part of the body of a patient (e.g., a vein) and the first sample reservoir **680**. As described in detail above, moving the sample reservoir **680** to the second configuration results in

the piercing member **655a** puncturing the vacuum seal of the sample reservoir **680** to be disposed inside the sample reservoir **680**. In this second configuration, the part of the body of a patient (e.g., a vein) is exposed to vacuum suction force from the sample reservoir **680** due to the negative pressure conditions (vacuum) therein. Thus, bodily-fluid can be urged to flow from the part of the body of a patient through the inlet port **621**, the second flow channel **644**, the second outlet port **631**, and the flow channel **638a** and piercing member **655a** of the first coupling member **637a**, and into the first sample reservoir **680**, as indicated by the arrow JJ in FIG. 33.

[0144] Once a desired volume of bodily-fluid (e.g., the second amount) is collected in the sample reservoir **680**, the user can actuate (rotate) the flow controller **640** to the third position and/or move the sample reservoir **680** back to its first configuration to isolate the first sample reservoir **680** from the second flow channel **644**. When the sample reservoir **680** is back in the first configuration, the piercing member **655a** is removed from the sample reservoir **680** and the seal of the sample reservoir **680** (e.g., a self sealing septum) fluidically isolates the first sample reservoir **680** from the flow channel **635a**. Filling the other sample reservoirs is done in a similar manner with the flow controller **640** being placed in the third, fourth and fifth configurations respectively.

[0145] FIGS. 34-40 present a collection device **700** according to an embodiment. The collection device **700** includes a diversion mechanism **720**, a flow controller **740**, and sample reservoirs **780** and **790** (although there are holders present for four sample reservoirs, only two sample reservoirs are included in the figures for purposes of clarity and additional sample reservoirs (e.g. a fifth, sixth and so on) may be included as part of the collection device **700**). As further described herein, the collection device **700** can be moved between a first, a second, a third, a fourth, and a fifth configuration to deliver a flow of a bodily-fluid that is substantially free from microbes exterior to the body, such as, for example, dermally residing microbes and/or other undesirable external contaminants. The collection device **700** can be any suitable shape, size, or configuration. For example, aspects and/or portions of the collection device **700** can be substantially similar in form and/or function as corresponding aspects and/or portions of any of the collection devices **100**, **200**, **300**, **400**, **500**, and/or **600** described above. Thus, such similar aspects and/or portions are not described in further detail herein. By way of example, in some embodiments, the sample reservoirs **780** and **790** of the collection device **700** can be substantially similar and/or the same in form and function as the sample reservoirs **480** and **490**, respectively, included in the collection device **400** of FIGS. 16-22.

[0146] The diversion mechanism **720** includes a housing **701**, a distribution member **729**, and a base plate **771**. As described above with reference to the collection device **400**, the housing **701** defines a first outlet aperture **703a**, a second outlet aperture **703b**, a third outlet aperture **703c**, a fourth outlet aperture **703d**, and a fifth outlet aperture **703e** that are each configured to be in fluid communication with a different portion of the distribution member **729**. More specifically, the distribution member **729** defines and/or forms at least a portion of a pre-sample reservoir **770** in fluid communication with the first outlet aperture **703a**, and a first fluid chamber **735a** in fluid communication with the second

outlet aperture **703b**, a second fluid chamber **735b** in fluid communication with the third outlet aperture **703b**, a third fluid chamber **735c** in fluid communication with the fourth outlet aperture **703d**, and a fourth fluid chamber **735d** in fluid communication with the fifth outlet aperture **703e**. Furthermore, the housing **701** defines a recess **766** that is configured to movably receive at least a portion of the flow controller **740**, as described in further detail herein.

[0147] As shown in FIG. 36, the distribution member **729** defines a chamber or volume that forms at least a portion of the pre-sample reservoir **770**. The pre-sample reservoir **770** is configured to contain bodily-fluids such as, for example, blood, plasma, urine, and/or the like. The first outlet aperture **703a** of the housing **701** can be substantially aligned with an open portion of the pre-sample reservoir **770** to allow the pre-sample reservoir **770** to receive a flow of bodily-fluid from the patient, as described in detail above. Expanding further, the distribution member **729** includes a set of walls **736** that can, for example, divide an inner volume of the distribution member **729** into portions and/or volumes that are fluidically isolated from one another. For example, as shown in FIG. 36, the set of walls **736** can divide an inner volume of the distribution member **729** into the pre-sample reservoir **770**, the first fluid chambers **735a**, the second fluid chamber **735b**, the third fluid chamber **735c**, and the fourth fluid chamber **735d**. In some embodiments, the walls **736** can define and/or form the pre-sample reservoir **770** and the fluid chambers **735a-735d** equally. In other embodiments, the pre-sample reservoir **770** can have define a volume that is different from a volume defined by the fluid chambers **735a-735d**.

[0148] The distribution member **729** further includes a first piercing member **755a**, a second piercing member **755b**, a third piercing member **755c**, and a fourth piercing member **755d** that are in fluid communication with the first fluid chamber **735a**, the second fluid chamber **735b**, the third fluid chamber **735c**, and the fourth fluid chamber **735d**, respectively. As such, the piercing members **755a-755d** can be used to puncture a vacuum seal of the sample reservoirs **780** and **790** (and corresponding sample reservoirs not shown in FIGS. 34-40) which can initiate a flow of bodily-fluid, as described in further detail herein.

[0149] The flow controller **740** of the collection device **700** includes a dial **745** and a seal member **741**. The seal member **741** is disposed in the recess **766** of the housing **701** (see e.g., FIGS. 38 and 40). More particularly, the flow controller **740** can be coupled to the housing **701** such that the seal member **741** is disposed between and in contact with a surface of the housing **701** defining the recess **766** and a surface of the dial **745**. The seal member **741** can be configured to form a substantially fluid tight seal with the surface of the dial **745** and the surface of the housing **701** that defines the recess **766**, as described in detail above. As shown in FIG. 35, the seal member **741** defines a first aperture **744a**, a second aperture **744b**, a third aperture **744c**, a fourth aperture **744d**, and a fifth aperture **744e**. The arrangement of the seal member **741** is such that when the seal member **741** is disposed in the recess **766**, the first aperture **744a**, the second aperture **744b**, the third aperture **744c**, the fourth aperture **744d**, and the fifth aperture **744e** are substantially aligned with the first outlet aperture **703a**, the second outlet aperture **703b**, the third outlet aperture **703c**, the fourth outlet aperture **703d**, and the fifth outlet aperture **703e** of the housing **701**, respectively.

[0150] The dial **745** of the flow controller **740** is rotatably coupled to the housing **701** and movable between a first position, a second position, a third position, a fourth position, and a fifth position relative to the housing **701**. The dial **745** includes an inlet port **721** that can be fluidically coupled to a medical device (either directly or indirectly via an adapter **704**) that defines a fluid flow pathway for withdrawing and/or conveying bodily-fluid from a patient to the collection device **700**. In this manner, the inlet port **721** can be configured to selectively place the pre-sample reservoir **770**, the first sample reservoir **780**, the second sample reservoir **780'**, the third sample reservoir **790**, and the fourth sample reservoir **790'** in fluid communication with the patient, as described in further detail herein. When the dial **745** is in the first position, the flow controller **740** is placed in a first configuration and the inlet port **721** can be substantially aligned with the first aperture **744a** of the seal member **741** and the first outlet aperture **703a** of the housing **701**. In this manner, first aperture **744a** of the seal member **741** establishes fluid communication between the inlet port **721** and the first outlet aperture **703a** while fluidically isolating the inlet port **721** from the outlet apertures **703b**, **703c**, **703d**, and **703e** which in turn, fluidically isolates the inlet port **721** from the fluid chambers **735a-735d**. When the dial **745** is rotated (or actuated) to the second position, the flow controller **740** is placed in a second configuration and the second outlet aperture **744b** establishes fluid communication between the inlet port **721** and the second outlet aperture **703b** while fluidically isolating the inlet port **721** from the pre-sample reservoir **770** and the fluid chambers **735b-735d**. The collection device **700** works in a similar manner when the dial **745** is rotated to the third, fourth and fifth positions. Thus, when the inlet port **721** is placed in fluid communication with the patient (e.g., via the medical device coupled to the inlet port **721**), the first outlet aperture **703a**, the second outlet aperture **703b**, the third outlet aperture **703c**, the fourth outlet aperture **703d**, and the fifth outlet aperture **703e** can be selectively placed in fluid communication with the inlet port **721** to allow all the bodily-fluid to flow into at least one of the pre-sample reservoir **770**, first sample reservoir **780**, or the second sample reservoir **790** (or any other fluid reservoir coupled thereto).

[0151] In some embodiments, the housing **701** can selectively limit movement of the dial **745** from its first position to its second, third, fourth, and fifth positions. In some other embodiments, the housing **701** can be configured to prevent movement of the dial once it has been moved to the fifth position. Said another way, the housing **701** can include a locking mechanism to that prevents the dial **745** from being moved from the fifth position back to the first position. This feature can reduce the risk of contaminating the bodily-fluid collected in the flow chambers **735a-735d** and/or sample reservoirs **780** and **790** from the bodily-fluid contained in the pre-sample reservoir **770** (which has a high risk of containing surface bound microbes and/or other undesirable external contaminants). This locking mechanism can also protect health care practitioners from exposure to blood-borne pathogens in patient samples which can include HIV, Hepatitis C, etc. The dial **745** and/or the housing **701** can also include mechanical detents and/or other indicators that provide visual or tactile feedback to ensure precise positioning of the dial **745** with respect to the outlet port **703a** and outlet apertures **703a-703d** in the housing **701**.

[0152] Similar to the embodiments of the collection device 400 presented in FIGS. 16-22, the collection device 700 includes a pre-sample reservoir 770 that is a chamber contained within the distribution member 729. The pre-sample reservoir 770 can contain bodily-fluids such as, for example, blood, plasma, urine, and/or the like. The pre-sample reservoir 770 is configured to be fluidically coupled to the first outlet port 703a of the collection device 700 (located in the housing 701). During operation of the collection device 700, when the flow controller 740 is in the first position, bodily-fluid is drawn from a part of the body of a patient (e.g., a vein) into the pre-sample reservoir 770, the aperture for the pre-sample reservoir 744a located in the seal member 741, and the first outlet port 703a, via the inlet port 721. The pre-sample reservoir 770 is configured to contain the first amount of the bodily-fluid withdrawn from the patient, where the first amount of bodily-fluid can be a pre-determined or undetermined amount, such that the first amount of bodily-fluid is fluidically isolated from a second and/or third and/or fourth and/or fifth amount of the bodily-fluid that is subsequently withdrawn from the patient.

[0153] In operation, the collection device 700 can be used to collect bodily-fluids (e.g., blood, plasma, urine, etc.) from a patient with reduced contamination. For example, the inlet port 721 of the collection device 700 can be fluidically coupled to a needle or other lumen-defining device (e.g., flexible sterile tubing). Following venipuncture, the dial 745 is rotated until it reaches the first position, as shown in FIGS. 37 and 38. Alternatively, the dial 745 can be pre-set in the first position and the collection device 700 can be otherwise sealed to preserve the sterility of the collection device 700, as described above. With the dial 745 in the first position, the flow controller 740 is placed in a first configuration and the first outlet aperture 744a of the seal member 741 establishes fluid communication between the inlet port 721 and the first outlet port 703a (contained within the housing 701) while fluidically isolating the inlet port 721 from the four sample flow channels 735a-735d. In this first configuration, the bodily-fluid flows from the portion of the body of the patient through the inlet port 721, the first outlet aperture 744a of the seal member 741, the first outlet port 703a of the housing 701, and into the pre-sample reservoir 770 defined by the distribution member 770, as indicated by the arrow KK in FIG. 38. Thus, a first amount (pre-determined or undetermined) of bodily-fluid can be received into the pre-sample reservoir 770 immediately after venipuncture and isolated from subsequent samples, as described in detail above.

[0154] Following collection of the bodily-fluid pre-sample in the pre-sample reservoir 770, the dial 745 can be actuated (or rotated) until it reaches the second position as shown in FIGS. 39 and 40. When the dial 745 is in the second position, the flow controller 740 is placed in a second configuration and the second outlet aperture 744a of the seal member 741 establishes fluid communication between the inlet port 721 and the first fluid chamber 735a while fluidically isolating the pre-sample reservoir 770 from the inlet port 721. Once the first fluid chamber 735a is filled with the bodily-fluid, the flow controller 740 can be moved to a third position to isolate and seal the first fluid channel 735a from an external environment. Additionally, the sample reservoir 780 can be actuated from the first configuration to the second configuration to transfer the bodily-fluid from the first fluid chamber 735a to the sample reservoir 780. For example, the sample reservoir 780 can be actuated (pushed against the

piercing member 755a) from the first configuration to the second configuration by the user, or automatically, to establish fluid communication between a part of the body of a patient (e.g., a vein) and the sample reservoir 780. As described above, moving the sample reservoir 780 to the second configuration causes the piercing member 755a to puncture the vacuum seal of the sample reservoir 780, and be disposed inside the sample reservoir 780. In the second configuration, the part of the body of a patient (e.g., a vein) is exposed to vacuum suction from the sample reservoir 780 due to the negative pressure conditions (vacuum) that in certain embodiments exist inside the sample reservoir 780. Thus, bodily-fluid flows from the part of the body of the patient through the inlet port 721, the second outlet aperture 744b of the seal member 741, the second outlet aperture 703b of the housing 701, the second fluid chamber 735b, and into the first sample reservoir 780, as indicated by the arrow LL in FIG. 40.

[0155] Once a desired volume of bodily-fluid (e.g., the second amount) is collected in the sample reservoir 780, the user can actuate (rotate) the flow controller 740 to the third position and/or move the sample reservoir 780 back to its first configuration to isolate the first sample reservoir 780 from the inlet port 721. When the sample reservoir 780 is back in the first configuration, the piercing member 755a is removed from the sample reservoir 780 and the seal of the sample reservoir 780 (e.g., a self sealing septum) fluidically isolates the first sample reservoir 780 from the second fluid chamber 735b and the external environment. Filling the other sample reservoirs is done in an identical manner with the flow controller 740 in the third, fourth and fifth configurations respectively.

[0156] In some embodiments, the collection device 700 can be constructed such that the set of walls 736 separating the different fluid chambers 735a-735d in the distribution member 729 are not present (see detailed cross-sectional view in FIG. 16). In such embodiments, the distribution member 729 is divided between a pre-sample reservoir 770 and a combined fluid chamber 735 (i.e. the fluid chambers are not separated into four separate sections by the walls 736). In such embodiments, the user can fill all four sample reservoirs at one time by actuating (rotating) the dial 745 to either the second, third, fourth or fifth positions.

[0157] Any of the embodiments described herein can be used with, for example, a metering device that can be used to meter (e.g., quantify) a flow of bodily-fluid into a pre-sample reservoir and/or a sample reservoir. In some instances, laboratory standard practices do not ensure consistent compliance with accurate inoculation volumes of bodily-fluids (e.g., blood specimens) due to the fact that the fill volume is visually determined by the clinician and/or phlebotomist and is thus subject to human error. The fact that the volume indicators on the blood collection bottle are difficult to read when being held and that often the collection bottle is not held upright during the draw procedure can contribute to inaccurate volumes of a bodily-fluid sample received from a patient. Insufficient sample volumes (e.g., below the manufacturer's recommendation) can decrease the sensitivity of culture tests, leading to false-negative results. Additionally, fill volumes above manufacturer's recommendations can cause false-positivity as is indicated in overview materials and instructions for use for specific types of testing supplies and apparatuses (e.g., blood culture bottles designed for use with automated microbial detection

systems produced by manufacturers such as Becton Dickinson, Franklin Lakes, NJ). Thus, flow metering and volume display features can allow a lab technician and/or a health care practitioner (e.g. phlebotomist) to confirm the volume of bodily-fluid that is collected into each individual sample reservoir before placing the sample reservoirs in an incubator or into other laboratory test equipment depending on how the sample needs to be processed. The lab technician and/or phlebotomist can also record (e.g., in a medical record, database, spreadsheet, etc.) the precise volume information for a clinician to evaluate when results are received, thereby helping reduce the possibility of misinterpretation of false-negative and/or false-positive results.

[0158] By way of example, FIGS. 41-45 illustrate a collection device 800 that can include one or more metering devices. The collection device 800 includes a diversion mechanism 820, a flow controller 840, a display 875, and a sample reservoir 880. As further described herein, the collection device 800 can be moved between a first, a second, and a third configuration to deliver a flow of a bodily-fluid that is substantially free from microbes exterior to the body, such as, for example, dermally residing microbes and/or other undesirable external contaminants. The collection device 800 can be any suitable shape, size, or configuration. For example, aspects and/or portions of the collection device 600 can be substantially similar in form and/or function as corresponding aspects and/or portions of any of the collection devices 100, 200, 300, 400, 500, 600, and/or 700 described above. Thus, such similar aspects and/or portions are not described in further detail herein. By way of example, in some embodiments, the sample reservoir 880 of the collection device 800 can be substantially similar and/or the same in form and function as the sample reservoir 480 included in the collection device 400 of FIGS. 16-22.

[0159] As shown in FIGS. 41-43, the diversion mechanism 820 includes an actuator portion 822 (e.g., a first portion), a medial portion 823 (e.g., a second portion), and a coupling portion 824 (e.g., a third portion). The actuator portion 822 of the diversion mechanism 820 is substantially cylindrical including a set of annular walls that define an inner volume 806. More specifically, the actuator portion 822 includes a first end portion that is substantially closed and a second end portion, opposite the first end portion, that is substantially open to allow access to the inner volume 806. In this manner, the actuator portion 822 can movably receive at least a portion of the flow controller 840, as described in further detail herein. The actuator portion 822 further includes an inlet port 821 and an outlet port 831. The inlet port 821 can be fluidically coupled to a medical device (either directly or indirectly via an adapter 804) that defines a fluid flow pathway for withdrawing and/or conveying bodily-fluid from a patient to the collection device 800, as described in detail above.

[0160] The outlet port 831 of the actuator portion 822 can selectively place a portion of the inner volume 806 of the actuator portion 822 in fluid communication with an inner volume 807 defined by the medial portion 823. As shown in FIG. 43, the medial portion 823 is disposed between the actuator portion 822 and the coupling portion 824. Although not shown in FIGS. 41-45 the medial portion 823 can include a metering device that can be configured to meter a volume of bodily-fluid that is transferred, for example, to the sample reservoir 880. For example, in some embodiments, the flow metering device can be fluidically coupled to the

outlet port 831 to meter a flow of bodily-fluid therethrough. As shown in FIGS. 41 and 42, the medial portion 823 includes a display 875 that can provide, to a user, a visual indicator and/or information that is associated with, for example, a volume of bodily-fluid that has flowed through the outlet port 831. In other embodiments, the flow metering device can be positioned at any other suitable position in or along the diversion mechanism 820.

[0161] The coupling portion 824 can be physically and fluidically coupled to the medial portion 823. For example, in some embodiments, the coupling portion 824 can be partially disposed in the inner volume 807 of the medial portion 823 and at least temporarily coupled thereto via a friction fit, a press fit, a snap fit, a threaded coupling, an adhesive, and/or the like. The coupling portion 824 is configured to receive a portion of the sample reservoir 880 and includes a piercing member 855 that can be used to puncture a vacuum seal of the sample reservoir 880 which can initiate a flow of bodily-fluid, as described in detail above.

[0162] The flow controller 840 of the collection device 800 is at least partially disposed in the inner volume 806 defined by the actuator portion 822 and is movable between a first configuration, a second configuration, and a third configuration. As shown in FIGS. 42 and 43, the flow controller 840 includes a movable member 850 having a first seal member 861, a second seal member 862, and a third seal member 863, and a bias member 859 (e.g., a spring or the like). The seal members 861, 862, and 863 are in contact with an inner surface of the actuator portion 822 that defines the inner volume 806. As such the seal members 861, 862, and 863 can each form a substantially fluid tight seal with the inner surface that can, for example, divide the inner volume 806 of the actuator portion 822 into fluidically isolated portions, as described in further detail herein.

[0163] The movable member 850 is movable within the inner volume 806 between a first position, a second position, and a third position. The arrangement of the movable member 850 can be such that as the movable member 850 is moved between its first, second, and third positions, the seal members 861, 862, and 863 are selectively moved within the inner volume 806. More specifically, the first seal member 861 can be moved concurrently with the movable member 850 as the movable member 850 is moved between its first position, second position, and third position. The second seal member 862 and the third seal member 863 can be fixedly coupled to each other (e.g., disposed at a fixed distance from each other) and slidably disposed about a portion of the movable member 850 which can allow the movable member 850 to move from its first position (see e.g., FIG. 43) to its second position (see e.g., FIG. 44), while the second seal member 862 and the third seal member 863 remain in a substantially fixed position relative to the actuator portion 822. For example, the second seal member 862 and the third seal member 863 can remain in a substantially fixed position as the movable member 850 is moved between its first position and the second position such that the inlet port 821 is disposed on a first side of the second seal member 862, while the outlet port 831 is disposed on a second side, opposite the first side, of the second seal member 862. Thus, when the movable member 850 is in its first position (FIG. 43) and its second position (FIG. 44), the inlet port 821 is in fluid communication with a portion of the inner volume 806 defined between a first seal member 861

and the second seal member **862** and the outlet port is in fluid communication with the a portion of the inner volume **806** defined between the second seal member **862** and the third seal member **863**, as described in further detail herein.

[0164] The arrangement of the flow controller **840** can be such that the first seal member **861** is moved relative to the second seal member **862** and the third seal member **863** when the movable member **850** is moved from its first position to its second position. The movement of the first seal member **861** relative to the second seal member **862** can be such that a space defined therebetween is increased, which can form and/or otherwise define a pre-sample reservoir **870**. Moreover, with the seal members **861** and **862** forming substantially fluid tight seals with the inner surface of the actuator portion **822**, the pre-sample reservoir **870** defined between the first seal member **861** and the second seal member **862** is fluidically isolated from other portions of the inner volume **806**. Thus, the inlet port **821** can be in fluid communication with the pre-sample reservoir **870** when the movable member **850** is moved from its first position to its second position. When the movable member **850** is moved from its second position (see e.g., FIG. 44) to its third position (see e.g., FIG. 45), a portion of the movable member **850** can contact the third seal member **863** to move the first seal member **861**, the second seal member **862**, and the third seal member **863** substantially concurrently within the inner volume **806**. As such, the second seal member **862** can be moved relative to the inlet port **821** such that both the inlet port **821** and the outlet port **831** are in fluid communication with the portion of the inner volume **806** defined between the second seal member **862** and the third seal member **863**, as described in further detail herein.

[0165] In operation, the collection device **800** can be used to collect bodily-fluids (e.g., blood, plasma, urine, etc.) from a patient with reduced contamination. For example, the inlet port **821** of the collection device **800** can be fluidically coupled to a needle or other lumen-defining device (e.g., flexible sterile tubing). With the inlet port **821** coupled to the lumen-defining device, the flow controller **840** can be moved from its first configuration to its second configuration. In this manner, a user can exert a force to move the movable member **850** from its first position to its second position, as indicated by the arrow MM in FIG. 44. As described above, the first seal member **861** is moved concurrently with the movable member **850** such that a space defined between the first seal member **861** and the second seal member **862** is increased, thereby forming and/or defining the pre-sample reservoir **870**. With the first seal member **861** and the second seal member **862** forming a substantially fluid tight seal with the inner surface of the actuator portion **822** that defines the inner volume **806**, the increase in volume between the first seal member **861** and the second seal member **862** produces a negative pressure in the pre-sample reservoir **870**. Thus, once fluid communication is established between a portion of the body of the patient (e.g., a vein) and the pre-sample reservoir **870** (e.g., via the inlet port **821** in FIG. 44), the negative pressure differential between the pre-sample reservoir **870** and the portion of the body of the patient draws the bodily-fluid through the inlet port **821** and into the pre-sample reservoir **870**, as indicated by the arrow NN in FIG. 44. In this first configuration, the flow controller **840** also fluidically isolates the pre-sample reservoir **870** from the outlet port **831**. Therefore, a first amount (predetermined or undetermined) of bodily-fluid can

be received into the pre-sample reservoir **870** immediately after venipuncture (for example) and isolated from subsequent samples. In this manner, the collection device **800** can be used to prevent the first amount of bodily-fluid, which is most likely to contain bodily surface microbes and/or other undesirable external contaminants, from contaminating subsequent amounts of the bodily-fluid samples that are collected and used for diagnostic or other testing that can be impacted by the contaminants. In some embodiments, the metering device can meter the volume of bodily-fluid disposed in the pre-sample reservoir **870** and present a value associated with the volume on the display **875**.

[0166] Following collection of the volume of bodily-fluid pre-sample in the pre-sample reservoir **870**, the movable member **850** can be moved from its second position to its third position to place the flow controller in its third configuration, as indicated by the arrow OO in FIG. 45. As described above, when the movable member **850** is moved from its second position to its third position, the portion of the movable member **850** is placed in contact with the third seal member **863**. Thus, the movable member **850** moves the first seal member **861**, the second seal member **862**, and the third seal member **863** substantially concurrently within the inner volume **806**. As such, the second seal member **862** can be moved relative to the inlet port **821** such that both the inlet port **821** and the outlet port **831** are in fluid communication with the portion of the inner volume **806** defined between the second seal member **862** and the third seal member **863**. Moreover, with the volume of bodily-fluid fluidically isolated in the pre-sample reservoir **870**, movement of the second seal member **862** and the third seal member **863** in the direction of the first seal member **861** is limited (i.e., the bodily-fluid is a substantially incompressible fluid). In this manner, the pre-sample volume of bodily-fluid is sequestered in the pre-sample reservoir **870** and the space defined between the second seal member **862** and the third seal member **863** defines a fluid flow path between the inlet port **821** and the outlet port **831**. In addition, the arrangement of the flow controller **840** is such that when in its third configuration, the first seal member **861** is placed in contact with the bias member **859** and at least a portion of a force exerted by a user on the movable member **850** is operable in deforming, compressing, bending, and/or otherwise reconfiguring the bias member **859**. Thus, the bias member **859** can exert a reaction force on the first seal member **861** that resists the movement of the flow controller **840** from its second configuration to its third configuration, as described in further detail herein.

[0167] The sample reservoir **880** can be positioned relative to the collection device **800** such that the piercing member **855** punctures the vacuum seal of the sample reservoir **880** to be disposed inside the sample reservoir, as described in detail above. The pressure differential between the sample reservoir **880** (e.g., vacuum or negative pressure) and the portion of the body draws the bodily-fluid into the sample reservoir **880**. Said another way, in the second configuration, the flow controller **840** and the diversion mechanism **820** establish a fluid flow path such that bodily-fluid can drawn from the patient, through the inlet port **821**, the portion of the inner volume **806** defined between the second seal member **862** and the third seal member **863**, and the outlet port **831** of the actuator portion **822**, through the medial portion **823** and the piercing member **855** of the coupling portion **824** and into the sample reservoir **880** as indicated by the arrow

PP in FIG. 45. As described above, the metering device (not shown) can meter the volume of bodily-fluid transferred through, for example, the outlet port 831 and can present a value associated with the volume of the bodily-fluid on the display 875.

[0168] Once a desired volume of bodily-fluid (e.g., the second amount) is collected in the sample reservoir 880, the user can remove and/or decrease the force exerted on the movable member 850, thereby allowing the bias member 859 to move the first seal member 861 and the movable member 850 from their third positions towards their second positions. Moreover, with the bodily-fluid disposed in the pre-sample reservoir 870 being substantially incompressible, the movement of the first seal member 861 transfers a force through the volume of bodily-fluid to move the second seal member 862 and the third seal member 863 from their third positions towards their second positions. In some embodiments, the bias member 859 can exert a force on the first seal member 861 that can be operable in moving the second seal member 862 to a fourth position relative to the actuator portion 822 that can, for example, substantially obstruct the inlet port 821. Thus, the inlet port 821 can be fluidically isolated from the inner volume 806 of the actuator portion 822. Furthermore, the piercing member 855 can be removed from the sample reservoir 880 and a seal (e.g., a self sealing septum) can fluidically isolate the bodily-fluid sample from a volume outside of the sample reservoir 880. Filling subsequent sample reservoirs can be similarly performed by disposing the piercing member 855 into a sample reservoir and moving the flow controller 840 to the third configuration to allow a flow of bodily-fluid from the patient to the sample reservoir.

[0169] FIGS. 46-53 illustrate a collection device 900 according to an embodiment. The collection device 900 includes a diversion mechanism 920, a flow controller 940, and sample reservoirs 980, 980', 990 and 990'. As further described herein, the collection device 900 can be moved between a first, a second, a third, a fourth, and a fifth configuration to deliver a flow of a bodily-fluid that is substantially free from microbes exterior the body, such as, for example, dermally residing microbes and/or other undesirable external contaminants. The collection device 900 can be any suitable shape, size, or configuration. For example, aspects and/or portions of the collection device 900 can be substantially similar in form and/or function as corresponding aspects and/or portions of any of the collection devices 100, 200, 300, 400, 500, 600, 700, and/or 800 described above. Thus, such similar aspects and/or portions are not described in further detail herein. By way of example, in some embodiments, the sample reservoirs 980, 980', 990, and 990' of the collection device 900 can be substantially similar and/or the same in form and function as the sample reservoirs 680, 680', 690, and 690', respectively, included in the collection device 600 of FIGS. 26-33.

[0170] The diversion mechanism 920 includes a housing 901, a distribution member 929, and movable members 950a, 950b, 950c, and 950d. The housing 901 is physically and fluidically coupled to the distribution member 929, and provides and/or defines a set of fluid flow pathways for collecting bodily-fluids from the patient. The housing 901 includes a set of displays 975' (e.g., liquid crystal displays (LCDs) or the like) that can be included in and/or otherwise coupled (e.g., electrically and/or mechanically) to a flow metering device, as described in further detail herein. The

housing 901 defines a recess 966, outlet apertures 903a, 903b, 903c, 903d, 903e, and movable member openings 950a, 950b, 950c, 950d (also referred to herein as “openings”). The recess 966 is configured to receive a seal member 941 included in the flow controller 940, as described in further detail herein. The first outlet aperture 903a, the second outlet aperture 903b, the third outlet aperture 903c, the fourth outlet aperture 903d, and the fifth outlet aperture 903e are each configured to define a different fluid flow path in fluid communication with different portions of the distribution member 929. More specifically, the distribution member 929 defines and/or forms at least a portion of a pre-sample reservoir 970 in fluid communication with the first outlet aperture 903a, and a first flow channel 935a in fluid communication with the second outlet aperture 903b, second flow channel 935b in fluid communication with the third outlet aperture 903b, a third flow channel 935c in fluid communication with the fourth outlet aperture 903d, and a fourth flow channel 935 in fluid communication with the fifth outlet aperture 903e.

[0171] As shown in FIGS. 47 and 48, the distribution member 929 defines a chamber or volume that defines at least a portion of the pre-sample reservoir 970. The pre-sample reservoir 970 is configured to contain bodily-fluids such as, for example, blood, plasma, urine, and/or the like. The first outlet aperture 903a of the housing 901 can be substantially aligned with an open portion of the pre-sample reservoir 970 to allow the pre-sample reservoir 970 to receive a flow of bodily-fluid from the patient, as described in detail above with reference to the pre-sample reservoir 470 in FIGS. 16-22. The flow channels 935a-935d extend radially from a center of the distribution member 929 and are arranged such that each flow channel 935a, 935b, 935c, and 935d is fluidically isolated from the pre-sample reservoir 970 and the other flow channels. In this manner, the flow channels 935a, 935b, 935c, and 935d can direct and/or otherwise define a fluid flow path between a first end portion that defines an opening substantially aligned with the outlet apertures 903b, 903c, 903d, and 903e, respectively, and a second end portion that defines an opening or port configured to receive the movable members 950a, 950b, 950c, and 950d, respectively. Although the distribution member 929 is shown in FIGS. 47 and 48 as including flow channels 935a-935d that are substantially closed, in other embodiments, the flow channels 935a-935d can be substantially open as shown and described above with reference to the distribution member 429 of FIGS. 17 and 18. As such, the distribution member 929 of the collection device 900 can function in a substantially similar manner as the distribution member 429 of the collection device 400.

[0172] The movable members 950a, 950b, 950c, and 950d are movably disposed in the openings 905a, 905b, 905c, and 905d, respectively, of the housing 901 and the corresponding openings defined by the second end portion of the distribution member 929. Although not shown in FIGS. 46-53, in some embodiments, the movable members 950a, 950b, 950c, and 950d can be operably coupled to a bias member or the like, as described in detail above with reference to the movable members 250 and 250' of the collection device 200. In this manner, the movable members 950a, 950b, 950c, and 950d can be actuated (e.g., moved) by the user from a first position and a second position relative to the housing 901 and distribution member 929 to direct fluid flow into the first sample reservoir 980, the second fluid reservoir 980', the

third fluid reservoir 990, and the fourth sample reservoir 990', respectively. The movable members 950a, 950b, 950c, and 950d are substantially the same and therefore are described with reference to a single movable member 950 in FIG. 49. Moreover, portions of the movable member 950 can be substantially similar to the movable members 250 and 350 described above. Thus, portions of the movable member 950 are not described in further detail herein. The movable member 950 defines an inner cavity 952 that is in fluid communication with an inlet port 953 and a piercing member 955. The piercing member is substantially similar to those described in detail above. The inlet port 953 extends through a set of walls that defines the inner chamber 952 to selectively place the inner volume 952 of the movable member 950 in fluid communication with the corresponding flow channel 935a, 935b, 935c, or 935d.

[0173] As shown in FIG. 49, the movable member 950 includes a flow control mechanism 967 rotatably disposed in the inner volume 952 and in substantially direct fluid communication with the inlet port 953. The flow metering mechanism 967 can be, for example, a wheel or the like that can include a set of spokes or fins. In this manner, bodily-fluid can enter the inlet port 953 of the movable member 950 and flow past the flow metering device 967, which in turn, can result in a rotation of the flow metering device 967 relative to the movable member 950. Thus, characteristics of the rotation of the flow metering device 967 can be operable in determining a volume of bodily-fluid transferred to the inner volume 952 of the movable member 950, a volumetric flow rate, and/or the like. Although not shown in FIGS. 46-53, the flow control mechanism 967 of the movable member 950 is operably coupled to the display 975' of the housing 901. Thus, as bodily-fluid is transferred, for example, to the sample reservoirs 980, 980', 990, and/or 990', volumetric information associated with the flow of bodily-fluid can be presented on the displays 975'. In this manner, a user can manipulate the collection device 900 to collect a bodily-fluid sample from a patient and can visualize at least one of the displays 975' to determine a precise volume of the bodily-fluid sample transferred to, for example, the sample reservoir 980.

[0174] The flow controller 940 of the collection device 900 includes a dial 945 and a seal member 941. The seal member 941 is disposed in the recess 966 of the housing 901. More particularly, the flow controller 940 can be coupled to the housing 901 such that the seal member 941 is disposed between and in contact with a surface of the housing 901 defining the recess 966 and a surface of the dial 945. The seal member 941 can be configured to form a substantially fluid tight seal with the surface of the dial 945 and the surface of the housing 901 that defines the recess 966, as described in detail above. As shown in FIG. 47, the seal member 941 defines a first aperture 944a, a second aperture 944b, a third aperture 944c, a fourth aperture 944d, and a fifth aperture 944e. The arrangement of the seal member 941 is such that when the seal member 941 is disposed in the recess 966, the first aperture 944a, the second aperture 944b, the third aperture 944c, the fourth aperture 944d, and the fifth aperture 944e are substantially aligned with the first outlet aperture 903a, the second outlet aperture 903b, the third outlet aperture 903c, the fourth outlet aperture 903d, and the fifth outlet aperture 903e of the housing 901, respectively.

[0175] The dial 945 of the flow controller 940 is rotatably coupled to the housing 901 and movable between a first position, a second position, a third position, a fourth position, and a fifth position relative to the housing 901. The dial 945 includes an inlet port 921 that can be fluidically coupled to a medical device (either directly or indirectly via an adapter 904) that defines a fluid flow pathway for withdrawing and/or conveying bodily-fluid from a patient to the collection device 900. In this manner, the inlet port 921 can be configured to selectively place the pre-sample reservoir 970, the first sample reservoir 980, the second sample reservoir 980', the third sample reservoir 990, and the fourth sample reservoir 990' in fluid communication with the patient, as described in further detail herein. The dial 945 can be configured to rotate through the first position, the second position, the third position, the fourth position, and the fifth position in a substantially similar manner as described above with reference to the dial 445 of the collection device 400 and is therefore, not described in further detail herein.

[0176] As shown, the dial 945 can further include a display 975 that can be configured to present volumetric information associated with a flow of bodily-fluid. For example, although not shown in FIGS. 46-53, the dial can include a flow metering device or the like such as the flow metering device 967 included in the movable member 950. In this manner, the flow metering device can meter a flow of bodily-fluid through, for example, the inlet port 921 and can be operably coupled to the display 975 such that volumetric information associated with the flow of bodily-fluid through the inlet port 921 is presented on the display 975 of the dial 945.

[0177] In operation, the collection device 900 can be used to collect bodily-fluids (e.g., blood, plasma, urine, and/or the like) from a patient with reduced contamination. For example, the inlet port 921 of the collection device 900 can be fluidically coupled to a needle or other lumen-defining device (e.g., flexible sterile tubing). Following venipuncture (or other method of accessing bodily-fluid), the dial 945 is actuated (or rotated) until it reaches the first position, as shown in FIGS. 50 and 51. Alternatively, the dial 945 can be pre-set in the first position and the collection device 900 can be otherwise sealed to preserve the sterility of the collection device 900.

[0178] As described above, when the dial 945 is in the first position, the flow controller 940 is placed in the first configuration and the first aperture 944a of the seal member 941 establishes fluid communication between the inlet port 921 and the first outlet port 930 (contained within the housing 901) while fluidically isolating the inlet port 921 from the four flow channels 935a-935d. Additionally, the sample reservoirs 980, 980', 990 and 990' are fluidically isolated from the inlet port 921 in the first configuration and a fluid flow path is defined between a portion of the body of a patient (e.g. a vein) and the pre-sample reservoir 970 as indicated by the arrow QQ in FIG. 51. In this first configuration, the bodily-fluid flows (e.g., by gravitation force, vacuum, etc.) from the portion of the body of the patient through the inlet port 921, the first aperture 944a of the seal member 941, the first outlet port 903a of the housing 901, and into the pre-sample reservoir 970. In the first configuration, the flow controller 940 also fluidically isolates the pre-sample reservoir 970 from the flow channels 935a-935d. Thus, a first amount (predetermined or undetermined) of

bodily-fluid can be received into the pre-sample reservoir **970** immediately after venipuncture and isolated from subsequent samples. In this manner, the collection device **900** can be used to prevent the first amount of bodily-fluid, which is most likely to contain bodily surface microbes and/or other undesirable external contaminants, from contaminating subsequent amounts of the bodily-fluid samples that are collected and used for diagnostic or other testing that can be impacted by the contaminants. Moreover, the display **975** can present, for example, information received from the flow control mechanism (not shown) that is associated with a volume of bodily-fluid transferred to the pre-sample reservoir **970**. Thus, a precise volume of bodily-fluid can be transferred to and fluidically isolated within the pre-sample reservoir.

[0179] Following collection of the bodily-fluid pre-sample in the pre-sample reservoir **970**, the dial **945** can be actuated (or rotated) until it reaches the second position as shown in FIGS. **52** and **53**. When the dial **945** is in the second position, the flow controller **940** is placed in the second configuration and the second aperture **944b** of the seal member **941** establishes fluid communication between the inlet port **921** and the flow channel **935a**, while fluidically isolating the pre-sample reservoir **970** from the inlet port **921**. With the flow controller **940** in the second configuration, the movable member **950a** can be actuated (i.e., depressed) from the first position to the second position by the user to establish fluid communication between the patient (e.g., a vein) and the first sample reservoir **880**. More specifically, the movable member **950** is moved from its first position to its second configuration to pass the piercing member **955** through a vacuum seal of the first sample reservoir **980** to be disposed therein, as indicated by the arrow RR in FIG. **53**.

[0180] While in the second position, the inlet port **953** of the movable member **950** is substantially aligned with, and in fluid communication with, the first flow channel **935a**, which allows the bodily-fluid to flow from the first flow channel **935a**, into the inner cavity **952** of the movable member **950**, and out of the piercing member **955** into the first sample reservoir **980**. The pressure differential between the sample reservoir **980** (e.g., vacuum or negative pressure) and the first flow channel **935a** draws the bodily-fluid into the sample reservoir **980**. Said another way, in the second configuration, the flow controller **940** and the movable member **950a** establish a fluid flow path between the inlet port **921** of the dial **945** and the first sample reservoir **980**, as indicated by the arrow SS in FIG. **53**. Moreover, the flow of bodily-fluid through the movable member **950a** rotates the flow metering mechanism **967** relative to the movable member **950**. Thus, the rotation of the flow metering mechanism **967** can be operable in determining a volume of bodily-fluid sample transferred to the sample reservoir **980**. In addition, the display **975'** can present, for example, information received from the flow control mechanism **967** that is associated with a volume of bodily-fluid transferred to the sample reservoir **980**. Therefore, a precise volume of bodily-fluid can be transferred to the sample reservoir **980**. For example, in some instances, the collection device **900** can be used to collect three sample volumes of 20 mL each in the first sample reservoir **980**, the second sample reservoir **980'**, and the third sample reservoir **990** (i.e., 60 mL of total sample volume collected).

[0181] Once a desired volume of bodily-fluid (e.g., the second amount) is collected in the first sample reservoir **980**, the user can release the movable member **950** allowing the bias member (not shown) to move the back to its first position. With the movable member **950** back in its first position, the piercing member **955** is removed from the first sample reservoir **980** and the seal (e.g., a self sealing septum) fluidically isolates the first sample reservoir **980** from the inner flow channel **935**. The collection device **900** can be used to transfer a second sample volume to the second sample reservoir **980'**, a third sample volume to the third sample reservoir **990**, and a fourth sample volume to the fourth sample reservoir **990'** in the same manner by rotating the dial **945** to its third position, fourth position, and fifth position, respectively.

[0182] In some instances, the bodily-fluid collection device **900** can allow a clinician and/or a phlebotomist to open the package containing the bodily-fluid collection device **900** and remove only the housing **901** (that contains the distribution member **929**) and take the housing **901** to a patient's bedside. The clinician and/or a phlebotomist can perform venipuncture (or employ any other method of accessing patient's bodily-fluid) on the portion of the body of a patient (e.g. a vein) using any standardized technique. Following venipuncture, the clinician and/or a phlebotomist can collect the total blood volume required for all samples. For example, the clinician and/or a phlebotomist can collect a 2.5 mL pre-sample diversion volume and a 10 mL sample volume for each of the four sample reservoirs that amounts to a total of 42.5 mL of collected bodily-fluid (e.g., blood). Following collection of the desired amount of bodily-fluid, the hypodermic needle can be removed from the portion of the body of a patient (e.g. a vein) and the clinician and/or a phlebotomist can place the housing **901** (that contains the bodily-fluid) on top of a 4-pack (or 2-pack) of pre-sterilized sample reservoirs with septum tops that are pre-positioned in a custom tray that matches the geometry of housing **901**. By using such a pre-sterilized pack of sample reservoirs, the clinician does not need to perform the process step of "wiping" the top of the sample reservoirs with a sterilizing agent, thereby reducing the likelihood of contamination if, for example, the reservoir tops are improperly and/or insufficiently sterilized. The clinician and/or a phlebotomist can then activate the automated inoculation of the sample reservoirs with the bodily-fluid with precise volume control. In certain embodiments, after the inoculation of the sample reservoirs is complete, the entire device **900** with volume information displayed for each individual sample reservoir can be sent to the laboratory for analysis. In other embodiments, sample reservoirs **980** and/or **990** can be removed individually and sent to the laboratory for analysis.

[0183] Although, the collection device **900** is shown and described with reference to FIGS. **46-53** as including a set of displays **975** and **975'** that can present volumetric data associated with a volume of bodily-fluid transferred through a portion of the collection device, in other embodiments, a collection device can include any suitable flow metering mechanism having any suitable output indicator. For example, FIGS. **54** and **55** illustrate a diversion mechanism **1020** and a flow controller **1040** according to an embodiment. The diversion mechanism **1020** and the flow controller **1040** can be substantially similar in form and function as the diversion mechanism **920** and the flow controller **940**, respectively. Therefore, similar portions are not described in

further detail herein. The diversion mechanism **1020** and the flow controller **1040** can differ, however, in the arrangement of a set of displays **1075**. For example, the diversion mechanism **1020** includes a housing **1001** that is configured to movably receive a set of movable members **1050a**, **1050b**, **1050c**, and **1050d** that can each include a flow metering mechanism as described above with reference to the movable member **950**. Thus, the movable members **1050a**, **1050b**, **1050c**, and **1050d** can be used to determine a precise volume of bodily-fluid transferred therethrough. As shown in FIG. **55**, the displays **1075** of the housing **1001** can include a set of three lights with a first light with low volume (e.g., 5 mL), a second light associated with medium volume (e.g., 20 mL), and a third light associated with acceptable and/or high volume (e.g., 40 mL). In this manner, as a flow of bodily-fluid is transferred through the flow controller **1040** and the diversion mechanism **1020**, and into, for example, the first movable member **1050a**, the flow metering mechanism included therein can send a signal or the like to the display that is operable in lighting the first light, the second light, and/or the third light according to a volume of bodily-fluid that is transferred through the movable member **1050**.

[**0184**] In other embodiments, the movable members **1050a**, **1050b**, **1050c**, and **1050d** can be moved from a first position to a second, third, or fourth position, relative to the housing **1001**. In such embodiments, the positions can be associated with, for example, an intended volume of bodily-fluid to be transferred to a sample reservoir. For example, in some embodiments, a user can actuate (e.g., move) the movable member **1050a** from its first position to its second position. In such embodiments, the second position can be associated with, for example, a low volume of bodily-fluid (e.g., 10 mL) to be transferred to a sample reservoir. In some embodiments, the housing **1001** and/or the movable member **1050a** can include a detent, lock, catch, protrusion, recess, and/or the like that can temporarily retain the movable member **1050a** in the second position until the low volume amount of sample has been transferred to the sample reservoir. Moreover, once placed in the second position, the display **1075** can be configured to illuminate the first light associated with the low volume to indicate to the user the preset volume of bodily-fluid to be transferred to the sample reservoir. Once the desired volume of bodily fluid is transferred to and fluidically isolated in the sample reservoir, the diversion mechanism **1020** can be configured to automatically return the movable member **1050a** back to its first position. In this manner, the diversion mechanism **1020** and the flow controller **1040** can be physically and fluidically coupled to any number of sample reservoirs and used to transferred a precise volume of bodily-fluid to each sample reservoir.

[**0185**] FIG. **56** is a flowchart illustrating a method **1190** of using a flow-metering transfer device to obtain a predetermined sample volume of a bodily-fluid from a patient. The flow metering transfer device can be any of the transfer devices (also referred to herein as “collection devices”) described herein. By way of example, in some embodiments, the transfer device can be the collection device **900** described above with reference to FIGS. **46-53**. As such, the transfer device can include a diversion mechanism with an inlet port configured to be selectively placed in fluid communication with the patient, a pre-sample reservoir and a sample reservoir, and a flow-metering mechanism config-

ured to meter a flow of bodily-fluid from the patient to the pre-sample reservoir and to the sample reservoir. The method **1190** includes establishing fluid communication between the patient and the port of the flow-metering transfer device, at **1191**. For example, the port can be fluidically coupled to a needle or other lumen-defining device (e.g., flexible sterile tubing), which in turn can be inserted into the patient (e.g., a venipuncture event or other method of accessing bodily-fluid).

[**0186**] With the port in fluid communication with the patient, fluid communication between the port and the pre-sample reservoir is established, at **1192**. In some embodiments, the flow-metering transfer device can include a flow controller or the like (e.g., such as the flow controller **940** included in the collection device **900**) that can be actuated and/or manipulated (e.g., rotated) to a position that establishes fluid communication between the port and the pre-sample reservoir (e.g., a first position). In some embodiments, the actuating of the flow controller can be such that the flow controller and the diversion mechanism collectively define at least a portion of a fluid flow path between the port and the pre-sample reservoir. In some embodiments, the pre-sample reservoir can include a negative pressure or the like that can, for example, initiate a flow of bodily-fluid from the patient to the pre-sample reservoir. In other embodiments the flow of bodily-fluid can be initiated in any other suitable manner (e.g., gravity or the like).

[**0187**] The flow of bodily-fluid transferred from the patient to the pre-sample reservoir is metered, at **1193**. For example, in some embodiments, the port can include the flow control mechanism which can be meter a flow of bodily-fluid that passes through the port (e.g., in a similar manner as described above with reference to the flow control mechanism **967** of the collection device **900**). Thus, a pre-sample volume of bodily-fluid is transferred to the pre-sample reservoir. The method **1190** includes verifying the pre-sample volume of bodily-fluid disposed in the pre-sample reservoir is a predetermined pre-sample volume of bodily-fluid via the flow metering mechanism of the flow-metering transfer device, at **1194**. For example, the flow metering mechanism can include and/or can be operably coupled to a display of the like (e.g., the display **975** and/or **975'** of the collection device **900**). The flow metering mechanism can be configured to present on the display volumetric information, as described above.

[**0188**] Once the pre-sample volume of bodily-fluid is disposed in the pre-sample reservoir, the pre-sample reservoir is fluidically isolated from the port to sequester the pre-sample volume of bodily-fluid in the pre-sample reservoir, at **1195**. For example, in some instances, the flow controller and/or the diversion mechanism can be actuated (or rotated) from the first position and/or configuration to a second position and/or configuration. With the flow controller and/or diversion mechanism in the second configuration, the pre-sample reservoir is fluidically isolated from a volume outside of the pre-sample reservoir. In some embodiments, when the flow controller and/or diversion mechanism is actuated to its second position and/or configuration, fluid communication is established between the port and a sample reservoir, at **1196**. For example, in some embodiments, the flow-metering transfer device can include a movable member (e.g., the movable member **950**) or the like that can include a piercing member configured to pierce a portion of the sample reservoir (e.g., a septum or the like). Therefore,

with the flow controller and/or diversion mechanism in its second position and/or configuration, the piercing of the portion of the sample reservoir places the sample reservoir in fluid communication with the port. As described above, the sample reservoir can include a negative pressure or the like that can, for example, initiate a flow of bodily-fluid from the patient to the sample reservoir.

[0189] The flow of bodily-fluid transferred from the patient to the pre-sample reservoir is metered, at **1197**. For example, as described above, the port can include the flow control mechanism which can be meter a flow of bodily-fluid that passes through the port (e.g., in a similar manner as described above with reference to the flow control mechanism **967** of the collection device **900**). In some embodiments, the flow control mechanism can be included in, for example, a movable member or the like such as the movable member **950** of FIG. **49**. Thus, a sample volume of bodily-fluid is transferred to the sample reservoir. The method **1190** includes verifying the sample volume of bodily-fluid disposed in the sample reservoir is a predetermined sample volume of bodily-fluid via the flow metering mechanism of the flow-metering transfer device, at **1198**. For example, the display or the like can be configured to present volumetric information, as described above.

[0190] In this manner, the predetermined pre-sample volume of bodily-fluid is collected that can contain, for example, externally residing microbes. For example, in some embodiments, the predetermined pre-sample volume can be about 0.1 mL, about 0.3 mL, about 0.5 mL, about 1.0 mL, about 2.0 mL, about 3.0 mL, about 4.0 mL, about 5.0 mL, about 10.0 mL, about 20 mL, about 50 mL, and/or any volume or fraction of a volume therebetween. In other embodiments, the pre-sample volume can be greater than 50 mL or less than 0.1 mL. In other embodiments, the predetermined pre-sample volume can be between about 2 mL and about 5 mL. In one embodiment, the predetermined pre-sample volume can be about 3 mL. Furthermore, by collecting the predetermined pre-sample volume, the predetermined sample volume disposed in one or more sample reservoirs can be substantially free-from externally residing microbes. In some embodiments, the predetermined sample volume can be between 10 mL and 60 mL. In other embodiments, the predetermined sample volume can be between 30 mL and 60 mL. In still other embodiments, the predetermined sample volume can be 60 mL. Although described above as transferring the sample volume of the bodily-fluid to a single sample reservoir, in other embodiments, the flow-metering transfer device can be used to transfer a predetermined sample volume to more than one sample reservoir. For example, in some embodiments, a pre-determined pre-sample volume of bodily-fluid can be collected and fluidically isolated in a pre-sample reservoir, as described above. With the pre-sample volume fluidically isolated, the flow-metering transfer device can be used to transfer a predetermined sample volume to a first sample reservoir, the predetermined sample volume to a second sample reservoir, and the predetermined sample volume to a third sample reservoir. In such instances, the predetermined sample volume can be, for example, 20 mL such that a total sample volume disposed in the first, second, and third sample reservoirs is 60 mL.

[0191] The various embodiments of the bodily-fluid collection devices described herein can allow the collection of two (or more) sets of bodily-fluids (e.g., blood) samples

from a single venipuncture. The current standard of care dictates that certain tests (e.g. blood cultures) be conducted with samples procured from distinct, separate bodily-fluid access points (e.g. via two separate venipunctures, via a catheter+venipuncture and/or any combination thereof). Embodiments described herein can facilitate the procurement of multiple samples for specific diagnostic testing (e.g. blood culture test) from a single bodily-fluid access point (e.g. venipuncture), which can reduce the annual number of venipunctures required for procurement of these samples by a factor of 2. This benefits both patients and health care practitioners alike. A reduction in the number of venipunctures (and/or other bodily-fluid access procedures) can significantly reduce the risk of needle stick injury to health care practitioners and reduce patient associated complications which result from these procedures (e.g. hematoma, thrombosis, phlebitis, infection, etc.). Additionally, reducing the number of bodily-fluid access procedures (e.g. venipunctures) reduces the utilization of supplies, labor and waste associated with these procedures. The decreased costs realized by the healthcare system are material and represent an opportunity to drive more efficient consumption of resources as well as enhanced patient outcomes due to improved sample integrity which results in more accurate patient diagnoses which inform development and implementation of treatment plan(s). The bodily-fluid collection devices also significantly reduce the occurrence of false-positives from post-collection analysis. The bodily-fluid collection devices described herein can also streamline the bodily-fluid collection process and reduce the number of manual steps and “touch points”, thereby decreasing opportunities for external contamination. The devices described herein can also minimize the risk for needle stick injuries and infection for the lab technicians and/or phlebotomists.

[0192] In some embodiments, the bodily-fluid collection devices described herein (e.g., **100**, **200**, **300**, **400**, **500**, **600**, **700**, **800**, and **900**) can include and/or be partially formed from antiseptics saturated materials (e.g., housing **401**). Current standards rely on health care practitioners placing individual antiseptics materials (e.g. isopropyl alcohol swabs) on the top of individual sample reservoirs (e.g., **480**, **480'**, **490**, and **490'**). To ensure compliance with this protocol, the device **400** (for example) can include antiseptics materials positioned in the device **400** such that when the housing **401** is placed on top of the 4-pack (or 2-pack) of bottles as illustrated in FIG. **16**, the first point of contact from the housing **401** and the tops of the sample reservoirs **480**, **480'**, **490**, **490'** is the antiseptics material. In this manner, the tops of the sample reservoirs **480**, **480'**, **490**, **490'** are assured to have an appropriate antiseptics applied prior to inoculation of the bodily-fluid into the sample reservoirs.

[0193] While various embodiments have been particularly shown and described, various changes in form and details may be made. For example, while the dial **445** (actuator) is shown and described with respect to FIGS. **19-22** as being rotated in a single direction, in other embodiments, the dial **445** (actuator) can be rotated in a first direction and a second direction, opposite the first. In such embodiments, the rotation in the second direction can be configured to move a collection device through any number of configurations. In other embodiments, the rotation of the actuator in the second direction can be limited. In some embodiments, the dial can include a mechanical stop or lock to fluidically isolate the first volume of bodily-fluid received from the patient (i.e.,

the contaminated sample). Said another way, once the first reservoir (pre-sample reservoir) is filled with a predetermined volume of bodily-fluid and the user has rotated the dial (actuator) to begin drawing additional sample, the dial (actuator) cannot be moved back to establish fluid communication with the first sample volume (contained in the pre-sample reservoir).

[0194] While embodiments have been described above, it should be understood that they have been presented by way of example only, and not limitation. Although various embodiments have been described as having particular features and/or combinations of components, other embodiments are possible having any combination or sub-combination of any features and/or components from any of the embodiments described herein. For example, while the collection device 700 is shown and described with respect to FIGS. 34-40 as having a first, second, third, fourth, or fifth configuration, in other embodiments, the collection devices described herein may have more or fewer configurations. In addition, while the collection device 200 is shown and described with respect to in FIGS. 2-13 as having a vacuum based collection tube as the pre-sample reservoir 270, in other embodiments, the collection device 200 can have a chamber contained within the housing 201 similar to the collection device 400 of the embodiment presented in FIGS. 16-22, which includes a pre-sample reservoir 470 that is a chamber contained within the distribution member 429, and vice versa.

[0195] The specific configurations of the various components can also be varied. For example, the size and specific shape of the various components can be different than the embodiments shown, while still providing the functions as described herein. More specifically, the size and shape of the various components can be specifically selected for a desired rate of bodily-fluid flow into a fluid reservoir. Furthermore, while the flow metering mechanism 967 is particularly shown in FIG. 49, any of the collection devices described herein can be used with any suitable flow metering mechanism. For example, in some embodiments, a collection device can include a flow metering mechanism and/or any other mechanism, device, or method configured to measure volumetric characteristics of a bodily-fluid such as, for example, a pressure sensor, a voltage sensor, a photo sensor, a velocity sensor, a flow meter, a strain gauge, a valve, a turbine, a float, displacement analysis, density analysis, weight analysis, optical analysis, ultrasound analysis, thermal analysis, Doppler analysis, electromagnetic field (emf) analysis, reflection analysis, obstruction analysis, area analysis, venturi analysis, coriolis analysis, visual analysis, and/or any other suitable sensor, analysis, and/or calculation (e.g., applying and/or using, for example, Boyle's law, ideal gas law, force calculation ($\text{force} = \text{mass} * \text{acceleration}$), and/or the like).

1-41. (canceled)

42. An apparatus, comprising:

a housing defining an inlet configured to be placed in fluid communication with a patient, an outlet configured to be placed in fluid communication with a sample reservoir, and a pre-sample reservoir configured to contain blood from the patient;

a seal member disposed in an inner volume of the housing and defining a portion of the pre-sample reservoir, the seal member configured to transition from a first configuration to a second configuration, the seal member

further configured to facilitate a negative pressure differential in the pre-sample reservoir to draw a pre-sample volume of blood into the pre-sample reservoir such that the pre-sample volume of blood is contained; and

a movable member disposed in the inner volume of the housing, the movable member configured to move, after the seal member transitions to the second configuration, to establish fluid communication between the inlet and the outlet such that a sample volume of blood flows from the inlet into the sample reservoir.

43. The apparatus of claim 42, wherein the pre-sample volume of blood flows from the inlet toward the seal member.

44. The apparatus of claim 42, wherein the movable member prevents blood from flowing from the inlet to the outlet until the pre-sample volume of blood is contained in the pre-sample reservoir.

45. The apparatus of claim 42, wherein the pre-sample volume of blood in the pre-sample reservoir includes dermally-residing contaminants and the sample volume of blood transferred to the sample reservoir is mostly free from dermally-residing contaminants.

46. The apparatus of claim 42, wherein the movable member is configured to form a portion of a fluid flow path separate from the pre-sample reservoir such that the sample volume of blood bypasses the pre-sample reservoir containing the pre-sample volume of blood therein.

47. The apparatus of claim 42, wherein the pre-sample volume of blood flows on a first side of the seal member, the sample volume of blood flows on a second side of the seal member.

48. The apparatus of claim 42, wherein the movable member moves substantially perpendicular to the inlet and the outlet.

49. The apparatus of claim 42, wherein the pre-sample reservoir is defined between the seal member and a portion of the movable member.

50. The apparatus of claim 42, wherein the seal member contacts an inner wall of the housing and prevents blood from flowing from the inlet to the outlet before the seal member is in the second configuration.

51. The apparatus of claim 42, wherein the negative pressure differential is a first negative pressure differential across a first portion of the inner volume of the housing, the seal member and the movable member configured to facilitate a second negative pressure differential across a second portion of the inner volume of the housing when the movable member establishes fluid communication between the inlet and the outlet.

52. An apparatus, comprising:

a housing defining an inlet configured to be placed in fluid communication with a patient, an outlet configured to be placed in fluid communication with a sample reservoir, and a pre-sample reservoir configured to contain blood from the patient; a seal member disposed in the housing and configured to transition from a first configuration to a second configuration, the seal member further configured to facilitate a negative pressure differential in the pre-sample reservoir to draw a pre-sample volume of blood into the pre-sample reservoir; and

a movable member movable in an inner volume of the housing, the movable member configured to move,

after the pre-sample volume of blood is disposed in the pre-sample reservoir, through a portion of the inner volume of the housing such that a flow path is established between the inlet and the outlet that bypasses the pre-sample reservoir and the pre-sample volume of blood therein.

53. The apparatus of claim **52**, wherein the pre-sample volume of blood flows from the inlet toward the seal member.

54. The apparatus of claim **52**, wherein the movable member prevents blood from flowing from the inlet to the outlet until the pre-sample volume of blood is contained in the pre-sample reservoir.

55. The apparatus of claim **52**, wherein the pre-sample volume of blood in the pre-sample reservoir includes dermally-residing contaminants and the sample volume of blood transferred to the sample reservoir is mostly free from dermally-residing contaminants.

56. The apparatus of claim **52**, wherein the pre-sample reservoir is defined at least in part by the seal member and inner walls of the housing.

57. The apparatus of claim **56**, wherein the pre-sample volume of blood is a predetermined volume, the pre-sample reservoir is configured to have an inner volume corresponding to the predetermined volume.

58. The apparatus of claim **52**, wherein the pre-sample volume of blood flows on a first side of the seal member, and the sample volume of blood flows on a second side of the seal member.

59. The apparatus of claim **52**, wherein the movable member moves substantially perpendicular to the inlet and the outlet.

60. The apparatus of claim **52**, wherein the pre-sample reservoir is defined between the seal member and a portion of the movable member.

61. The apparatus of claim **52**, wherein the seal member contacts an inner wall of the housing and prevents blood from flowing from the inlet to the outlet before the seal member is in the second configuration.

62. An apparatus, comprising:

- a housing including an inlet configured to be placed in fluid communication with a patient, an outlet configured to be placed in fluid communication with at least one sample reservoir, and a pre-sample reservoir configured to contain blood from the patient;
- a seal member defining a portion of the pre-sample reservoir; and
- a movable member disposed in an inner volume of the housing,

the seal member and the movable member in a first configuration configured to facilitate formation of a first negative pressure differential across a first portion of the inner volume of the housing between the inlet and the outlet to draw a pre-sample volume of blood from the inlet into the pre-sample reservoir,

the seal member and the movable member configured to transition to a second configuration to facilitate formation of a second negative pressure differential across a second portion of the inner volume of the housing between the inlet and the outlet to draw a sample volume of blood from the inlet, past the pre-sample reservoir, and into the at least one sample reservoir.

63. The apparatus of claim **62**, wherein the pre-sample volume of blood flows from the inlet toward the seal member.

64. The apparatus of claim **62**, wherein the movable member prevents blood from flowing from the inlet to the outlet until the pre-sample volume of blood is contained in the pre-sample reservoir.

65. The apparatus of claim **62**, wherein the pre-sample volume of blood in the pre-sample reservoir includes dermally-residing contaminants and the sample volume of blood transferred to the at least one sample reservoir is mostly free from dermally-residing contaminants.

66. The apparatus of claim **62**, wherein the pre-sample reservoir is defined at least in part by the seal member and inner walls of the housing.

67. The apparatus of claim **66**, wherein the pre-sample volume of blood is a predetermined volume, the pre-sample reservoir is configured to have an inner volume corresponding to the predetermined volume.

68. The apparatus of claim **62**, wherein the pre-sample volume of blood flows on a first side of the seal member, the sample volume of blood flows on a second side of the seal member.

69. The apparatus of claim **62**, wherein the movable member moves substantially perpendicular to the inlet and the outlet.

70. The apparatus of claim **62**, wherein the pre-sample reservoir is defined between the seal member and a portion of the movable member.

71. The apparatus of claim **62**, wherein the seal member contacts an inner wall of the housing and prevents blood from flowing from the inlet to the outlet before the seal member is in the second configuration.

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